Foundation Trilogy

Selection, sharing and return

Complete Collected Volume

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Introduction

A structure that can be described need not be a structure that can be realized or shared. The Foundation papers carry this distinction through three stages: what a source and its noise let a reader select, what several readers actually share, and what a returned action changes while the old content and history are retained.

A shared algebra does not imply a shared state, and agreement does not establish correctness. A physically implemented return is stronger than updating an external estimate. The papers supply constructions and explicit conditions for these transitions, including preparation, calibration and receiver renewal rather than assuming them from an algebraic name.

One finite semiclassical mean-action scalar/JT construction joins old-content protection, the apparatus's own source and a second reading on the same retained history. An acquired record installs protection for old quantum content, changes that source, and receives its geometric response in a later X-basis reading of the old probe into a spent cell. Initial mass and geometric constraints are held fixed; task benefit and return occur in the same finite window. It does not simultaneously implement every Type III hull or a multi-reader public structure. Sustained thermal renewal is a further requirement, not part of this finite certificate.

Reading map

F-I *Foundation I: From Source to Manifestation. Accessible Algebras from a Source, Its Noise and a Readable Seed.* A source, its noise and a readable seed define an accessible hull. Selection and finite positive-energy preparation require their declared dynamical permissions.

F-II *Foundation II: Shared Manifestation. Shared Structure, Public Calibration and Different Histories.* Shared structure does not impose identical states or reliable common beliefs. Public references distinguish agreement, correctness and shared error.

F-III *Foundation III: The Perfuming Loop. Held Content, Learned Frames and Physical Return.* A return preserves old content while learned frames, physical action and source response change what can be read next. Receiver renewal is not renewal of an entire apparatus.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.

Foundation I: From Source to Manifestation

Accessible Algebras from a Source, Its Noise and a Readable Seed

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Abstract

A source, its noise and a physical reader supply different constraints on the structure in which a world can be read. We separate finite measurement permissions from the ideal algebra generated by their effects. A partially mixed record probe generates the same pointer algebra at every nonzero gain, although its sharp-test error approaches one half as the gain vanishes. Repeated fresh nondemolition probes give a finite instrument-level approximation to the sharp measurement, including old references; this supplies a resource-sensitive bridge to finite tasks.

For a stationary source with a faithful state, state-preserving noise has a modular-invariant fixed algebra and a unique state-preserving expectation. Its Cesàro projection need not be a long-time limit of individual trajectories. A stable seed has a modular hull inside that fixed algebra. If the ambient source has positive translations with only the vacuum invariant, every nonscalar hull closed under forward translations inherits a proper Cut on its vacuum space. A minimal compact symmetry identifies any hull containing the symmetry-fixed algebra with the fixed algebra of its seed stabilizer, provided the closure maps preserve it. In the SU(2) level-one net, a positive log-bump stress seed generates the Virasoro half-line algebra under modular evolution alone and its one-sided orbit forms a standard Cut on the Virasoro vacuum space. One bounded current seed generates the current algebra under modular flow alone; its character distinguishes the stabilizer up to conjugacy. Axis dephasing leaves three hulls for the stated stress/current seeds, and weak symmetry breaking determines their passive lifetimes. Stress excitations have the same relative entropy across these hulls, whereas current excitations distinguish them. These results derive the source–access–Cut chain in its net-theoretic geometry. The ambient net, available couplings and preparation permissions remain explicit inputs; forward translations are unnecessary for the stated log-bump stress seeds.

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1 Introduction

Consciousness I [2] works with an accessible algebra A inside a global algebra M with vacuum Ω. When A contains a Cut, a subalgebra stable under the modular evolution in one direction only, the translations of a chiral conformal net are computed from modular data, and for a standard Cut the net and the modular data determine each other. In the SU(2) level-one current net the same paper finds that the algebra fixed by an explicit rotation noise, the $c = 1$ Virasoro algebra of a half-line, coincides with the algebra generated from one bounded stress observable by dilations and forward translations. For a positive log-coordinate bump with monotone ends, the construction needs only modular evolution: its one-sided orbit forms a proper standard Cut on the Virasoro vacuum space, and its full modular hull is the Virasoro half-line algebra. The origin of the ambient structure and of physically available seed operations remains open; the present paper asks which accessible algebra a source and seed select.

Foundation I is the source end of that chain. Its question is what an actual source, physical access to it and persistent organization fix on the way to an accessible algebra, and what they leave free. The HAFF papers supply part of the answer in finite dimension. Paper A [3] shows that two channels can generate the same algebra and still resolve different things, so that one state and one reconstruction rule give different geometries. Paper B [4] derives the exact fixed algebra in the commuting-coupling class and the unique largest block algebra passing its stated window criteria. Block-constant recording couplings expose central labels, while protected relations, classical sector correlations and record-conditioned formation have different roles. Its collective recording and Hamiltonian-link constructions show how actual process resources create relations; the noise graph alone does not determine all correlations. Paper E [6] separates what a probe can implement, what the dynamics leaves fixed and what fragments record, and computes all three in one collision model. For its Gaussian linear-response task with quadratic loss and an unrestricted rank-k projector menu, T-DOME II [8] identifies a top-k posterior-task eigenspace as an optimal reading subspace. A restricted physical instrument menu need not realize that optimizer. What these results do not say is how a seed obtained from access grows into an algebra that can carry geometry, when the result is proper, how it depends on the seed, and whether different seeds give different geometries. HAFF-D lists this as the missing selection step: with a faithful ambient state the ambient algebra itself passes every modular and semigroup stability test, and a nontrivial selection needs a seed and a minimal hull [5].

The objects are as follows. A *source* is a von Neumann algebra $\mathcal{A}_U$ with a cyclic and separating vector Ω, its state ω, and a semigroup $\{\mathcal{E}_t\}$ of normal unital completely positive maps describing the environment in the Heisenberg picture. The modular group σ_s of ω is fixed by the source and needs no further input; for a thermal state it is the time evolution in units of the inverse temperature, and for the vacuum of a wedge it is the boost group. *Access* supplies actual effects through physical probes; a reader may keep a set of them as a *seed*. Algebraic functional calculus then describes the ideal distinctions encoded by that seed. It is a further closure, not an additional measurement permission. The *stable hull* of the seed is the smallest von Neumann algebra that contains it and is closed under the modular flow and environmental dynamics. When an ambient positive translation structure is available, we also study its closure under forward translations. These are distinct closures: translation stability must be imposed or proved before a Cut can be inherited.

The argument proceeds from physical access to persistent structure:

1. Section 3 gives the finite pointer-instrument bridge, including old references and adaptive task error.

2. Sections 4–5 derive the state-preserving expectation and modular stable hull from stationarity, and separate this projection from dynamical relaxation.

3. Section 6 proves proper Cut inheritance for translation-stable nonscalar hulls with unique invariant vacuum.

4. Section 7 identifies the classified hull by its seed stabilizer. Current and positive log-bump stress seeds admit modular-only generation; characters distinguish stabilizers and symmetry-breaking noise fixes passive lifetimes.

5. Sections 8–9 separate ideal seed closure from finite coding budgets and distinguish seed-independent stress response from seed-dependent current response.

The ambient algebra, its translations, the coupling to the environment and the task that fixes the foreground remain inputs; Section 10 lists them with the conjectures that would remove some of them. The finite-dimensional statements are checked by scripts described in Appendix B.

2 Source, access, seed and hull

Definition 2.1 (Source)**.** A *source* is a quadruple $(\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$: a von Neumann algebra $\mathcal{A}_U$ on $\mathcal{H}_U$, a vector Ω cyclic and separating for $\mathcal{A}_U$, and a semigroup of normal unital completely positive maps $\mathcal{E}_t : \mathcal{A}_U \to \mathcal{A}_U$, pointwise continuous in the σ-weak topology. We write $\omega = \langle\Omega, \cdot\,\Omega\rangle$, Δ, J for the modular operator and conjugation of $(\mathcal{A}_U, \Omega)$, $\sigma_s = \mathrm{Ad}\,\Delta^{is}$ for the modular group, and

$$\mathrm{Fix}(\mathcal{E}) = \{x \in \mathcal{A}_U : \mathcal{E}_t(x) = x \text{ for all } t \geq 0\}.$$

The source is *stationary* if $\omega \circ \mathcal{E}_t = \omega$ for all t. An *ambient positive translation structure* (a Borchers pair with $\mathcal{A}_U$) is a strongly continuous unitary group $U(a) = e^{iaP}$ with $P \geq 0$, $U(a)\Omega = \Omega$ and $U(a)\mathcal{A}_U U(a)^* \subseteq \mathcal{A}_U$ for $a \geq 0$.

The ambient positive translation structure is the structure of a half-line algebra of a chiral net in its vacuum, or of a wedge algebra with a lightlike translation. By Borchers' theorem it satisfies $\Delta^{it}U(a)\Delta^{-it} = U(e^{-2\pi t}a)$ and $JU(a)J = U(-a)$ [12], and $U(1)\mathcal{A}_U U(1)^* \subset \mathcal{A}_U$ is then a half-sided modular inclusion whose translation group, in the sense of Wiesbrock's theorem, is U [31, 10]. We use it only where it is stated.

Access and seed. Paper E of HAFF describes access by a context: allowed probe channels $\mathcal{R}_\alpha^* : \mathcal{B}(\mathcal{K}_\alpha) \to \mathcal{A}_U$, whose images of probe effects are the implementable effects Eff, the environmental dynamics with its fixed set, and a record algebra [6]. Paper B identifies central record labels for its block-constant recording couplings, not for every possible environment instrument [4]. A seed $S \subseteq \mathcal{A}_U$ is a set of bounded self-adjoint observables. A *physical effect seed* is selected from the actual probe effects and retained under a declared budget (Section 8); an *ideal spectral seed* may additionally contain bounded Borel functions of those effects. Both enter the same mathematical hull, but the latter need not be implementable by the former probe menu. A seed is *stable* if $S \subseteq \mathrm{Fix}(\mathcal{E})$. The hull definitions do not depend on the mechanism selecting the seed. Section 3 supplies a nonempty physical bridge and exhibits the information lost by identifying permissions with this ideal closure.

Definition 2.2 (Stable hull)**.** Let Γ be a family of normal maps of $\mathcal{A}_U$ into itself that contains the modular group, $\{\sigma_s\}_{s\in\mathbb{R}} \subseteq \Gamma$. A von Neumann subalgebra $\mathcal{N} \subseteq \mathcal{A}_U$ is Γ-*admissible* if $\gamma(\mathcal{N}) \subseteq \mathcal{N}$ for every $\gamma \in \Gamma$. The Γ-*hull* of a seed S is

$$\mathrm{Hull}_\Gamma(S) = \bigcap\{\mathcal{N} : \ \mathcal{N} \ \Gamma\text{-admissible}, \ S \subseteq \mathcal{N}\}.$$

We use three families: $\Gamma_\sigma = \{\sigma_s\}$ (the *modular hull*), $\Gamma_2 = \{\sigma_s\} \cup \{\mathcal{E}_t\}$ (the *stable hull*), and, in the presence of an ambient positive translation structure, $\Gamma_3 = \Gamma_2 \cup \{\mathrm{Ad}\, U(a)\}_{a\geq 0}$ (the *translation-stable hull*). Existence of an ambient positive translation structure alone does not identify the Γ_2- and Γ_3-hulls.

Since $\sigma_{-s} \in \Gamma$, an admissible algebra satisfies $\sigma_s(\mathcal{N}) = \mathcal{N}$. For a seed of exact pointer observables realising a foreground frame, Hull_{Γ_2} is the minimal modular–decoherence hull of that frame. The (σ, U)-hull of Consciousness I is Hull_{Γ_3} without the noise maps [2]. HAFF-D calls the Γ_2-admissible algebras modular-stable geometric candidates and notes that the class contains the ambient algebra, so that a selection needs a seed [5].

3 From a physical effect to a finite task

A von Neumann algebra closes under bounded spectral functions, products and operator limits. Probe permissions need not close under these operations. Even the linear inverse of a noisy effect may require a negative coefficient, so it cannot in general be performed by randomizing permitted tests.

Proposition 3.1 (Same algebra, different sharp permissions)**.** *Let P_0, P_1 be the pointer projections of a qubit. A fresh memory $\eta_\alpha = (\mathbf{1} + \alpha Z)/2$, $0 \leq \alpha \leq 1$, interacts through*

$$V = P_0 \otimes \mathbf{1} + P_1 \otimes X.$$

A Z readout of the memory implements

$$F_\alpha = \tfrac{1+\alpha}{2} P_0 + \tfrac{1-\alpha}{2} P_1.$$

For $\alpha > 0$, $W^*(F_\alpha) = W^*(P_0)$, and every hull in Definition 2.2 gives the same result for these two seeds. With arbitrary memory effects but only this single collision, the closest implementable effect to P_0 has operator-norm error $(1-\alpha)/2$. Thus equality of the ideal hulls does not give equality of their sharp-measurement permissions.

Proof. For $\alpha > 0$, $P_0 = [F_\alpha - (1-\alpha)\mathbf{1}/2]/\alpha$; each seed belongs to the algebra of the other. Paper E computes the exact single-port permission set as the parallelogram with vertices $(0,0), (1,1), ((1+\alpha)/2, (1-\alpha)/2)$ and its transpose, where (a,b) denotes $aP_0 + bP_1$ [6]. In that set $a - b \leq \alpha$, so $\max(1-a, b) \geq (1-\alpha)/2$; F_α attains the bound. At $\alpha = 0$ all effects are scalar and the seed algebra is trivial. □

All nonzero gains therefore have the same ideal structure, while their finite-read abilities approach the uninformative probe continuously as $\alpha \downarrow 0$. The spectral inverse is an algebraic identity. It neither sharpens a one-shot probe nor specifies its back-action.

There is an abstract bounded coupling for an unsharp version of any self-adjoint contraction O. Put

$$\vartheta(O) = \tfrac{1}{2}\arccos(\alpha O), \qquad V_O = \exp[-i\vartheta(O) \otimes Y].$$

A pure qubit port prepared in $|0\rangle$ and read in Z yields $(\mathbf{1}+\alpha O)/2$. For a chosen gate duration t_g, $H_O = (\overline{h}/t_g)\vartheta(O)\otimes Y$ is bounded, and adding $\pi\overline{h}\mathbf{1}/(2t_g)$ makes it positive without changing the instrument. For $\alpha > 0$, this effect generates $W^*(O)$; if O is fixed by stationary noise, its spectral functions are fixed by Theorem 4.1(a). This construction specifies the required interaction, preparation and readout. It does not derive a local field Hamiltonian implementing $\vartheta(O)$, or grant that coupling to a reader whose original menu lacks it. In particular, the current and stress seeds below require a physical bounded-function readout or another justified representation of that readout.

Proposition 3.2 (Finite sharp-instrument approximation)**.** *Fix $0 < \alpha \leq 1$. Let $m \geq 1$ fresh memories of polarization α undergo the preceding controlled-X collision with the* same *source qubit. Read each memory in Z, retain the majority label, and use a fair independent tie breaker if m is even. No rotation of the source occurs between collisions. Let e_m be the majority error of m independent binary tests of error $(1-\alpha)/2$. After the fine records are discarded, the coarse instrument, including its output source, is*

$$\mathcal{I}_m(\rho) = \sum_{y=0,1} |y\rangle\langle y| \otimes \sum_{z=0,1} q_m(y|z) P_z \rho P_z, \qquad q_m(z|z) = 1 - e_m, \quad q_m(1-z|z) = e_m.$$

For the sharp Lüders instrument $\mathcal{I}_{\mathrm{sh}}(\rho) = \sum_z |z\rangle\langle z| \otimes P_z \rho P_z$,

$$\tfrac{1}{2}\|\mathcal{I}_m - \mathcal{I}_{\mathrm{sh}}\|_\diamond = e_m \leq e^{-m\alpha^2/2}.$$

This bound holds on arbitrary old references of the source.

Proof. The memory state is diagonal. After one collision, projection onto either memory Z outcome removes the source off-diagonal blocks: the partial maps are $\rho \mapsto \sum_z q_1(y|z) P_z \rho P_z$. Fresh memories give the product probabilities conditional on the same unchanged pointer label, as a consequence of this joint instrument. Grouping all fine outcomes by their majority gives the displayed map. For a joint input with a reference, put $R_z = (P_z \otimes \mathbf{1})\rho(P_z \otimes \mathbf{1})$. The two output-label blocks of the difference are $e_m(R_1 - R_0)$ and its negative. Their orthogonal source supports give trace norm $2e_m \operatorname{Tr}\rho$. Equivalently the map is the sharp instrument followed by a binary label flip, whose diamond distance from the identity is $2e_m$; a pointer input attains it. Hoeffding's inequality applied to the conditional $\{\pm 1\}$ samples of mean α gives the final bound. □

This is a physical finite route from an ideal pointer distinction to a task permission. To reach error $\epsilon > 0$, it suffices to use $m \geq 2\alpha^{-2}\log(1/\epsilon)$. The allowance must contain the controlled-X coupling, fresh independently prepared memories, m readouts, classical majority processing and storage; if orientation or gain is unknown, Paper E's paid references or another calibration are also needed. Discarding the fine records means restricting the task output channel; it does not physically reset their memories or erase externally retained copies. The first read destroys pointer coherence exactly as does the comparison sharp instrument. The claim concerns that instrument, not undisturbed reading of an arbitrary unknown quantum state.

Corollary 3.3 (Finite task transfer)**.** *Consider a finite adaptive circuit with arbitrary retained systems and references. Suppose its q sharp pointer-instrument slots can each be replaced by Proposition 3.2, with half-diamond errors at most e_i, uniformly over earlier records. Keep all other allowed operations the same in both circuits. Their output states have trace distance at most* $\min(1, \sum_i e_i)$*; a loss in $[0, L]$ changes in expectation by at most $L \min(1, \sum_i e_i)$.*

Proof. Include the classical history in the retained state. Replacing one slot at a time and using contractivity of all intervening physical channels bounds the full diamond distance by $2\sum_i e_i$. Trace distance bounds the expectation of a loss in the stated interval. □

The same argument incorporates time-window or gate-error certificates at their corresponding circuit slots; Paper B gives such finite-time risk bounds for its channel class [4]. Each slot requires an instrument-level bound, not just nearby effects. A Γ-hull supplies candidate ideal observables for a task. A physical compiler supplies their actual channels and costs. The corollary connects these layers for a finite protocol meeting its conditions; it does not make all modular times, products, translations or von Neumann limits available at a fixed seed budget.

For example, applying X to the output source after a sharp pointer instrument leaves its effects unchanged but sends a P_0 input to P_1. Its output is at trace distance one from the unmodified instrument on that input. Equal probabilities, equal seed algebras and even equal sharp effects do not certify the physical task after reading.

4 Stationary sources

Three structural conditions on the fixed set are usually imposed separately: that $\mathrm{Fix}(\mathcal{E})$ be closed under products, that it be invariant under the modular group, and that it carry an ω-preserving conditional expectation. A persistent Nakajima–Zwanzig projection is a fourth. For a stationary source all of them follow.

Theorem 4.1 (Stationary source)**.** *Let the source be stationary. Then:*

(a) every $a \in \mathrm{Fix}(\mathcal{E})$ lies in the multiplicative domain of each $\mathcal{E}_t$, so $\mathcal{E}_t(axb) = a\,\mathcal{E}_t(x)\,b$ for $a, b \in \mathrm{Fix}(\mathcal{E})$, $x \in \mathcal{A}_U$, and $\mathrm{Fix}(\mathcal{E})$ is a von Neumann subalgebra;

(b) there is a unique normal conditional expectation $E_{\mathrm{fix}} : \mathcal{A}_U \to \mathrm{Fix}(\mathcal{E})$ with $\omega \circ E_{\mathrm{fix}} = \omega$; it is faithful and $E_{\mathrm{fix}} \circ \mathcal{E}_t = \mathcal{E}_t \circ E_{\mathrm{fix}} = E_{\mathrm{fix}}$;

(c) $\sigma_s(\mathrm{Fix}(\mathcal{E})) = \mathrm{Fix}(\mathcal{E})$ for all s, and the modular group of $\omega|_{\mathrm{Fix}(\mathcal{E})}$ is the restriction of σ;

(d) with $\mathcal{P} = E_{\mathrm{fix}}$ and $\mathcal{Q} = \mathrm{id} - \mathcal{P}$, $\mathcal{P}\mathcal{E}_t\mathcal{Q} = \mathcal{Q}\mathcal{E}_t\mathcal{P} = 0$ and $\mathcal{P}\mathcal{E}_t\mathcal{P} = \mathcal{P}$: the projected dynamics on the fixed algebra is the identity, and the Nakajima–Zwanzig memory kernel between the fixed algebra and its complement vanishes.

Proof. (a) For $a \in \mathrm{Fix}(\mathcal{E})$ the Kadison–Schwarz inequality gives $\mathcal{E}_t(a^*a) \geq \mathcal{E}_t(a)^*\mathcal{E}_t(a) = a^*a$, and $\omega(\mathcal{E}_t(a^*a) - a^*a) = 0$ by stationarity. Since ω is faithful, $\mathcal{E}_t(a^*a) = a^*a$; the same holds for aa^* because a^* is also fixed. By Choi's theorem [16] a belongs to the multiplicative domain of $\mathcal{E}_t$, on which $\mathcal{E}_t$ is a bimodule map. Products of fixed elements are therefore fixed, and normality of $\mathcal{E}_t$ makes the fixed set σ-weakly closed.

(b) The idea is the mean ergodic theorem of Kovács and Szűcs [24]; for semigroups of normal unital positive maps with a faithful invariant state see Watanabe [23]. By Kadison–Schwarz and stationarity, $\|\mathcal{E}_t(x)\Omega\|^2 \leq \omega(\mathcal{E}_t(x^*x)) = \|x\Omega\|^2$, so $V_t x\Omega = \mathcal{E}_t(x)\Omega$ defines a contraction semigroup on $\mathcal{H}_U$ with $V_t\Omega = \Omega$. Its Cesàro means converge strongly to the orthogonal projection P onto its fixed vectors. The means $x_T = T^{-1}\int_0^T \mathcal{E}_t(x)\,dt$ are bounded by $\|x\|$ and satisfy $x_T\Omega \to Px\Omega$; every σ-weak cluster point y has $y\Omega = Px\Omega$, and since Ω is separating the cluster point is unique, so $x_T \to y$ σ-weakly. Put $E_{\mathrm{fix}}(x) = y$. Since $\|\mathcal{E}_s(x_T) - x_T\| \leq 2s\|x\|/T$, $y \in \mathrm{Fix}(\mathcal{E})$; on $\mathrm{Fix}(\mathcal{E})$ the map is the identity, so E_{fix} is a projection of norm one onto a von Neumann algebra, hence a conditional expectation (Tomiyama). The

expectation preserves the state because $\omega(E_{\text{fix}}(x)) = \langle\Omega, Px\Omega\rangle = \omega(x)$, as $P\Omega = \Omega$. To prove normality, let $0 \le x_i \uparrow x$ and $y = \sup_i E_{\text{fix}}(x_i) \le E_{\text{fix}}(x)$. Normality of ω gives

$$\omega(y) = \sup_i \omega(E_{\text{fix}}(x_i)) = \sup_i \omega(x_i) = \omega(x) = \omega(E_{\text{fix}}(x)).$$

Faithfulness forces $y = E_{\text{fix}}(x)$; thus the expectation is normal, and state preservation makes it faithful. The means of $\mathcal{E}_t(x)$ differ from those of x by at most $2t\|x\|/T$ in norm, which gives $E_{\text{fix}} \circ \mathcal{E}_t = E_{\text{fix}}$, and $\mathcal{E}_t \circ E_{\text{fix}} = E_{\text{fix}}$ because the range is fixed. A normal ω-preserving conditional expectation onto a given subalgebra is unique, since it is determined by $E(x)\Omega = e\,x\Omega$ with e the projection onto $\overline{\text{Fix}(\mathcal{E})\Omega}$ [29].

(c) By Takesaki's theorem [29], a von Neumann subalgebra carries a faithful normal ω-preserving conditional expectation if and only if it is globally invariant under σ, and then σ restricts to the modular group of the restricted state.

(d) follows from (b). □

Stationarity supplies all four conditions. HAFF-E proves the fixed-algebra part [6]; the vacuum-preserving internal noise of Consciousness I [2] and the block-diagonal stationary states of HAFF-B [4] give concrete instances.

What fails without stationarity is (c). If a faithful normal state $\varphi \neq \omega$ is invariant, (a) and (b) hold with φ in place of ω and $\text{Fix}(\mathcal{E})$ is invariant under the modular group of φ, which need not preserve it for ω. An entangled faithful state of $M_2 \otimes M_3$ with noise acting ergodically on the second factor is an example (check 3 of Appendix B): the fixed algebra is $M_2 \otimes \mathbf{1}$, and the modular group of the entangled state moves it.

Example 4.2 (A non-stationary qubit)**.** Let $\mathcal{A}_U = M_2$, let $\mathcal{E}_t$ multiply the off-diagonal entries in the Z-basis by $e^{-\gamma t}$, and let $\omega = \text{Tr}(\rho\,\cdot)$ with ρ faithful and not diagonal. Then $\text{Fix}(\mathcal{E})$ is the diagonal algebra, but $\sigma_s(P_0) = \rho^{is}P_0\rho^{-is}$ is a rank-one projection different from $P_0 = |0\rangle\langle 0|$ for small $s \neq 0$. Two distinct nonorthogonal rank-one projections generate M_2, so the hull of the single pointer projection P_0 for any family $\Gamma \supseteq \Gamma_\sigma$ is all of M_2. For the diagonal state $\rho_\infty = \text{diag}(\rho)$, which is stationary, the hull of P_0 is the two-dimensional algebra it generates.

An invariant expectation supplies a stationary partner with the same values on the persistent sector, whether or not individual trajectories relax.

Proposition 4.3 (Stationary partner)**.** *Suppose there is a faithful normal conditional expectation E_∞ of $\mathcal{A}_U$ onto* $\text{Fix}(\mathcal{E})$ *with $E_\infty \circ \mathcal{E}_t = E_\infty$. Then $\omega_\infty = \omega \circ E_\infty$ is faithful, normal and stationary, agrees with ω on* $\text{Fix}(\mathcal{E})$*, and Theorem 4.1 applies to $(\mathcal{A}_U, \omega_\infty, \mathcal{E})$.*

Proof. $\omega_\infty \circ \mathcal{E}_t = \omega \circ E_\infty \circ \mathcal{E}_t = \omega_\infty$; faithfulness and normality are inherited from ω and E_∞; on $\text{Fix}(\mathcal{E})$, E_∞ is the identity. In standard form ω_∞ is a vector state of a cyclic and separating vector. □

For the commuting couplings of HAFF-B, E_∞ is the pinching onto the blocks; for the noises of Consciousness I it is the Haar average. The partner state defines its own modular hull. When these semigroups converge to their averages it is also the relaxed state, but Proposition 4.3 needs only the invariant expectation. The change of hull in Example 4.2 illustrates why the partner and original source must remain distinguished.

Example 4.4 (A mean without relaxation)**.** On M_2, let $\mathcal{E}_t(x) = e^{itZ}xe^{-itZ}$, with the faithful stationary source state $\tau = \mathbf{1}/2$. The fixed algebra is diagonal and its Cesàro expectation is

pinching. For $\rho = (\mathbf{1} + aX)/2$, $0 < a < 1$, the Schrödinger trajectory is $e^{-itZ}\rho e^{itZ}$, whose mean tends to $\mathbf{1}/2$, but

$$\tfrac{1}{2}\|e^{-itZ}\rho e^{itZ} - \mathbf{1}/2\|_1 = a/2$$

at every time. The fixed-sector projection in Theorem 4.1 is exact; it is not a claim that the complementary dynamics decays.

Finite windows. In its commuting class, HAFF-B selects a unique largest admissible block algebra $\mathcal{B}_\pi$ when the stated window and tolerance criteria are met; smaller admissible block algebras can also exist. It proves that $\|\Phi_t^*(Y) - Y\| \le t\,\Gamma_{\text{in}}(\pi)\|Y\|$ for $Y \in \mathcal{B}_\pi$, with Γ_{in} the block-width bound of HAFF-B [4]:

$$\Gamma_{\text{in}}(\pi) = \max_{C\in\pi} \frac{1}{2}\sum_k \gamma_k \big(\max_{x\in C} a_k(x) - \min_{x\in C} a_k(x)\big)^2.$$

The maximum is taken after summing the squared widths of each block; it is not generally the largest pairwise decoherence rate within that block. The pinching E_π onto $\mathcal{B}_\pi$ commutes with the Schur multiplier Φ_t^*, so

$$\|E_\pi \circ \Phi_t^* - E_\pi\| = \|\Phi_t^* \circ E_\pi - E_\pi\| \le t\,\Gamma_{\text{in}}(\pi),$$

and the generator has no matrix elements between $\mathcal{B}_\pi$ and its complement. If the source state lies in $\mathcal{B}_\pi$, then $\omega \circ E_\pi = \omega$ and $\mathcal{B}_\pi$ is invariant under the modular group. On the window, the selected algebra plays the part of $\mathrm{Fix}(\mathcal{E})$ and the pinching is a Nakajima–Zwanzig projection that persists up to $t\Gamma_{\text{in}}$. In this class the memory kernel between the selected algebra and the rest vanishes exactly; a persistent record process with genuine memory needs non-commuting couplings or environments that are revisited, as in Paper F of HAFF [7].

5 Structure of the stable hull

Theorem 5.1 (The hull)**.** *Let $\Gamma \supseteq \Gamma_\sigma$ and let $S \subseteq \mathcal{A}_U$ be a seed. Write $\mathcal{H} = \mathrm{Hull}_\Gamma(S)$.*

(a) $\mathcal{H}$ is Γ-admissible and is contained in every Γ-admissible algebra that contains S. With $M_0 = W^*(S)$ and $M_{n+1} = W^*\big(M_n \cup \bigcup_{\gamma\in\Gamma}\gamma(M_n)\big)$, $\mathcal{H} = \big(\bigcup_n M_n\big)''$.

(b) There is a unique normal conditional expectation $E_\mathcal{H} : \mathcal{A}_U \to \mathcal{H}$ with $\omega \circ E_\mathcal{H} = \omega$; it is faithful. The modular group of $\omega|_\mathcal{H}$ is $\sigma|_\mathcal{H}$. The projection $e_\mathcal{H}$ onto $\overline{\mathcal{H}\Omega}$ commutes with Δ^{it} and J, and $\Delta|_{\overline{\mathcal{H}\Omega}}$, $J|_{\overline{\mathcal{H}\Omega}}$ are the modular objects of $(\mathcal{H},\Omega)$ on that subspace.

(c) $S \subseteq S'$ implies $\mathrm{Hull}_\Gamma(S) \subseteq \mathrm{Hull}_\Gamma(S')$, and $W^*(S) \subseteq \mathcal{H}$; in particular $\mathcal{H} \neq \mathbb{C}\mathbf{1}$ as soon as S contains a nonscalar element.

Proof. (a) An intersection of admissible von Neumann algebras is admissible, and $\mathcal{A}_U$ is admissible. Every admissible algebra containing S contains each M_n by induction. Each $\gamma \in \Gamma$ is σ-weakly continuous and maps the $*$-algebra $\bigcup_n M_n$ into itself, hence maps its σ-weak closure, which is $(\bigcup_n M_n)''$, into that closure. (b) $\mathcal{H}$ is σ-invariant, so Takesaki's theorem gives $E_\mathcal{H}$ and the restriction property [29]. Since $\Delta^{it}\mathcal{H}\Omega = \sigma_t(\mathcal{H})\Omega = \mathcal{H}\Omega$, $e_\mathcal{H}$ commutes with Δ^{it}, hence with every function of Δ. For $x \in \mathcal{H}$, $Jx^*\Omega = \Delta^{1/2}x\Omega = \Delta^{1/2}e_\mathcal{H}x\Omega = e_\mathcal{H}\Delta^{1/2}x\Omega$, so J maps the dense subspace $\mathcal{H}\Omega$ of $\overline{\mathcal{H}\Omega}$ into it. The restricted operators are the modular objects of $(\mathcal{H},\Omega)$ because the Tomita operator of $\mathcal{H}$ is the restriction of that of $\mathcal{A}_U$ to a core that is invariant under Δ^{it}. (c) is immediate from (a). □

By part (b) the hull is never an arbitrary subalgebra. It carries its own modular data, and those data are the source's. When the hull is later read for geometry, the modular flow that enters is the restriction of the flow of the source state; nothing is added.

For stable seeds of a stationary source the hull has a simple form.

Theorem 5.2 (Stable seeds)**.** *Let the source be stationary and let $S \subseteq \mathrm{Fix}(\mathcal{E})$. All statements concerning Γ_3 also assume an ambient positive translation structure and $\mathcal{E}_t \circ \mathrm{Ad}\,U(a) = \mathrm{Ad}\,U(a) \circ \mathcal{E}_t$ for $a \geq 0$.*

(a) $\mathrm{Hull}_{\Gamma_2}(S) = \mathrm{Hull}_{\Gamma_\sigma}(S) = W^*\big(\bigcup_s \sigma_s(S)\big) \subseteq \mathrm{Fix}(\mathcal{E})$. *If there is an ambient Cut with the stated covariance, then* $\mathrm{Hull}_{\Gamma_3}(S)$ *is the smallest von Neumann algebra containing S that is invariant under σ and* $\mathrm{Ad}\,U(a)$*, $a \geq 0$, and it lies in* $\mathrm{Fix}(\mathcal{E})$.

(b) $W^*(S) \subseteq \mathrm{Hull}(S) \subseteq \mathrm{Fix}(\mathcal{E})$ *for* $\mathrm{Hull} = \mathrm{Hull}_{\Gamma_2}$ *or* Hull_{Γ_3}. *The hull is nontrivial if S contains a nonscalar element, and proper if* $\mathrm{Fix}(\mathcal{E}) \neq \mathcal{A}_U$.

(c) $\mathrm{Hull}(S \cup S') = \mathrm{Hull}(S) \vee \mathrm{Hull}(S')$ *for stable seeds S, S'.*

(d) $E_{\mathcal{H}} \circ \mathcal{E}_t = \mathcal{E}_t \circ E_{\mathcal{H}} = E_{\mathcal{H}}$ *for* $\mathcal{H} = \mathrm{Hull}(S)$*: the expectation onto the hull is a persistent projection with no back-flow.*

Proof. (a) $\mathrm{Fix}(\mathcal{E})$ is a σ-invariant von Neumann algebra (Theorem 4.1) containing S, so it contains the modular hull. The noise acts as the identity on $\mathrm{Fix}(\mathcal{E})$, so the modular hull is Γ_2-admissible, and it is contained in every Γ_2-admissible algebra containing S. If $\mathcal{E}_t$ commutes with $\mathrm{Ad}\,U(a)$, then $U(a)xU(a)^*$ is fixed whenever x is, so $\mathrm{Fix}(\mathcal{E})$ is invariant under forward translations and the same argument applies. (b) follows from (a). (c) The algebra generated by two admissible subalgebras of $\mathrm{Fix}(\mathcal{E})$ is invariant under the automorphisms σ_s and the normal endomorphisms $\mathrm{Ad}\,U(a)$, and the noise is the identity on it. (d) For $a \in \mathcal{H} \subseteq \mathrm{Fix}(\mathcal{E})$ and $x \in \mathcal{A}_U$, using bimodularity, Theorem 4.1(a) and stationarity,

$$\omega\big(a\,E_{\mathcal{H}}(\mathcal{E}_t(x))\big) = \omega(a\,\mathcal{E}_t(x)) = \omega(\mathcal{E}_t(ax)) = \omega(ax) = \omega\big(a\,E_{\mathcal{H}}(x)\big).$$

Hence $E_{\mathcal{H}}(\mathcal{E}_t(x)) - E_{\mathcal{H}}(x) \in \mathcal{H}$ is orthogonal to $\mathcal{H}\Omega$ and vanishes because Ω is separating. The second identity holds because the range of $E_{\mathcal{H}}$ is fixed. □

Part (c) answers one of the obstructions listed in HAFF-D: a completely positive map that is not multiplicative can preserve two subalgebras without preserving their join [5]. For stable seeds of a stationary source the noise is multiplicative on the relevant algebra, and the hull is a closure operator that preserves joins.

Proposition 5.3 (Records that commute with the state)**.** *Let the source be stationary and let $S \subseteq \mathrm{Fix}(\mathcal{E})$ lie in the centralizer $\mathcal{A}_U^{\omega} = \{x : \sigma_s(x) = x\ \forall s\}$. Then* $\mathrm{Hull}_{\Gamma_2}(S) = W^*(S)$. *In the commuting-coupling class of HAFF-B with a stationary faithful state, the block projections, which generate the record algebra, lie in the centralizer; the stable hull of any set of record projections is the abelian algebra they generate, and it contains no proper half-sided modular inclusion.*

Proof. The centralizer is a von Neumann algebra on which σ acts trivially, so $W^*(S)$ is σ-invariant; it lies in $\mathrm{Fix}(\mathcal{E})$, where the noise is trivial. In the commuting class a stationary state has vanishing entries between basis vectors with different eigenvalue tuples of the couplings [4], so it commutes with the block projections. The modular group of a faithful state on an abelian algebra is trivial, so the translation generator of a half-sided modular inclusion inside it vanishes [10] and the inclusion is an equality. □

These centralizer records do not grow under Γ_2. The thermal dephasing chain of HAFF-D at zero transverse field, where the diagonal algebra is fixed and the modular flow acts trivially on it, is an instance [5]. For a stable seed in a stationary source, Γ_2-growth requires elements moved by the modular flow. Consciousness I gives the basic example: in $M_2 \otimes M_2$ with state $\mathrm{diag}(0.8, 0.2) \otimes \frac{1}{2}\mathbf{1}$ and depolarising noise on the second factor, $X \otimes \mathbf{1}$ is fixed by the noise and its hull is $M_2 \otimes \mathbf{1}$ [2].

In finite dimension the growth is bounded by the state.

Proposition 5.4 (Finite dimension)**.** *If $\mathcal{A}_U = M_n$, $\omega = \mathrm{Tr}(\rho\,\cdot)$ is stationary and $S \subseteq \mathrm{Fix}(\mathcal{E})$, then $W^*(S) \subseteq \mathrm{Hull}_{\Gamma_2}(S) \subseteq W^*(S \cup \{\rho\}) \cap \mathrm{Fix}(\mathcal{E})$.*

Proof. $\sigma_s(x) = \rho^{is} x \rho^{-is} \in W^*(S \cup \{\rho\})$; combine with Theorem 5.2. □

For three qubits under the collective dephasing $X_1 + X_2 + X_3$ of HAFF-B, with a stationary Gibbs-type state in the block algebra and the in-block seed $X_1 - X_2$, the seed generates a 3-dimensional algebra, its hull has dimension 10, equal to that of $W^*(S \cup \{\rho\})$, and the fixed algebra has dimension 20 (check 4 of Appendix B). In type III the picture changes.

Proposition 5.5 (Chiral vacuum)**.** *Let $\mathcal{A}_U = \mathcal{A}(0, \infty)$ be the half-line algebra of a Möbius covariant net with unique vacuum Ω. Then the centralizer of ω is $\mathbb{C}\mathbf{1}$, and every σ-invariant von Neumann subalgebra other than $\mathbb{C}\mathbf{1}$ is infinite-dimensional.*

Proof. By the Bisognano–Wichmann property the modular group is the dilation group $D(t)$ [13], and $D(t)U(a)D(t)^* = U(e^{-2\pi t}a)$ for the translations. If $D(t)\xi = \xi$ for all t, then $\langle \xi, U(a)\xi \rangle = \langle \xi, U(e^{-2\pi t}a)\xi \rangle \to \|\xi\|^2$ as $t \to \infty$, so $U(a)\xi = \xi$ and ξ is a multiple of Ω. For x in the centralizer, $x\Omega$ is dilation invariant, hence $x\Omega = c\Omega$ and $x = c\mathbf{1}$. On the orthogonal complement of Ω the representation of the translation–dilation group has positive translation generator and no invariant vectors, so it is a multiple of the unique irreducible representation of this kind, in which the dilation generator has absolutely continuous spectrum. A finite-dimensional σ-invariant subalgebra $\mathcal{N}$ would make $\mathcal{N}\Omega$ a finite-dimensional subspace invariant under Δ^{it}, hence spanned by eigenvectors of $\log \Delta$; these are multiples of Ω, so $\mathcal{N} = \mathbb{C}\mathbf{1}$. □

In a chiral vacuum every nonscalar seed is moved by the modular flow and generates an infinite-dimensional hull. Thus in this chiral vacuum the only centralizer seeds of Proposition 5.3 are scalar. Other type III source states can have nontrivial centralizers.

6 Cuts in the hull

Consciousness I needs a Cut inside the accessible algebra. A hull closed under forward translations inherits a half-sided modular inclusion from the ambient positive translation structure; it is proper when the hull is nonscalar and the invariant vector is unique.

Theorem 6.1 (Cut inheritance)**.** *Let the source carry an ambient positive translation structure U, and let $\mathcal{H} \subseteq \mathcal{A}_U$ be a von Neumann subalgebra with $\sigma_s(\mathcal{H}) = \mathcal{H}$ for all s and $U(a)\mathcal{H}U(a)^* \subseteq \mathcal{H}$ for $a \geq 0$; every Γ_3-hull has these properties. Put $\mathcal{K} = \overline{\mathcal{H}\Omega}$ and $\mathcal{N}_\mathcal{H} = U(1)\mathcal{H}U(1)^*$.*

(a) $\mathcal{K}$ reduces $U(a)$ for every $a \in \mathbb{R}$, as well as Δ^{it} and J.

(b) On $\mathcal{K}$, $(\mathcal{N}_\mathcal{H} \subset \mathcal{H}, \Omega)$ is a half-sided modular inclusion, and its translation group is $U|_\mathcal{K}$.

(c) $\mathcal{N}_{\mathcal{H}} \neq \mathcal{H}$ *if and only if* $U|_{\mathcal{K}} \neq \mathbf{1}$*. If* Ω *is, up to scalars, the only translation-invariant vector, then* $\mathcal{N}_{\mathcal{H}} \neq \mathcal{H}$ *whenever* $\mathcal{H} \neq \mathbb{C}\mathbf{1}$*.*

In particular, if Ω *is, up to scalars, the only* U*-invariant vector, as in the vacuum representation of a chiral net, then for every seed containing a nonscalar element the* Γ_3*-hull contains a Cut in its vacuum-cyclic representation, and its translations are the restricted ambient translations.*

Proof. (a) For $a \geq 0$ and $x \in \mathcal{H}$, $U(a)x\Omega = U(a)xU(a)^*\Omega \in \mathcal{H}\Omega$. By Theorem 5.1(b), $J\mathcal{K} = \mathcal{K}$ and $\Delta^{it}\mathcal{K} = \mathcal{K}$. Borchers' relation $U(-a) = JU(a)J$ then gives invariance under negative translations.

(b) $\mathcal{N}_{\mathcal{H}}\Omega = U(1)\mathcal{H}\Omega$ is dense in $U(1)\mathcal{K} = \mathcal{K}$, and Ω is separating for every subalgebra of $\mathcal{A}_U$. The modular group of $(\mathcal{H}, \Omega)$ on $\mathcal{K}$ is $\Delta^{it}|_{\mathcal{K}}$ by Theorem 5.1(b). For $t \geq 0$, Borchers' relation $\Delta^{-it}U(1)\Delta^{it} = U(e^{2\pi t})$ gives

$$\Delta^{-it}\mathcal{N}_{\mathcal{H}}\Delta^{it} = U(e^{2\pi t})\mathcal{H}U(e^{2\pi t})^* = U(1)\big[U(e^{2\pi t} - 1)\mathcal{H}U(e^{2\pi t} - 1)^*\big]U(1)^* \subseteq \mathcal{N}_{\mathcal{H}},$$

which is the half-sided condition. The modular operator of $\mathcal{N}_{\mathcal{H}}$ is $U(1)\Delta U(1)^*$ on $\mathcal{K}$, so $\Delta^{it}_{\mathcal{N}_{\mathcal{H}}}\Delta^{-it}_{\mathcal{H}} = U(1 - e^{-2\pi t})|_{\mathcal{K}}$. This is the relation that defines the translation group of the inclusion in the Wiesbrock–Araki–Zsidó theorem [31, 10], which is therefore $U|_{\mathcal{K}}$.

(c) If $\mathcal{N}_{\mathcal{H}} = \mathcal{H}$, applying σ_t gives $U(b)\mathcal{H}U(b)^* = \mathcal{H}$ for all $b > 0$, hence for all b. On $\mathcal{K}$ the group U then implements automorphisms of $\mathcal{H}$ and of its commutant $\mathcal{H}' \cap \mathcal{B}(\mathcal{K})$, with positive generator and fixed vector Ω. Borchers' theorem applied to $\mathcal{H}$ and to $\mathcal{H}'$, whose modular operator is $\Delta_{\mathcal{H}}^{-1}$, gives both $\Delta^{it}_{\mathcal{H}}U(a)\Delta^{-it}_{\mathcal{H}} = U(e^{-2\pi t}a)$ and $\Delta^{it}_{\mathcal{H}}U(a)\Delta^{-it}_{\mathcal{H}} = U(e^{2\pi t}a)$ on $\mathcal{K}$. Hence $U(a) = U(e^{4\pi t}a)$ for all t, and letting $t \to -\infty$ gives $U|_{\mathcal{K}} = \mathbf{1}$. Conversely, if $U|_{\mathcal{K}} = \mathbf{1}$ then $\mathcal{N}_{\mathcal{H}}$ and $\mathcal{H}$ act identically on $\mathcal{K}$. If Ω is the only invariant vector, $U|_{\mathcal{K}} = \mathbf{1}$ forces $\mathcal{K} = \mathbb{C}\Omega$ and $\mathcal{H} = \mathbb{C}\mathbf{1}$. □

In Consciousness I the Virasoro algebra B of the half-line is fixed by the noise, Ω is not cyclic for B on the ambient space, and the Cuts used for the geometry are the subalgebras $\mathrm{Vir}_1(a, \infty)$ of B on its own vacuum space [2]. Under the unique-invariant-vector hypothesis of Theorem 6.1, the Γ_3-hull of any nonscalar seed carries a Cut on its vacuum-cyclic subspace, and the modular orbit of that Cut generates the hull by the modular-orbit generation lemma of Consciousness I [2]. If the inclusion is standard, the Guido–Longo–Wiesbrock correspondence turns it into a strongly additive conformal net with translations $U|_{\mathcal{K}}$ and dilations $\Delta^{it}|_{\mathcal{K}}$ [19].

Two points limit what this achieves. The translations are the ambient ones; the hull selects a subspace and an algebra on which they act, and does not produce them. Standardness is not automatic; it holds in the symmetric examples of the next section. Since these translation-stable hulls carry the same translation and dilation groups, restricted to different subspaces, the translation–dilation group alone cannot distinguish the geometries of different seeds. An invariant of the whole triple is needed, as Consciousness I already notes [2].

7 Seeds and symmetry

This section answers three questions about the hull: whether it is unique relative to the seed, how it changes with the seed, and whether different seeds give different geometries. The answer is governed by symmetry.

7.1 A Galois law for hulls

Lemma 7.1 (Vacuum-space criterion)**.** *Let $\mathcal{B} \subseteq \mathcal{C} \subseteq \mathcal{A}_U$ be σ-invariant von Neumann algebras. Then $\mathcal{B} = \mathcal{C}$ if and only if $\overline{\mathcal{B}\Omega} = \overline{\mathcal{C}\Omega}$.*

Proof. Let $E_{\mathcal{B}}$ be the ω-preserving expectation onto $\mathcal{B}$. For $x \in \mathcal{A}_U$ and $y \in \mathcal{B}$, $\langle y\Omega, (x - E_{\mathcal{B}}(x))\Omega\rangle = \omega(y^*x) - \omega(E_{\mathcal{B}}(y^*x)) = 0$, so $e_{\mathcal{B}}x\Omega = E_{\mathcal{B}}(x)\Omega$. If the two subspaces agree and $x \in \mathcal{C}$, then $x\Omega = E_{\mathcal{B}}(x)\Omega$ and $x = E_{\mathcal{B}}(x) \in \mathcal{B}$. □

Symmetric sources. Let $\mathcal{A}_U$ be a factor with separable predual and G a compact group with a strongly continuous unitary representation u on $\mathcal{H}_U$ such that $u(g)\Omega = \Omega$ and $\alpha_g = \operatorname{Ad} u(g)$ maps $\mathcal{A}_U$ onto itself; if an ambient positive translation structure is present, $u(g)$ commutes with $U(a)$. Then $u(g)$ commutes with the Tomita operator, hence with Δ, and $\alpha_g \circ \sigma_s = \sigma_s \circ \alpha_g$. The action is *minimal* if $g \mapsto \alpha_g$ is injective and $(\mathcal{A}_U^G)' \cap \mathcal{A}_U = \mathbb{C}\mathbf{1}$. The *stabilizer* of a seed is the closed subgroup $\mathrm{Stab}(S) = \{g \in G : \alpha_g(x) = x \text{ for all } x \in S\}$. Izumi, Longo and Popa proved that for a minimal action of a compact group on a factor with separable predual, $K \mapsto \mathcal{A}_U^K$ is a bijection from the closed subgroups of G onto the von Neumann algebras between $\mathcal{A}_U^G$ and $\mathcal{A}_U$ [21].

Theorem 7.2 (Symmetry fixes the hull)**.** *Let G act minimally as above, let $\Gamma \supseteq \Gamma_\sigma$, and let S be a seed such that*

(i) every $\gamma \in \Gamma$ maps $\mathcal{A}_U^{\mathrm{Stab}(S)}$ into itself, and

(ii) $\mathrm{Hull}_\Gamma(S) \supseteq \mathcal{A}_U^G$.

Then $\mathrm{Hull}_\Gamma(S) = \mathcal{A}_U^{\mathrm{Stab}(S)}$. *Two seeds satisfying (i) and (ii) have the same hull if and only if they have the same stabilizer. Condition (i) holds if every $\gamma \in \Gamma$ commutes with every α_g; for a stable seed of a stationary source it also holds when Γ consists of σ, forward translations and a noise with* $\mathrm{Fix}(\mathcal{E}) = \mathcal{A}_U^{H_0}$ *for a closed subgroup H_0.*

Proof. By (i), $\mathcal{A}_U^{\mathrm{Stab}(S)}$ is Γ-admissible, and it contains S; hence $\mathrm{Hull}_\Gamma(S) \subseteq \mathcal{A}_U^{\mathrm{Stab}(S)}$. By (ii) and the Izumi–Longo–Popa theorem, $\mathrm{Hull}_\Gamma(S) = \mathcal{A}_U^{K'}$ for a closed subgroup K'. Since $S \subseteq \mathcal{A}_U^{K'}$, $K' \subseteq \mathrm{Stab}(S)$, so $\mathcal{A}_U^{\mathrm{Stab}(S)} \subseteq \mathcal{A}_U^{K'} = \mathrm{Hull}_\Gamma(S)$. The second statement is the injectivity of $K \mapsto \mathcal{A}_U^K$. For the last statement, if $S \subseteq \mathcal{A}_U^{H_0}$ then $\mathrm{Stab}(S) \supseteq H_0$, so $\mathcal{A}_U^{\mathrm{Stab}(S)} \subseteq \mathrm{Fix}(\mathcal{E})$, where the noise is the identity; σ and the translations commute with α_g. □

The hull is therefore unique relative to the seed in a strong sense: it depends on the seed only through the symmetry the seed leaves unbroken. Adding observables to a seed can only shrink the stabilizer and enlarge the hull, and observables that do not change the stabilizer add nothing.

The law has no counterpart in type I. Every automorphism of a type I factor is inner, so each α_g is implemented by a unitary $v(g) \in \mathcal{A}_U$; these unitaries commute with $\mathcal{A}_U^G$, so a nontrivial action has $(\mathcal{A}_U^G)' \cap \mathcal{A}_U \supseteq \{v(g)\}'' \neq \mathbb{C}\mathbf{1}$, and no nontrivial action is minimal. Distinct stabilizers can then give the same algebra: in M_2 with the adjoint action of $SO(3)$, both $O(2)$ and $SO(3)$ fix only the scalars (check B.1.6).

7.2 The SU(2) level-one current net

Let $\mathcal{A}$ be the $\mathrm{SU}(2)_1$ current net, $\mathcal{A}_U = M = \mathcal{A}(0,\infty)$ with the vacuum, the translations U as ambient positive translation structure and the dilations as modular group [13]. The group $SO(3)$ acts by the adjoint action, as in Assumption 4.1 of Consciousness I [2]. Write $J_n(f) = \sum_a n_a J^a(f)$ for a unit vector n and a real test function f, so that $\alpha_g(J_n(f)) = J_{R(g)n}(f)$, and $\mathcal{A}_{J_n}(I) = \{e^{iJ_n(h)} : \operatorname{supp} h \subset I\}''$ for the U(1) current subnet of J_n. We use four known facts.

(F1) $M^{SO(3)} = \mathrm{Vir}_1(0,\infty)$, the half-line algebra of the Virasoro net of the Sugawara stress tensor (Rehren [27]).

(F2) The Sugawara stress tensor equals the stress tensor of each current J_n: their difference commutes with J_n and is a positive-energy Virasoro field of central charge $1-1=0$, hence zero [18]. Thus $\mathrm{Vir}_1(I) \subseteq \mathcal{A}_{J_n}(I)$ [14].

(F3) On $\overline{\mathcal{A}_{J_n}(0,\infty)\Omega}$ the field J_n is a free current: the vacuum is quasi-free on $\mathcal{A}_{J_n}$, with one-particle space the completion of the real test functions in the norm $\|h\|^2 = c\int_0^\infty k\,|\hat{h}(k)|^2\,dk$, and Weyl relations $e^{iJ_n(h_1)}e^{iJ_n(h_2)} = e^{i\theta(h_1,h_2)}e^{iJ_n(h_1+h_2)}$ with real θ hold on the whole space.

(F4) Vir_1 and $\mathcal{A}$ are strongly additive [15, 32], and the fixed-point subnet of a strongly additive net under a compact group is strongly additive [32].

Proposition 7.3 (Minimality)**.** *$SO(3)$ acts minimally on M: the action is faithful and $\mathrm{Vir}_1(0,\infty)' \cap M = \mathbb{C}\mathbf{1}$. This is known [14, 32]; we include a proof.*

Proof. Let $x \in M$ commute with $\mathrm{Vir}_1(0,\infty)$. For an interval I inside the half-line, $\mathrm{Vir}_1(I) \subseteq \mathrm{Vir}_1(0,\infty)$ commutes with x; for I in the complement, $\mathrm{Vir}_1(I) \subseteq \mathcal{A}(I) \subseteq M'$ by locality. An interval that contains 0 or ∞ splits, after removing these points, into intervals of the first two kinds, and by strong additivity of Vir_1 its algebra is generated by theirs. So x commutes with $\mathrm{Vir}_1(I)$ for every interval, hence with $V(\varphi)$ for every diffeomorphism supported in an interval, hence, since such diffeomorphisms generate $\mathrm{Diff}_+(S^1)$, with the Möbius group. Then $x\Omega$ is Möbius invariant, $x\Omega = c\Omega$ and $x = c\mathbf{1}$. Faithfulness: $\alpha_g(J_n(f)) = J_{R(g)n}(f)$ and the vectors $J^a(f)\Omega$, $a = 1,2,3$, are orthogonal and nonzero, so $\alpha_g = \mathrm{id}$ forces $R(g) = 1$. □

Lemma 7.4 (A current seed)**.** *For every unit vector n and every nonzero real $g \in C_c^\infty((0,\infty))$, the modular hull of $O_{n,g} = \frac{2}{\pi}\arctan J_n(g)$ is $\mathcal{A}_{J_n}(0,\infty)$.*

Proof. $\mathcal{A}_{J_n}(0,\infty)$ is invariant under dilations and contains $O_{n,g}$, so it contains the modular hull. Conversely, the hull contains $e^{itJ_n(g)}$ for all real t, and dilations act by $e^{iJ_n(h)} \mapsto e^{iJ_n(h_\lambda)}$, $h_\lambda(x) = h(x/\lambda)$. By the Weyl relations the hull contains $e^{iJ_n(h)}$ for every h in the real span L of the dilates of g; let $\mathcal{B}_L$ be the von Neumann algebra these generate. It is dilation invariant and lies in $\mathcal{A}_{J_n}(0,\infty)$.

$L + iL$ is dense in the one-particle space. With $k = e^u$ and $\psi(u) = e^u\hat{g}(e^u)$, the one-particle norm is the $L^2(\mathbb{R}, du)$ norm and a dilation is a translation of ψ. The Fourier transform $\tilde{\psi}(s) = \int_0^\infty k^{-is}\hat{g}(k)\,dk$ is the restriction to $\mathrm{Re}\,z = 1$ of the Mellin transform $\int_0^\infty k^{z-1}\hat{g}(k)\,dk$, which is analytic for $\mathrm{Re}\,z > 0$ because $\hat{g}$ is bounded and rapidly decreasing, and is not identically zero because $g \neq 0$. Its zeros on the line are isolated, so $\tilde{\psi} \neq 0$ almost everywhere, and by Wiener's theorem the translates of ψ span a dense subspace [30].

The vectors $e^{iJ_n(h)}\Omega$, $h \in L$, are multiples of the coherent vectors $\exp(ih)$. Differentiating $\exp(i\sum_j t_j h_j)$ in the real parameters t_j at zero produces all symmetric tensor products of elements of L, whose complex span is dense in each n-particle space because $L + iL$ is dense. Hence $\overline{\mathcal{B}_L\Omega}$ is the Fock space $\overline{\mathcal{A}_{J_n}(0,\infty)\Omega}$, and Lemma 7.1 gives $\mathcal{B}_L = \mathcal{A}_{J_n}(0,\infty)$. □

Lemma 7.5 (A stress seed under modular evolution)**.** *Let $0 < a < b$, $\alpha = \log a$, $\beta = \log b$, and $g \in C_c^\infty(\mathbb{R}, \mathbb{R})$ have support $[\alpha, \beta]$. Suppose $g > 0$ on (α, β), is strictly increasing near α, and strictly decreasing near β. For $f(x) = xg(\log x)$ and $O_f = \frac{2}{\pi} \arctan T(f)$,*

$$\mathrm{Hull}_{\Gamma_\sigma}(\{O_f\}) = \mathrm{Vir}_1(0, \infty).$$

The one-sided orbit $N_{O_f}^- = W^*(\sigma_s(O_f) : s \le 0)$, restricted to the Virasoro vacuum space, is a proper standard Cut and satisfies $\mathrm{Vir}_1(b, \infty) \subseteq N_{O_f}^- \subseteq \mathrm{Vir}_1(a, \infty)$.

Proof. These are the single-orbit Cut theorem and full modular-hull corollary of Consciousness I [2]. The ambient modular group restricts to the Virasoro vacuum modular group by (F1). Every stress flow used in that construction belongs to the ambient algebra; its linear combinations, conjugations and fixed-compact-support smooth limits retain their strong closure there. The construction therefore puts $\mathrm{Vir}_1(b, \infty)$ in the one-sided orbit algebra without an independent forward-translation permission. Modular invariance of the full hull then puts $\mathrm{Vir}_1(\lambda b, \infty)$ in it for every $\lambda > 0$. These algebras generate $\mathrm{Vir}_1(0, \infty)$, which is also an upper bound because it contains the seed and is dilation invariant. Standardness and properness refer to the Virasoro vacuum space, not the larger $\mathrm{SU}(2)_1$ Hilbert space. □

Thus a current seed always has modular-only generation, while a stress seed has it under the stated bump conditions. For an arbitrary nonzero real stress smearing, the general stress-generation theorem of Consciousness I still uses dilations and forward translations. None of these closure results supplies a physical stress-flow gate from an unsharp effect.

Theorem 7.6 (Hulls in the $\mathrm{SU}(2)_1$ net)**.** *Let S be a seed containing at least one of the following:*

(i) a stress observable $O_f = \frac{2}{\pi} \arctan T(f)$ with arbitrary $0 \neq f \in C_c^\infty((0, \infty), \mathbb{R})$, with $\Gamma = \Gamma_3$;

(ii) a stress observable O_f satisfying Lemma 7.5, with $\Gamma \in \{\Gamma_\sigma, \Gamma_2, \Gamma_3\}$;

(iii) a current observable $O_{n,g}$, with $\Gamma \in \{\Gamma_\sigma, \Gamma_2, \Gamma_3\}$.

Let the noise be absent, the Haar noise $D^{SO(3)}$, or, for a stable seed, a Haar noise D^{H_0} of Consciousness I. Put $K = \mathrm{Stab}(S)$. Then:

(a) $\mathrm{Hull}_\Gamma(S) = M^K$;

(b) $\overline{M^K \Omega} = \mathcal{H}_U^K$, the K-invariant vectors, and its character is

$$\mathrm{Tr}_{\mathcal{H}_U^K} q^{L_0} = \frac{1}{\phi(q)} \sum_{j \ge 0} m_j(K) \big(q^{j^2} - q^{(j+1)^2}\big), \qquad m_j(K) = \dim V_j^K,$$

where V_j is the spin-j representation of $SO(3)$ and $\phi(q) = \prod_{n \ge 1}(1 - q^n)$;

(c) $U(1)M^K U(1)^* \subset M^K$ is a standard Cut on $\mathcal{H}_U^K$, and the net it reconstructs is the fixed-point subnet $I \mapsto \mathcal{A}(I)^K$ with the restricted Möbius representation.

Proof. (a) Every Γ-admissible algebra is invariant under σ; a Γ_3-admissible algebra is also stable under forward translations. In case (i), the general stress-generation theorem of Consciousness I [2] puts $\mathrm{Vir}_1(0, \infty)$ in the hull. In case (ii), Lemma 7.5 gives the same inclusion using only modular evolution. In case (iii), it follows from Lemma 7.4 and (F2). By (F1) this is

$M^{SO(3)}$. The modular group and the translations commute with α_g; the Haar noise $D^{SO(3)}$ commutes with every α_g by invariance of Haar measure; a noise D^{H_0} has fixed algebra M^{H_0} [2]. Proposition 7.3 and Theorem 7.2 give the claim.

(b) Since $u(k)\Omega = \Omega$, the projection $P_K = \int_K u(k)\,dk$ onto $\mathcal{H}_U^K$ maps $x\Omega$ to $Q_K(x)\Omega \in M^K\Omega$, where Q_K is the Haar average; as $M\Omega$ is dense, $M^K\Omega$ is dense in $\mathcal{H}_U^K$. The vacuum module decomposes under $SO(3) \times \mathrm{Vir}$ as $\bigoplus_{j\geq 0} V_j \otimes L(1, j^2)$, and the character of the irreducible $c = 1$ module $L(1, j^2)$ is $q^{j^2}(1 - q^{2j+1})/\phi(q)$ [27, 28, 22]; taking K-invariants gives the formula.

(c) $\mathcal{A}^K$ is a Möbius covariant subnet, strongly additive by (F4), and Ω is cyclic on $\mathcal{H}_U^K$ for each $\mathcal{A}(I)^K$ by the Reeh–Schlieder property of the subnet in its vacuum representation. Since $M^K = \mathcal{A}^K(0, \infty)$ and $U(1)M^KU(1)^* = \mathcal{A}^K(1, \infty)$, the relative commutant contains $\mathcal{A}^K(0, 1)$, so the Cut is standard, and the Guido–Longo–Wiesbrock reconstruction returns $\mathcal{A}^K$ [19]. The Möbius representation of a Möbius covariant net is fixed by the modular groups of its interval algebras [13], so it is the restriction of the ambient one. □

Theorem 7.7 (Seeds are distinguished by geometry)**.** *For two seeds S_1, S_2 as in Theorem 7.6, the Cut triples $(U(1)\mathcal{H}_iU(1)^* \subset \mathcal{H}_i, \Omega)$ on $\overline{\mathcal{H}_i\Omega}$, $\mathcal{H}_i = \mathrm{Hull}_\Gamma(S_i)$, are isomorphic if and only if* $\mathrm{Stab}(S_1)$ *and* $\mathrm{Stab}(S_2)$ *are conjugate in $SO(3)$.*

Proof. If $\mathrm{Stab}(S_2) = g\,\mathrm{Stab}(S_1)g^{-1}$, then $u(g)$ maps $M^{\mathrm{Stab}(S_1)}$ onto $M^{\mathrm{Stab}(S_2)}$, commutes with U and Δ, and fixes Ω. Conversely, an isomorphism of standard triples induces an isomorphism of the reconstructed nets intertwining their Möbius representations [19, 13], so the two characters of Theorem 7.6(b) agree. Multiplying by $\phi(q)$ gives $\sum_j (m_j - m_{j-1})q^{j^2}$, from which every $m_j(K)$ is read off. With $\chi_j(\theta) = \sum_{m=-j}^{j} e^{im\theta}$ the character of V_j at rotation angle θ, $m_j - m_{j-1} = \int(\chi_j - \chi_{j-1})\,d\mu_K = 2\int \cos(j\theta)\,d\mu_K(\theta)$ for $j \geq 1$, where μ_K is the image of Haar measure of K under the rotation angle $\theta \in [0, \pi]$. The cosine moments determine μ_K, and μ_K determines the conjugacy class of K (Lemma A.1). □

The classified triples use the ambient-translation inclusion $U(1)M^KU(1)^* \subset M^K$. The seed's one-sided orbit $N_{O_f}^-$ $\subset \mathrm{Vir}_1(0, \infty)$ in Lemma 7.5 is a different standard inclusion. Its reconstructed translations need not equal U. Consciousness I's Conjecture T1(b) asserts that this one-sided orbit is not an ordinary translated half-line; that noncoincidence remains open. The character invariant here is proved for the ambient-translation triples.

The identification compares the entire standard Cut triples and their conformal subnets on the same ambient light ray. Their nonisomorphism does not assert that their underlying spacetime metrics differ. The whole character is needed: the multiplicities of C_n are $2\lfloor j/n \rfloor + 1$, so its hull and the U(1) hull have the same character below level n^2. A reader resolving finitely many conformal levels distinguishes only stabilizers whose characters already differ below that level.

Table 1 lists the first coefficients. Some symmetries show up late: the hulls of seeds with octahedral and icosahedral stabilizers agree with the Virasoro hull up to level 15, and the icosahedral hull differs from the Virasoro hull first at level 36, where the first invariant harmonic of degree six appears.

Corollary 7.8 (Dephasing about one axis)**.** *Let the noise be D^{H_z}, with H_z the rotations about e_3. The seed and operation family in the hull statements below satisfy Theorem 7.6: arbitrary stress smearings use Γ_3; the stated log-bump stress seeds and current seeds also admit Γ_σ and Γ_2.*

(a) $\mathrm{Fix}(D^{H_z}) = M^{H_z} = \mathcal{A}_{J_{e_3}}(0, \infty)$*, the U(1) current algebra of the half-line.*

stabilizer K	$m_j(K)$	character coefficients, levels 0–12
$SO(3)$	δ_{j0}	1, 0, 1, 1, 2, 2, 4, 4, 7, 8, 12, 14, 21
$O(2)$	1 for even j, else 0	1, 0, 1, 1, 3, 3, 6, 7, 12, 14, 22, 27, 40
$SO(2)$	1	1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, 56, 77
D_2		1, 0, 1, 1, 4, 4, 8, 10, 17, 21, 33, 42, 62
T		1, 0, 1, 1, 2, 2, 4, 4, 7, 9, 13, 16, 24
C_2	$2\lfloor j/2 \rfloor + 1$	1, 1, 2, 3, 7, 9, 15, 21, 32, 44, 64, 86, 121
$\{1\}$	$2j + 1$	1, 3, 4, 7, 13, 19, 29, 43, 62, 90, 126, 174, 239

Table 1: Vacuum characters of the hulls M^K in the SU(2)$_1$ net. The rows for $SO(3)$, $SO(2)$ and $\{1\}$ are the Virasoro, U(1) current and SU(2)$_1$ vacuum characters; the $O(2)$ row counts partitions into an even number of parts. Computed in check B.2 of Appendix B.

(b) A current observable $O_{n,g}$ is stable if and only if $n = \pm e_3$. Every stable seed that contains a stress or a current observable has one of exactly three hulls: $\mathcal{A}_{J_{e_3}}(0,\infty)$, its $\mathbb{Z}_2$-orbifold $M^{O(2)_z}$ under $J_{e_3} \mapsto -J_{e_3}$, or $\mathrm{Vir}_1(0,\infty)$. They are attained by $\{O_{e_3,g}\}$, by $\{O_f, x\}$ with $x \in M^{O(2)_z} \setminus \mathrm{Vir}_1(0,\infty)$, and by $\{O_f\}$, and their characters differ at levels 1 and 4.

(c) Under the Haar noise $D^{SO(3)}$ every stable seed containing a stress observable has the hull $\mathrm{Vir}_1(0,\infty)$. Without noise every conjugacy class of closed subgroups occurs as a stabilizer, and two current observables along non-parallel axes generate all of M under the modular flow alone.

Proof. (a) $\mathrm{Fix}(D^{H_z}) = M^{H_z}$ [2]. Since $J_n(g)\Omega = \sum_a n_a J^a(g)\Omega$ with orthogonal nonzero vectors $J^a(g)\Omega$, α_g fixes $O_{e_3,g}$ exactly when $R(g)e_3 = e_3$; so $\mathrm{Stab}(O_{e_3,g}) = H_z$, and Lemma 7.4 with Theorem 7.6 gives $\mathcal{A}_{J_{e_3}}(0,\infty) = M^{H_z}$. (b) The same argument shows that $O_{n,g} \in M^{H_z}$ only for $n = \pm e_3$. A stable seed has $\mathrm{Stab}(S) \supseteq H_z$. The closed subgroups of $SO(3)$ containing H_z are H_z, its normalizer $O(2)_z$ and $SO(3)$: the identity component of such a subgroup is H_z or $SO(3)$, and in the first case the subgroup normalizes H_z. The rotation by π about e_1 acts on M^{H_z} by $J_{e_3} \mapsto -J_{e_3}$, so $M^{O(2)_z}$ is the orbifold. An element $x \in M^{O(2)_z}$ outside $\mathrm{Vir}_1(0,\infty)$ exists by the Galois correspondence, and $\mathrm{Stab}(\{O_f, x\})$ contains $O(2)_z$ and is not $SO(3)$, so it equals $O(2)_z$. The characters follow from $m_j(H_z) = 1$, $m_j(O(2)_z) = 1$ for even j and 0 otherwise, and $m_j(SO(3)) = \delta_{j0}$. (c) The Haar noise fixes exactly $M^{SO(3)}$. Without noise, the self-adjoint part of M^K is a seed with stabilizer K by the Galois correspondence, and $\mathrm{Stab}(\{O_{n,g}, O_{n',g'}\})$ is trivial for non-parallel n, n'. □

Under axis dephasing, stable seeds in this stress/current-containing class therefore have exactly three ideal hulls, and they are told apart by the spectrum of the conformal Hamiltonian at the first and fourth levels. A reader whose ideal seed contains a bounded injective function of $J_{e_3}(g)$ has the whole fixed algebra as its hull; a reader that registers only quantities insensitive to the sign of J_{e_3}, together with the energy density, gets the orbifold; a reader that registers only energy gets the Virasoro algebra.

Lifetime of a hull. Real noise breaks a symmetry slightly. Let $\mathcal{E}_t = D_t^{H_z} \circ D_{st}^{SO(3)}$, with rates κ and $s\kappa$, $0 < s \ll 1$. The two semigroups commute, because $Q_{SO(3)}Q_{H_z} = Q_{H_z}Q_{SO(3)} = Q_{SO(3)}$, and the fixed algebra of the product is $\mathrm{Vir}_1(0,\infty)$ [2].

Proposition 7.9 (A plateau of length $1/s$)**.** *For every closed $K \supseteq H_z$ with $M^K \neq \mathrm{Vir}_1(0,\infty)$,*

$$\sup_{x \in M^K,\ \|x\| \le 1} \|\mathcal{E}_t(x) - x\| = c_K\big(1 - e^{-s\kappa t}\big), \qquad 1 \le c_K \le 2.$$

Thus M^K is ε-stable on $[0,T]$ in the sense of HAFF-B exactly when $c_K(1-e^{-s\kappa T}) \le \varepsilon$, which for small ε is a window of length $\varepsilon/(c_K s\kappa)$.

Proof. On M^{H_z} the first factor is the identity and $D_{st}^{SO(3)}(x) = e^{-s\kappa t}x + (1-e^{-s\kappa t})Q_{SO(3)}(x)$, so $\mathcal{E}_t(x) - x = -(1-e^{-s\kappa t})(x - Q_{SO(3)}(x))$. Put $c_K = \sup\{\|x - Q_{SO(3)}(x)\| : x \in M^K, \|x\| \le 1\} \le 2$. For $y \in M^K \setminus \mathrm{Vir}_1(0,\infty)$ the element $x = (y - Q_{SO(3)}y)/\|y - Q_{SO(3)}y\|$ has $Q_{SO(3)}(x) = 0$, so $c_K \ge 1$. □

The U(1) hull of a current reader thus lives for a time inversely proportional to the strength of the symmetry-breaking noise, after which only the Virasoro algebra remains stable. This is the type III counterpart of the plateau that HAFF-B computes for the GHZ geometry [4]. A possible maintenance mechanism would correct the symmetry-breaking jumps on their characteristic timescale $(s\kappa)^{-1}$. The plateau formula establishes the passive lifetime; it supplies neither a universal correction-rate bound nor the energy cost of maintaining this type III hull. Q-RAIF B gives a field-coupled correcting actuator in a finite model [9]; extending it to the present hull is a concrete test of HAFF-D's conjecture that maintained access carries a gravitating stress-energy flux [5].

8 Ideal seeds and finite reading budgets

The symmetry classification concerns ideal operator seeds. A finite processing budget first restricts a physical reader's instrument menu; it does not assign a cost to the spectral closure of a current field. Appendix C gives a separate finite-coordinate coding model. No dictionary from its coordinate projections to the current or stress observables is supplied here.

Corollary 8.1 (A single ideal current seed)**.** *Under the hypotheses of Corollary 7.8, the ideal spectral seed $\{O_{e_3,g}\}$, $g \neq 0$, has the whole fixed algebra M^{H_z} as its hull. No stable seed has a larger hull. Without noise, two ideal seeds containing bounded injective functions of nonparallel current components have M as their hull.*

Proof. Lemma 7.4 and Theorem 7.6 identify the one-axis hull. Theorem 5.2 bounds every stable hull by the fixed algebra. Two nonparallel current components have trivial common stabilizer, so Theorem 7.6(a) gives M. □

This corollary uses ideal bounded functional calculus, all modular times and von Neumann closure. Proposition 3.2 instead prices an actual finite pointer instrument. Neither statement converts a one-coordinate processing allowance into permission to implement the whole current algebra. When a physical menu implements a specified spectral seed, its ideal hull depends on the stabilizer of that seed; the menu and its finite cost remain separate inputs.

9 How a hull responds to excitations

HAFF-D connects a selected algebra to gravity through the response of its modular data to excitations of the state [5], and Consciousness I computes the relative entropy lost when a Cut is displaced [2]. Whether these responses depend on the seed is decided by the following facts. For a normal state ψ and a von Neumann subalgebra $\mathcal{N}$, write $S_{\mathcal{N}}(\psi\|\omega)$ for Araki's relative entropy of the restrictions, and $\omega_W = \omega(W^* \cdot W)$ for a unitary W.

Proposition 9.1 (Response and the hull)**.** *Let $\mathcal{H} \subseteq \mathcal{H}'$ be σ-invariant von Neumann subalgebras.*

(a) For a unitary $W \in \mathcal{H}$, $S_{\mathcal{H}}(\omega_W\|\omega) = S_{\mathcal{A}_U}(\omega_W\|\omega)$, and the same holds with the arguments exchanged.

(b) For every normal state ψ, $S_{\mathcal{H}}(\psi\|\omega) \le S_{\mathcal{H}'}(\psi\|\omega)$. If the right side is finite, equality holds if and only if $\psi|_{\mathcal{H}'} = \psi|_{\mathcal{H}} \circ E_{\mathcal{H}}|_{\mathcal{H}'}$.

Proof. (a) Since $W \in \mathcal{H}$ and $E_{\mathcal{H}}$ is $\mathcal{H}$-bimodular and ω-preserving, $\omega_W \circ E_{\mathcal{H}} = \omega_W$ and $\omega \circ E_{\mathcal{H}} = \omega$. Relative entropy is monotone under restriction to $\mathcal{H}$ and under the channel $\varphi \mapsto \varphi \circ E_{\mathcal{H}}$, which maps the two restrictions back to the two states, so the two values coincide. (b) The inequality is monotonicity under restriction. The equality condition is Petz's characterization of sufficient subalgebras [26, 25]: with $\omega \circ E_{\mathcal{H}} = \omega$, the recovery map for ω is $E_{\mathcal{H}}$ itself. □

Corollary 9.2 (Stress excitations do not see the seed; current excitations do)**.** *In the SU(2)$_1$ net:*

(a) for $W = e^{iT(f)}$ with $f \in C_c^\infty((0,\infty),\mathbb{R})$, every hull M^K gives the same relative entropy, so the coherent-state formula of Hollands and the Cut-displacement theorem of Consciousness I [20, 2] hold verbatim in every hull of Theorem 7.6;

(b) for $W = e^{iJ_{e_3}(h)}$ with $0 \neq h \in C_c^\infty((0,\infty),\mathbb{R})$,

$$S_{\mathrm{Vir}_1(0,\infty)}(\omega_W\|\omega) < S_{\mathcal{A}_{J_{e_3}}(0,\infty)}(\omega_W\|\omega) = 2\pi\langle W\Omega, KW\Omega\rangle,$$

where $K = -\frac{1}{2\pi}\log\Delta$ is the dilation generator.

Proof. (a) $e^{iT(f)} \in \mathrm{Vir}_1(0,\infty) \subseteq M^K$; apply Proposition 9.1(a). (b) $W \in \mathcal{A}_{J_{e_3}}(0,\infty)$, so the middle term equals the ambient value, which for a unitary of M is $-\langle W\Omega, \log\Delta\, W\Omega\rangle$; it is finite because the coherent state has finite dilation energy. On $\mathcal{H}' = M^{H_z}$ the expectation onto $\mathrm{Vir}_1(0,\infty)$ is the Haar average $Q = Q_{SO(3)}$. For a real $k \in C_c^\infty((0,\infty))$ and the odd function arctan, the rotation by π about e_1 maps $\arctan J_{e_3}(k)$ to its negative and leaves Q unchanged, so $Q(\arctan J_{e_3}(k)) = 0$. In the state ω_W, $J_{e_3}(k)$ has the vacuum law, a centred Gaussian, shifted by a real number μ proportional to $\int hk'\,dx$, because the commutator of the current is a multiple of the identity; $\mu \neq 0$ for $k = h'$. Since arctan is odd and strictly increasing, $\omega_W(\arctan J_{e_3}(k)) \neq 0 = \omega_W(Q(\arctan J_{e_3}(k)))$. By Proposition 9.1(b) the inequality is strict. □

For a stress excitation lying in every hull, its relative entropy and modular-energy response agree across the hulls, by Proposition 9.1(a). The modular operators are restrictions of the ambient operator to different vacuum-cyclic spaces. The equality applies to excitations lying in every hull; arbitrary states can have different relative entropies on those spaces. Relative entropy can differ: the hull of an energy reader underestimates the information in a current excitation that the hull of a current reader sees in full. Whatever role a selected algebra plays in a gravitational response, its dependence on the seed enters through excitations outside the Virasoro algebra; stress excitations alone cannot detect which seed was selected. In higher dimensions Consciousness II shows that selected algebras with the required stress and boundary readings, on a conformal net in Minkowski space and conditional on its supplied MC$_\eta$ ball-consistency law and source regularity, give the same matter-sourced linearized Einstein tensor when they use one source and normalization; their relative entropies can differ [1]. That paper also constructs an exact neutral scalar preparation inside a proper stable hull: a seed containing the stress readings and one invariant quadratic observable contains a local unitary that produces nonzero Fourier weight. With permission to implement this pulse, the observer can test the intrinsic response without a charged-field phase reference. This construction is in the higher-dimensional matter theory; the chiral Cut classification here does not supply its spacetime or its modular-consistency law.

10 What is still supplied, and three conjectures

The construction derives the fixed algebra, its expectation and its modular invariance from stationarity; the hull and its expectation from a seed; a proper Cut in every nonscalar translation-stable hull from an ambient positive translation structure with a unique invariant vector; and, under the hypotheses of Theorem 7.2, the hull from the seed's stabilizer. The following remain inputs.

(1) The ambient algebra, its state and, for Cuts, its translations. Hulls inherit Cuts; they do not create the ambient one. An inclusion of two hulls is never a Cut, since both are invariant under the modular group in both directions; one-sided objects such as the past modular orbit of Consciousness I are needed for that.

(2) The coupling to the environment. HAFF-B derives the selected algebra from the couplings, and Consciousness I from the noise group; neither derives the couplings.

(3) Stationarity, or a stationary partner supplied by an invariant expectation (Proposition 4.3). Relaxation requires an additional dynamical statement (Example 4.4).

(4) The task that fixes the foreground. T-DOME II gives the optimal reading directions for a given task; the task is supplied.

(5) A minimal compact symmetry for the classification. In the $\mathrm{SU}(2)_1$ net it is verified; elsewhere the hull is still determined by Theorems 5.1 and 5.2, but its identification with a fixed-point algebra is open.

Conjecture 10.1 (Quantum symmetry fixes the hull)**.** *Let $\mathcal{A}$ be a completely rational conformal net and $\mathcal{B} \subset \mathcal{A}$ a Möbius covariant subnet of finite index with $\mathcal{B}(I)' \cap \mathcal{A}(I) = \mathbb{C}\mathbf{1}$. For every seed whose Γ_3-hull contains $\mathcal{B}(0,\infty)$, the hull is the half-line algebra of the intermediate net fixed by the generalized stabilizer of the seed in the quantum symmetry of $\mathcal{B} \subset \mathcal{A}$, and its vacuum character is the corresponding partial sum of branching functions.*

The mechanism is Theorem 7.2 with the compact group replaced by the symmetry that a finite-index inclusion carries. Support comes from the compact case proved here, from the extension of the Izumi–Longo–Popa correspondence to compact Kac algebras [21], and from the correspondence between intermediate extensions and intermediate Q-systems [11]. The strongest objection is that a Γ_3-hull is invariant under dilations and forward translations but is not known in general to be the half-line algebra of a local subnet; in the $\mathrm{SU}(2)_1$ case this follows from the Galois law, not from the hull construction. Without the trivial relative commutant the conjecture fails, since intermediate algebras then need not be fixed by any symmetry.

Conjecture 10.2 (Window selection in type III)**.** *For the noise $D_t^{H_z} \circ D_{st}^{SO(3)}$ and seeds containing a stress observable, a window with $s\kappa T \ll \varepsilon \ll \kappa T$ selects M^{H_z} as the unique largest stable hull in this class: it is ε-stable on $[0,T]$, and no hull M^K with $K \not\supseteq H_z$ is. As the window grows past $\varepsilon/(s\kappa)$ the selected hull drops to $\mathrm{Vir}_1(0,\infty)$, and the step is visible in the vacuum character.*

This is HAFF-B's window theorem [4] in type III. Proposition 7.9 proves the stability half. For $K \subsetneq H_z$, an element $x \in M^K$ with $Q_{H_z}(x) = 0$ and $\|x\| = 1$ has $\|\mathcal{E}_t(x) - x\| = 1 - e^{-(1+s)\kappa t}$, which proves the instability half for those K. The objection is that for other K the supremum of the instability over the unit ball of M^K has no evident lower bound independent of K; localized charged unitaries are the natural candidates for proving one.

Conjecture 10.3 (Every hull contains the Virasoro algebra)**.** *In the SU(2)*$_1$ *net every nonscalar* Γ_3*-hull contains* $\mathrm{Vir}_1(0,\infty)$*. Then Theorem 7.6 classifies all hulls of the net.*

The mechanism is Theorem 6.1: a nonscalar hull carries a Cut, and if the Cut is standard it reconstructs a conformal subnet $\mathcal{B}$. If $\mathcal{B}$ is diffeomorphism covariant with central charge one, the coset of $\mathcal{B}$ in $\mathcal{A}$ has central charge zero and is trivial, so the stress tensor of $\mathcal{B}$ is the Sugawara tensor and $\mathrm{Vir}_1 \subseteq \mathcal{B}$ [18]. A Möbius covariant subnet of central charge below one cannot occur: Carpi showed that every Möbius covariant subsystem of the $\mathrm{SU}(2)_1$ net is a fixed-point net $\mathcal{A}^H$ for a closed subgroup $H \subseteq SO(3)$, and all of these contain Vir_1 [14]. The conjecture therefore reduces to a lifting problem, which is the objection: the net reconstructed from a standard hull Cut must be shown to be a conformal subsystem of the ambient net, that is, its local algebras $\mathcal{H} \cap (U(1)\mathcal{H}U(1)^*)'$ must lie in the ambient local algebras and its Möbius representation must be the restriction of the ambient one. Standardness of the inherited Cut is also not automatic.

11 Conclusion

The physical reader and the ideal closure carry different information. A nonzero-gain probe can generate a sharp pointer algebra without being able to read that pointer sharply in one shot. In the explicit nondemolition model, repetition with fresh ports makes the ideal instrument available to any prescribed finite accuracy, with a budget depending on gain. The instrument-level estimate survives old references and finite adaptive composition. This is the task bridge; spectral closure alone cannot supply it.

Stationarity of the source makes the fixed algebra a modular-invariant expected subalgebra and supplies a persistent projection with no memory kernel, without requiring relaxation of every trajectory. The stable hull of a seed fixed pointwise by the noise carries the source's own modular data and lies between the seed and the fixed algebra. A nonscalar hull that is also closed under forward translations carries a proper Cut on its vacuum space when the ambient positive translation group leaves only the vacuum invariant. Translation covariance of the noise keeps this larger hull in the fixed algebra. Under a minimal compact symmetry, a hull containing the symmetry-fixed algebra is fixed by the stabilizer of the seed when its closure maps preserve the stabilizer-fixed algebra. Within this class the accessible algebra is unique relative to what is read and changes only when the symmetry left unbroken by the seed changes. In the $\mathrm{SU}(2)_1$ net the vacuum characters of the hulls tell the stabilizers apart; a single current reading generates the U(1) current algebra under the modular flow. For the positive log-bump stress class, modular flow alone generates the Virasoro half-line algebra, and the one-sided orbit constructs a proper standard Cut on its own vacuum space. In the stated stress/current seed class, dephasing about one axis leaves exactly three ideal hulls, and each lasts as long as the noise respects its symmetry. The response of a hull to stress excitations does not depend on the seed, while its response to a current excitation does.

The source–access–Cut chain has a derived net-theoretic middle. The retained seed fixes the ideal algebra through symmetry in the classified cases; physical protocols decide which distinctions can be used within a task budget. The translation-stable algebra inherits the source's modular data and a Cut. In the $\mathrm{SU}(2)_1$ realization that Cut is standard and reconstructs its conformal subnet on the inherited light ray. Different seeds can select nonisomorphic nets there without producing distinct spacetime metrics. The ambient net and physical reader remain inputs. The log-bump construction removes an independent forward-translation requirement from ideal stress-seed generation, but still presupposes the ambient stress tensor and modular evolution. Coupling, preparation permissions and the task remain physical inputs. Learned

access and retained-history mechanisms can change these inputs during an actual run; the hull theorem then identifies the structure supported by the resulting seed. It does not by itself explain the origin of the first available coupling.

Centred and uncentred task losses. A rank optimizer for centred omission loss need not optimize an uncentred squared loss: the latter also contains $(\sum_{i\notin I}\mathbb{E}\lambda_i)^2$. For three independent channels with means and variances $(0,1),(1,.1),(1,.1)$ and one retained channel, the uncentred minimizer keeps a channel with smaller centred weight. The coding result in Appendix C therefore retains its stated task assumptions.

A Rotation angles and conjugacy classes

Lemma A.1. *For a closed subgroup $K \subseteq SO(3)$ let μ_K be the distribution of the rotation angle $\theta \in [0,\pi]$ under normalized Haar measure. Then μ_K determines K up to conjugacy.*

Proof. Up to conjugacy the closed subgroups are $SO(3)$, $O(2)$, $SO(2)$, the icosahedral, octahedral and tetrahedral groups I, O, T of orders $60, 24, 12$, the dihedral groups D_n of order $2n$, $n \geq 2$, and the cyclic groups C_n, $n \geq 1$. The three infinite groups have μ_K with density $(1-\cos\theta)/\pi$, the density $1/\pi$, and half the density $1/\pi$ plus an atom $\frac{1}{2}$ at π; these correspond respectively to $SO(3)$, $SO(2)$, and $O(2)$. The finite ones are atomic. For finite K the atom at 0 is $1/|K|$, which fixes the order. Among groups of equal order, the atom at π separates them: it is $1/n$ for C_n with n even and 0 for n odd; $(n+1)/2n$ for D_n with n even and $\frac{1}{2}$ for n odd; $\frac{1}{4}$ for T, $\frac{3}{8}$ for O and $\frac{1}{4}$ for I. For order $2m$, C_{2m} has $1/2m$ and D_m at least $\frac{1}{2}$, equal only for $m=1$, where D_1 is conjugate to C_2. For orders 12, 24 and 60 the three values are $(\frac{1}{12},\frac{7}{12},\frac{1}{4})$, $(\frac{1}{24},\frac{13}{24},\frac{3}{8})$ and $(\frac{1}{60},\frac{31}{60},\frac{1}{4})$ for C, D, and T, O, I respectively. □

B Numerical checks

The supplementary source map identifies the following diagnostics and their retained outputs. Finite samples test the stated identities; the infinite-dimensional conclusions use the proofs in the text.

B.0 Physical permission bridge. The controlled-X complete instrument is checked on an old Bell reference and eleven random mixed inputs at $\alpha = 0, .1, .6, 1$ and $m = 1, 2, 3, 5$, including the fair tie rule and post-read-flip counterexample. The sharp half-diamond identity is analytic (Proposition 3.2).

B.1 Finite hulls. Closure under products, adjoints, the modular generator and noise reproduces the nonstationary qubit hull M_2, its pinched-state two-dimensional hull, and the $M_2 \otimes 1$ stationary fixed sector. The three-qubit example has seed, hull and fixed dimensions $3, 10, 20$, respectively. Other checks distinguish equal type-I fixed algebras with different stabilizers, verify persistence and joins, and test Proposition C.2 on 200 task matrices.

B.2 Characters. The class data of C_n, D_n for $n \leq 40$, the three finite polyhedral groups and $SO(2), O(2), SO(3)$ give 85 distinct multiplicity sequences through $j = 60$. Their characters reproduce the table and the Virasoro, current, SU(2) level-one and even-Fock low-energy

coefficients. Finite-level agreement does not replace the analytic character and rotation-angle arguments.

C A separate finite-coordinate coding model

Here V is an abstract finite-dimensional task coordinate space. The propositions do not construct a net observable from a coordinate projection and are not premises of Corollary 8.1. T-DOME II supplies the Gaussian task optimizer used in this model.

Proposition C.1 (Rank under a processing budget)**.** *Let a reader hold a real coordinate space V of dimension D, let admissible encoders be orthogonal projections Q on V with processing cost $h_\mu \operatorname{rank} Q$, $h_\mu > 0$, and assume:*

(1) the full-rank encoder is infeasible, $h_\mu D > \dot{\mathcal{C}}_{\text{budget}}$;

(2) the distortion functional attains its minimum on the feasible projections and strictly decreases whenever a feasible projection is enlarged by one coordinate;

(3) a group G acts irreducibly and orthogonally on V, the distortion is G-invariant, and $0 < \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor < D$.

Then every minimizing encoder has rank $k^* = \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor < D$, its range is a k^*-dimensional foreground, and no such projection is G-invariant, so choosing a minimizer breaks the G-symmetry.

Proof. Feasibility gives $\operatorname{rank} Q \le k^*$, and strict improvement excludes smaller rank. A minimizing projection that commuted with the irreducible representation would have invariant range and kernel, hence would be 0 or $\mathbf{1}$, contrary to $0 < k^* < D$. □

The rank charge, strict improvement and irreducible action are inputs of this coding model. They are not consequences of persistence, and the proposition does not say that symmetry breaking is unavoidable outside this class of encoders.

Which k^* directions are kept is answered by T-DOME II. For a budgeted reader whose task is linear in a fresh Gaussian input and whose loss is quadratic, with all rank-k linear projectors allowed, the best risk of a rank-k reading policy is attained by the top-k eigenspace of the posterior task matrix $\widehat{W}_M$ built from its memory M, and the optimal subspace is unique when $\lambda_k(\widehat{W}_M) > \lambda_{k+1}(\widehat{W}_M)$ [8]. These directions are record channels when $\widehat{W}_M$ commutes with the projections onto the record channels, and approximately so otherwise.

Proposition C.2 (Foreground and record channels)**.** *Let the reader's D channels be record channels, and write $\widehat{W}_M = W_{\text{d}} + W_{\text{o}}$ with W_{d} diagonal in the channel basis and W_{o} its off-diagonal part. Let $\Delta_k > 0$ be the gap between the k-th and $(k+1)$-st largest diagonal entries and P_{ch} the projection onto the corresponding k channels. If $W_{\text{o}} = 0$, the optimal foreground is spanned by those k channels. If $\|W_{\text{o}}\| < \Delta_k$, any rank-$k$ optimal projection P_{opt} satisfies*

$$\|P_{\text{opt}} - P_{\text{ch}}\| \le \frac{\|W_{\text{o}}\|}{\Delta_k - \|W_{\text{o}}\|}.$$

Proof. The first statement is the capability theorem of T-DOME II for a diagonal matrix. For the second, Weyl's inequality separates the top k eigenvalues of $\widehat{W}_M$ from the remaining diagonal entries of W_{d} by at least $\Delta_k - \|W_{\text{o}}\|$, and the Davis–Kahan $\sin\Theta$ theorem [17] bounds the sine of the largest principal angle, which for projections of equal rank equals $\|P_{\text{opt}} - P_{\text{ch}}\|$, by $\|W_{\text{o}}\|$ divided by that separation. □

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Foundation II: Shared Manifestation Shared Structure, Public Calibration and Different Histories

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Abstract

Several readers of one process can have different accessible algebras while sharing a world. Their common structure is the intersection of their stable hulls. It is the largest algebra available to all of them, inherits the common modular flow and, with a supplied positive translation structure and a unique vacuum, contains a proper half-sided modular inclusion whenever it is non-scalar. No foreground overlap or commutation of noise channels is required. For compact symmetry-selected hulls the common algebra is fixed by the group generated by the readers' stabilizers; under the minimal-action Galois conditions their jointly generated algebra is fixed by the intersection of those stabilizers. The SU(2) level-one model realizes a shared Virasoro sector between two different current sectors, even with disjoint foreground seeds.

Agreement does not certify the truth of a shared description. A common symmetry change of instruments and preparations is indistinguishable, in every finite adaptive record law, from the corresponding change of actual state. We derive the effective groups of these errors and show that their kernels differ on individual and shared content. Conversely, a non-scalar translation-stable shared algebra contains a finite positive-energy pulse. With preparation and readout permission, a known vacuum baseline turns it into an energy-orientation reference with an explicit finite-record error bound. Under the additional source regularity and modular-consistency conditions of Consciousness II, the pulse also supplies a spectral witness for a common sourced geometric response. It does not fix a metric without boundary or initial data. For a finite family of common frame errors, legal reference blocks identify a public target exactly when indistinguishable reference laws have the same target value. The resulting finite-record bound permits residual private frame differences; an explicit reference protocol preserves different old histories. A two-observer calibration model measures common error and fault transfer. Independent-source compatibility is treated separately. Shared carrier, shared state, agreeing records and correct content are distinct requirements.

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1 Introduction

Foundation I asks how a source, its noise and a retained seed specify an accessible algebra [1]. This paper asks what several readers of that source share. The question has two parts. A common carrier must be constructed from the readers' access. A common description must also survive errors that all readers can make together. Agreement alone addresses neither part completely.

The main setting is one ambient process with different seeds and accessible hulls. Their intersection derives the shared algebra instead of assuming a candidate carrier. Its invariance needs no commutation of the noise channels. If a positive translation structure is supplied, a non-scalar intersection inherits a Cut on its own vacuum-cyclic space. Cuts sit inside that shared algebra; the algebra itself is not asserted to be a Cut of the ambient algebra. The translations remain physical inputs, as in Foundation I and Consciousness I [1, 5].

Symmetry makes the distinction between common and joint access precise. For hulls fixed by subgroups, the shared algebra is fixed by the group those subgroups generate. The joint algebra obeys a different law under the compact Galois hypotheses [3]. Two independent current seeds in the SU(2) level-one net have no foreground overlap yet share a Virasoro sector. The algebraic sharing mechanism is therefore stronger than foreground matching. Foundation III provides a noncommuting, repeated-noise realization of the same fixed intersection [2]; the present paper uses that result without repeating its convergence proof.

Readers can then agree and still share a displaced standard. Reference-frame restrictions and relational encodings are established subjects in quantum information [4]. Here an adaptive-protocol covariance identity identifies a class of common errors that no agreement statistic can expose. The group acting faithfully on individual content can be larger than the group acting faithfully on shared content. In the current-net model a common energy sector is insensitive to reversals of current orientation. This is an operational distinction, not a claim that a symmetry-invariant reading is automatically true.

A shared algebra can also contain a way to test a standard. Consciousness II's bounded-preparation construction gives a finite positive-energy pulse whenever the shared algebra is non-scalar and translation-stable [6]. If the pulse and vacuum baseline can actually be used at a common energy port, the mean difference determines an otherwise unknown energy-gain sign. The preparation changes the physical state, so it can also provide the source witness required to identify a geometric response, subject to the separate regularity and consistency conditions. This connects selected access to a testable world rather than identifying geometry from consensus alone.

The source need not have been described by one state in every problem. Independent source states, projections and memory kernels have a different compatibility question. Appendices A–D retain that theorem and its nonempty examples. It cannot be made unconditional by substituting the one-process construction for its independent states. Likewise, the two-observer simulation of Section 5 is a finite-resource calibration experiment, not a finite reconstruction of a Type III algebra.

Quantum Darwinism and spectrum broadcast structures study information held redundantly in environmental fragments [20, 21, 22]. An intersection of accessible algebras is not a redundancy certificate, and two carriers can encode the same bit even when their literal algebraic intersection is scalar. The paper keeps these distinctions while connecting the source-selection problem of HAFF to the reference and self-calibration problems of T-DOME [9, 10, 12, 13].

Section 2 derives shared structure and its symmetry laws. Section 3 identifies common frame ambiguities. Section 4 constructs the internal reference and its conditional geometric

test. Section 5 gives the calibration simulation, and Section 6 returns the results to the full theory.

2 One process, several accessible structures

Let M be a von Neumann algebra with a cyclic and separating vector Ω and faithful normal reference state $\omega(x) = \langle \Omega, x\Omega \rangle$. A family $\mathfrak{E}$ of normal unital completely positive maps preserves ω. Its members need not commute. Include the full reference modular group and put

$$\Gamma_2 = \{\sigma_s^\omega : s \in \mathbb{R}\} \cup \mathfrak{E}.$$

When a positive translation group $U(a) = e^{iaP}$ is supplied, with $P \geq 0$, $U(a)\Omega = \Omega$ and $U(a)MU(a)^* \subseteq M$ for $a \geq 0$, also put

$$\Gamma_3 = \Gamma_2 \cup \{\mathrm{Ad}\, U(a) : a \geq 0\}.$$

For either family, a subalgebra is admissible when every map in the family takes it into itself. A reader retains a seed $S_j \subseteq M$ and has the ideal accessible hull $B_j = \mathrm{Hull}_\Gamma(S_j)$: the intersection of all admissible von Neumann subalgebras containing S_j [1]. This closure describes an operator structure. It is not a finite-resource reconstruction algorithm or a licence to implement every unitary in the hull. The seed, noise, control permissions and reference state are physical inputs.

Theorem 2.1 (The shared structure of one process)**.** *For any family of readers with the same M, ω and Γ, set $B_\cap = \bigcap_j B_j$.*

1. *$B_\cap$ is the largest admissible von Neumann subalgebra contained in every B_j. It contains* $\mathrm{Hull}_\Gamma(T)$ *for every $T \subseteq B_\cap$, and carries a unique normal ω-preserving conditional expectation. Its modular group in $\overline{B_\cap \Omega}$ is $\sigma^\omega|_{B_\cap}$.*

2. *For every normal actual state ψ of M, the restrictions $\psi|_{B_j}$ agree on $B_\cap$. This says nothing about equality of the readers' joint records or about the accuracy of their decoding.*

3. *For Γ_3, suppose U is the translation group of an ambient half-sided modular inclusion and* $\ker P = \mathbb{C}\Omega$*. If $B_\cap \neq \mathbb{C}1$, then $U(a)B_\cap U(a)^* \subset B_\cap$ for $a > 0$ is a proper half-sided modular inclusion on $\overline{B_\cap \Omega}$. The recovered translations are the restricted ambient ones. If the inclusion is standard, Consciousness I's Cut–geometry correspondence applies there.*

Proof. An intersection of von Neumann algebras is a von Neumann algebra. If x is in every B_j, then $\gamma(x)$ is in every B_j for each $\gamma \in \Gamma$. For the modular maps, including both s and $-s$ gives equality rather than only inclusion. The intersection is therefore admissible, and every admissible subalgebra contained in all the B_j lies in it. If $T \subseteq B_\cap$, the intersection is itself a competitor in the definition of its hull. Modular invariance gives the expectation and restricted modular group by Takesaki's theorem [17, 16]. Restrictions of one state agree on an intersection. Finally $B_\cap$ inherits forward translation invariance; Foundation I's Cut-inheritance theorem supplies (3), including properness from the unique invariant vector hypothesis [1, section on Cuts in the hull]. □

This route needs neither a candidate shared carrier nor commuting projected memory kernels. It also does not require the foreground seeds themselves to overlap. It addresses readers of one process. Independently specified source states can still disagree on an overlap; their compatibility is treated in Appendices A–C. The reference ω determines the closure and the Cut. The actual state ψ may be different, even nonstationary. Its restrictions agree because they are restrictions of one state, not because the reference and actual states have been identified.

2.1 Shared, individual and jointly available algebras

Let a compact group G act continuously on M, preserving ω, and let K_j be closed subgroups such that $B_j = M^{K_j}$. Write

$$K_\cup = \overline{\langle \bigcup_j K_j \rangle}, \qquad K_\cap = \bigcap_j K_j.$$

The fixed-algebra identity

$$B_\cap = M^{K_\cup} \tag{1}$$

holds without a minimal-action assumption: an element fixed by every K_j is fixed by their generated group and, by continuity, its closure. If the action is minimal on a factor with separable predual, the compact-group Galois correspondence also gives

$$\bigvee_j B_j = M^{K_\cap}. \tag{2}$$

Here minimal means faithful with $(M^G)' \cap M = \mathbb{C}1$ [3]. Indeed the join is an intermediate algebra M^L. Containment of every M^{K_j} forces $L \subseteq K_\cap$, whereas the join lies in $M^{K_\cap}$, forcing the opposite inclusion. If $M^{K_\cap}$ is also Γ-admissible, the hull of the joint seed equals that join. Admissibility of a join is not automatic for arbitrary UCP maps, which need not preserve products of elements from different subalgebras.

Example 2.2 (Shared geometry without shared foreground)**.** In the SU(2) level-one current net let $M = \mathcal{A}(0,\infty)$, with its vacuum, modular dilations and forward translations. With no additional noise, the bounded current seeds $\frac{2}{\pi}\arctan J_{e_3}(f)$ and $\frac{2}{\pi}\arctan J_{e_1}(f)$ have hulls M^{H_z} and M^{H_x}, where H_z and H_x are the axial rotation groups. These seed-to-hull statements and the fixed subnet are established in Foundation I [1]. Here f is nonzero, real and smooth with compact support in the positive half-line. The two groups generate SO(3), so

$$B_\cap = M^{SO(3)} = \mathrm{Vir}_1(0,\infty), \qquad B_1 \vee B_2 = M.$$

Each individual hull is a current algebra; their common algebra is Virasoro; their joint algebra is the full current net. They are three different objects. The two seed lines have zero foreground intersection, yet their hulls share a standard Cut. Foreground overlap is therefore not necessary for sharing. The supplied vacuum net and translations remain inputs, not outputs of the sharing argument.

Nontriviality is not universal. In $M_2 \otimes M_2$ with the trace, trivial modular flow and no noise, the seed algebras $W^*(Z \otimes 1)$ and $W^*(1 \otimes Z)$ intersect only in the scalars. The two registers can nevertheless encode the same classical bit with high agreement. A shared subalgebra on one ambient carrier and matching content on different carriers are distinct ways of sharing; neither defines the other.

2.2 A dynamical realization without commuting noises

For a finite family $K_1, \ldots, K_m$ let Q_j be the Haar expectations and apply $D_n = (1-\theta_n)\mathrm{id} + \theta_n Q_{i_n}$, with $0 < \theta_n \leq 1$. Foundation III proves

$$\sum_{n:i_n=j} \theta_n = \infty \quad \text{for each } j \quad \Longrightarrow \quad D_n \cdots D_1(x) \longrightarrow Q_{K_\cup}(x) \quad \text{strong*}. \tag{3}$$

Thus $B_\cap$ can be the algebra retained by repeated noise, not only an intersection written after the fact [2]. There is no bounded-gap or commutation condition. Record-dependent schedules are covered when their realized sequence meets the exposure condition, but intervening measurement and control channels must be included separately. The finite-family restriction is essential; Foundation III supplies a countable family for which every noise is visited indefinitely but convergence fails. This constructs persistence for supplied noise actions. It does not derive why the source has those actions, or turn the ideal limit into a finite-time resource guarantee.

3 Common errors that agreement cannot expose

A reader's standard assigns record labels to instruments, preparations and decoders. Correlating standards is a reference-frame problem, not the equality of two restricted states [4]. Let α_g be a normal automorphism of M commuting with every uncontrolled evolution map used in a protocol. Each instrument branch $\mathcal{I}^y$ is a normal completely positive map, and its sum over outcomes is unital. Choices of later branches and preparations may depend on earlier labels and on recorded private randomness, with the same prescribed conditional coin law in the two configurations being compared.

Proposition 3.1 (An indistinguishable common change of standard)**.** *For any finite adaptive protocol, replacing every controlled operation by $\alpha_g \mathcal{I}^y \alpha_g^{-1}$ in the actual state ψ gives the same joint record law as the original protocol in $\psi \circ \alpha_g$. If $x = x^* \in M$, $\alpha_g(x) = -x$ and $\psi(x) \neq 0$, a binary report about the sign of $\psi(x)$ has correctness probabilities adding to one in these two indistinguishable configurations. At least one of them has correctness at most $1/2$, regardless of the agreement rate or the number of readers.*

Proof. For a fixed record string the probability is a state applied to the ordered composition of instrument branches and intervening evolution maps, evaluated at 1. Conjugating the controlled branches and commuting α_g through the uncontrolled maps leaves α_g on the outside and $\alpha_g^{-1}(1) = 1$ on the inside. This changes ψ to $\psi \circ \alpha_g$. Adaptivity changes the branch selected by the fixed string, not this identity. Averaging over private coins preserves it. The truth changes sign between the two configurations while the report law does not. A binary report is correct in exactly one of them, proving the sum and the bound. □

The ambiguity concerns two physical configurations: a displaced standard and a displaced state. It is not the assertion that the same state has two different true values. A calibrated reference left unchanged by the protocol can break the symmetry; conjugating that reference as well would remove the very information that makes it a reference. This result classifies a source of common error, not every statistical mistake. Two independent error bits with error probability e still agree wrongly with probability e^2.

3.1 The effective group of a common frame error

Let G be a compact group of allowed frame transformations, and let $H \leq G$ be its closed subgroup preserving the uncontrolled world laws. Suppose the individual algebras are $B_j = M^{K_j}$. We first consider errors that preserve each labelled reader's algebra rather than exchange the readers. Define

$$N = H \cap \bigcap_j N_G(K_j), \qquad L = N \cap \bigcap_j K_j, \qquad L_\cap = N \cap K_\cup.$$

Under the minimal compact-action hypotheses used in (2), the pointwise stabilizer of M^K inside G is exactly K [3, Remark 3.17].

Proposition 3.2 (Joint and shared errors have different kernels)**.** *In this regime the transformations in H preserving every B_j are exactly N. Their faithful action on the tuple of individual algebras, or on their jointly generated algebra, is N/L. Their faithful action on the shared algebra $B_\cap = M^{K_\cup}$ is $N/L_\cap$. In particular $L \subseteq L_\cap$, so a nontrivial error on individual content can be invisible on all shared content. For one reader the effective group is*

$$N_H(K)/(K \cap H),$$

where $N_H(K) := H \cap N_G(K)$ even when K is not contained in H. This reduces to $N_H(K)/K$ only when $K \subseteq H$. For one reader the observables unchanged by every such error form $M^{\overline{\langle K, N_H(K)\rangle}}$.

Proof. $\alpha_g(M^K) = M^{gKg^{-1}}$, so injectivity of the fixed-subgroup correspondence makes preservation of M^K equivalent to normalization of K. The pointwise kernel on each B_j is $N \cap K_j$; intersecting gives L. Fixing every individual algebra pointwise is equivalent to fixing its join pointwise. Every element of N also normalizes $K_\cup$, and the pointwise kernel on $M^{K_\cup}$ is $N \cap K_\cup = L_\cap$. Both are kernels of continuous group actions, hence closed normal subgroups of N, even if K_j is not contained in H. Also $K_\cap \subseteq K_\cup$. The invariant-algebra statement follows by imposing invariance under K and $N_H(K)$ together. □

Without the Galois hypothesis one can still restrict normalizing transformations to the reader's algebra, but the effective group is the group divided by its actual pointwise kernel. That kernel need not be $K \cap H$. The invariant algebra in Proposition 3.2 is immune to the specified frame errors. Its readings can still have statistical error or be decoded incorrectly; immunity is not correctness.

Example 3.3 (Energy is shared while a current can be reversed)**.** In the single-axis SU(2) level-one noise model the world-law symmetries inside SO(3) form the axial normalizer $O(2)_z$, while a current reader has $K = H_z \cong SO(2)$. The residual frame group is $\mathbb{Z}_2$: its nontrivial element reverses J_z but fixes the stress tensor. The immune algebra is the $O(2)_z$ orbifold, not the full current algebra [1]. For the two perpendicular current readers of Example 2.2, with no extra noise, N is the four-element group of identity and the three half-turns about the coordinate axes. Its action on the pair of private algebras is faithful, whereas it acts trivially on their shared Virasoro algebra. The energy sector is unchanged while either current orientation can be wrong. This gives an explicit nontrivial N/L with trivial $N/L_\cap$.

4 A reference made inside the shared structure

Agreement cannot break a symmetry of the whole protocol. A preparation with a known baseline can. The distinction matters for the connection from algebra selection to geometry: a changed reading is not a changed material source, whereas an implemented excitation is.

Proposition 4.1 (Shared positive-energy preparation and orientation)**.** *Suppose the readers' hulls have the common representation and Γ_3 translation closure of Theorem 2.1, with* $\ker P = \mathbb{C}\Omega$*. If $B_\cap$ is non-scalar, it contains a bounded self-adjoint non-scalar operator B obtained by smoothing a translation orbit, and a unitary $W_\theta = e^{i\theta B}$ with*

$$0 < E_\theta = \langle W_\theta\Omega, PW_\theta\Omega\rangle < \infty \quad \textit{for } 0 < |\theta|\|B\| < \pi.$$

All energy moments of this state are finite. If this unitary is an allowed preparation and the same energy port can be used on the known vacuum and on freshly prepared states, suppose its output is

$$Y = \epsilon g\, p + b + \nu, \qquad \epsilon \in \{-1, 1\}, \quad g > 0,$$

where p is the spectral energy outcome and the noise has the same finite mean in both uses. The difference of population means is $\epsilon g E_\theta$, so these record laws identify the sign ϵ.

Proof. The intersection inherits separation and the forward translation orbit of every one of its operators. Consciousness II's bounded-preparation proposition therefore applies [6, section on a bounded accessible preparation]: for a narrow real nonnegative $f \in C_c^\infty(0, \tau)$ with $\int f = 1$ and a bounded non-scalar $C = C^* \in B_\cap$, set $B = \int f(t) U(t) C U(t)^* dt$. As the support of f shrinks to t_0, $B\Omega \to U(t_0)C\Omega \notin \mathbb{C}\Omega$ by separation, so a sufficiently narrow f keeps B non-scalar. Its bounded iterated commutators with P then give finite energy moments of $W_\theta\Omega$. Separation and the unique ground vector give strictly positive energy for non-scalar W_θ. The vacuum has $p = 0$, so subtracting its output mean removes both the unknown offset and the common noise mean. The remaining sign is ϵ. □

P is the common translation generator of the model. In a null Cut it is null momentum, not automatically the time-translation Hamiltonian of a higher-dimensional source. Positivity alone does not supply this calibration: in a finite two-level system $P = \mathrm{diag}(0, 1)$ satisfies $XPX = 1 - P$. The reversal fails to preserve the known vacuum baseline, which the preceding protocol requires. Reading the baseline and implementing the excitation are real permissions, not consequences of algebra membership.

Corollary 4.2 (A finite-record orientation certificate)**.** *Assume independent fresh samples and additive noise independent of the energy outcome, with variance σ_ν^2 common to the two uses. From n vacuum reads and n prepared reads, the sign of their sample-mean difference has error probability at most*

$$\min\left\{1, \frac{g^2 \operatorname{Var}_\theta(P) + 2\sigma_\nu^2}{n g^2 E_\theta^2}\right\}. \tag{4}$$

This protocol uses $2n$ reads and n fresh preparations, whose prepared subsystem energy is nE_θ. Switching, detection, reference preparation and reset costs are separate. A uniform prescribed-confidence budget needs bounds on the gain, noise and energy variance.

Proof. The mean difference has mean $\epsilon g E_\theta$ and variance $(g^2 \operatorname{Var}_\theta(P) + 2\sigma_\nu^2)/n$. A wrong sign requires a deviation at least gE_θ; Chebyshev's inequality gives the bound. □

This reference breaks energy-orientation ambiguity but not every internal frame symmetry. A group commuting with P can still reverse a charge or current. Thus a common positive pulse calibrates one part of the world without supplying a universal external standard.

4.1 Public correctness without equal private histories

Calibrating a public target need not identify the whole frame. Let F be a finite set of possible common frame errors. A legal reference block comprises specified preparations, their source changes, instrument uses and recorded coins. It has a finite record alphabet and known law $p_f(z)$ under $f \in F$. These laws concern actual reference operations that are held fixed when comparing frames. They are not inferred from the agreement of the readers. Let $t(f)$ be a finite-valued public target.

Proposition 4.3 (Target-specific reference criterion)**.** *Repeated independent fresh reference blocks permit uniformly consistent estimation of $t(f)$ if and only if*

$$p_f = p_{f'} \implies t(f) = t(f').$$

When the criterion holds, put

$$r = \max_{t(f)\neq t(f')} \sum_z \sqrt{p_f(z)p_{f'}(z)} < 1.$$

For n blocks, a maximum-likelihood estimate $\hat{f}$, followed by the report $t(\hat{f})$, has target error at most

$$\min\{1, (|F|-1)r^n\}.$$

If no two hypotheses have different targets the error is zero. A single decoded public report transmitted without corruption makes every reader correct on the same event, irrespective of their private histories.

Proof. If two identical laws have different targets, their complete n-block laws are identical for every n. The probabilities of reporting their two different correct values sum to at most one, so one error is at least $1/2$. Conversely, under true f, selection of a wrong-target candidate f' requires its likelihood to be at least that of f. Markov's inequality applied to the square root likelihood ratio bounds this event by $\left(\sum_z \sqrt{p_f(z)p_{f'}(z)}\right)^n$. Zero-probability records are treated on the support of p_f. The union bound over wrong-target candidates gives the result. Distinct probability laws have affinity below one by Cauchy–Schwarz. Finiteness makes the maximum strictly below one. □

The testing inequality is standard; the structural consequence here is that public calibration is determined by the target's indistinguishability classes. A full frame reconstruction would require separation of every distinct frame. When frame symmetries are represented by a group, unremoved transformations may still act on private currents while fixing the public energy target. Identifying a reference law is also weaker than implementing the inverse frame transformation: reporting and control are separate permissions.

There is a concrete finite physical witness. Let a reference qubit have $H_R = \Delta|1\rangle\langle 1|, \Delta > 0$. Legal preparations $|0\rangle, |1\rangle$ and an energy pointer supply the known baseline and excited input. Suppose the pointer's two values are reversed by an unknown common sign $f = \pm 1$ and its binary read flips with probability $e < 1/2$. Reading the excited input gives Bernoulli laws with affinity $2\sqrt{e(1-e)}$, so n reads identify the common sign with error at most $[2\sqrt{e(1-e)}]^n$. An ideal controlled pointer interaction realizes the unflipped measurement; the read noise is a supplied physical channel. The source preparation uses $n\Delta$ in excited-subsystem energy. The pointer preparation, coupling, elapsed time, storage and disposal of spent systems have their own costs. Energy supplied is not net heat.

Let D contain all old private memory registers and their old references, in an arbitrary state ρ_D, with different histories at different readers. Prepare the reference blocks independently of D and apply the calibration instrument only to those blocks. The final state is exactly $\rho_D \otimes \rho_{\text{calibration}}$; all old correlations in D remain intact, even conditional on the calibration outcome. More generally, for initially correlated active and old registers, a trace-preserving map acting only on the active register preserves the old marginal nonselectively, whereas a selected new record can reweight the old history. This is the distinction needed by the retained-history return in Foundation III [2]. The tensor architecture is an existence witness, rather than a factorization theorem for Type III algebras.

The reference criterion fails if the excitation and its label are conjugated along with the faulty instruments: Proposition 3.1 then applies again. It also fails for a destroyed pointer with $e = 1/2$. A shared algebra supplies possible operators; it supplies neither an independently trusted reference nor the preparation and communication operations used here. For a reference block costing resource vector $\mathbf{c}$, n blocks cost $n\mathbf{c}$, in addition to the retained private storage and public communication. Those requirements make the claimed public correctness falsifiable.

4.2 From a shared preparation to a geometric test

Suppose an internal compact group G fixes the bounded functions of the required stress-tensor smearings. Every selected algebra of the form M^{K_j}, $K_j \leq G$, then contains those readings, as does $B_{\cap}$. This is a direct condition for a common stress sector; it is not a theorem about every non-scalar hull. Readers of separate decoupled fields can have disjoint stress sectors.

Proposition 4.4 (A shared test of the sourced geometric response)**.** *In Consciousness II's conformal Minkowski setting, suppose a shared positive-energy pulse is implemented, either the bounded pulse of Proposition 4.1 or an explicit scalar preparation of that paper. Its translation generator is the physical positive Hamiltonian, and its vacuum-subtracted source has an integrable energy density with integral $E_\theta > 0$. Suppose the source and response belong to the smooth tempered Radon Fourier class, and the full centre, open-frame and modular-consistency hypotheses of that paper hold. If every reader has the required stress and boundary observables, uses the same actual state and normalization, and applies the same geometric readout law, the pulse has spectral weight away from zero and identifies their common canonical response up to the law's normalization. They obtain the same matter-sourced linearized Einstein tensor. A unique metric requires additional common initial or boundary data and gauge choices.*

Proof. If a Radon Fourier measure were supported only at zero, the source would be constant. Integrability of its energy density on the infinite spatial slice would force that density to vanish, contrary to $E_\theta > 0$. The spectral identification theorem and common sourced response of Consciousness II now apply [6]. A homogeneous gravitational wave in dimension at least four can have zero linearized Einstein tensor and nonzero linearized curvature, so the source alone cannot fix the metric. □

Finite energy does not itself imply the Radon regularity used here, and the finite-amplitude pulse need not have a nonzero first-order stress tangent at zero amplitude. Nor does the chiral SU(2) model establish all the higher-dimensional hypotheses. The bridge states exactly which source and readout data must be common. A unitary used for calibration physically changes the state; merely choosing a different label or algebra does not. The implemented source and its costs are the link to HAFF-D [10]; bounded source feedback and persistent access are treated in Foundation III [2], rather than inferred from agreement.

Example 4.5 (Shared hulls and a physical time generator)**.** Take two independent massless conformal complex scalar fields in $3+1$ dimensions, on one vacuum Fock space, and let $M = \mathcal{A}(V_+) = \mathcal{A}_1(V_+)\bar{\otimes}\mathcal{A}_2(V_+)$. The future-cone vacuum algebra is standard and its modular group is dilation [8, Theorem 2.3(ii)]; the free massless-field modular structure was studied in [7]. Write $\mathcal{N}_j(O) = \mathcal{A}_j(O)^{U(1)}$ and give the two readers the selected nets $B_1(O) = \mathcal{N}_1(O)\bar{\otimes}\mathcal{A}_2(O)$, $B_2(O) = \mathcal{A}_1(O)\bar{\otimes}\mathcal{N}_2(O)$. For $O = V_+$ their shared algebra is $\mathcal{N}_1(V_+)\bar{\otimes}\mathcal{N}_2(V_+)$. The two vacuum-preserving Haar expectations commute, proving this intersection directly. Specify the noise family to be $\mathcal{E} = \{Q_1 \otimes \mathrm{id}, \mathrm{id} \otimes Q_2\}$, where Q_j is Haar projection on field j. Both maps preserve each $B_i(V_+)$: one acts identically on its fixed factor, and the other projects its

unrestricted factor into a subalgebra. The bounded self-adjoint parts of these algebras can serve as ideal seed families. Their invariance under this noise family, the full modular group and forward time translations makes their Γ_3 hulls exactly the displayed cone algebras.

Here $U(t) = e^{it(H_1+H_2)}$ is the physical time translation, $V_+ + te_0 \subset V_+$ for $t \geq 0$, and its only zero-energy vector is the vacuum. The latter follows from strictly positive one-particle energy $|p|$ almost everywhere. Gauge-fixed algebras are represented on their own vacuum-cyclic subspaces, not asserted cyclic on the ambient Fock space. Haar projection commutes with H; its restriction to each such subspace retains positivity, the same physical translation and the unique vacuum.

A compact neutral quadratic preparation in field 1, as specified in Consciousness II, lies in both readers' algebras, has strictly positive finite H energy and a smooth nonzero-frequency Radon source [6, Proposition on a neutral preparation]. Their selected algebras at every diamond contain the bounded stress and collar readings. Haar expectations preserve the common vacuum; permitted preparations, calibrated collar limits and common MC_η normalization remain inputs. The constraint-compatible linear response of that paper then instantiates Proposition 4.4. The reference hull construction uses the cone algebra; the geometric test uses the same gauge-fixed nets on all $D(B)$. This supplies the required diamond readings, not a claim that each diamond algebra is the cone's Γ_3 hull or that every ball operator belongs to $\mathcal{A}(V_+)$. The generic bounded pulse also exists here, but its source regularity is not inferred solely from its energy moments.

5 Agreement, calibration and fault transfer

A finite two-observer experiment tests the distinct roles of agreement and paid references. A bank of 4096 bits evolves under a hidden event $c_k = \pm 1$, flipping with probability $.01$ per epoch. Each bit is retained with probability $.65$, replaced by a fair coin with probability $.25$, or drawn with mean $.6c_k$ with probability $.10$. Each drifting binary instrument has mean response $g_i z + b_i$ and a 512-read budget per epoch. Known ± 1 references displace world reads; agreement exchanges current estimates and changes next-epoch bias calibration. Only the scorer accesses c_k. Appendix F and the supplementary protocol give the legal clipping, seeds, updates and full outputs.

For binary reports put $A = P(\hat{c}_1 = \hat{c}_2)$, $C_i = P(\hat{c}_i = c)$ and $E = P(\hat{c}_1 = \hat{c}_2 \neq c)$. Eventwise counting, without independence, gives

$$C_1 + C_2 = 1 + A - 2E, \qquad P(\text{correct} \mid \text{agree}) = 1 - E/A \quad (A > 0).$$

Scheme	A	C_1	E	$1 - E/A$
Initial references only	.647	.753	.067	.896
Agreement only	.837	.789	.125	.851
64 references per sign, no coupling	.860	.908	.020	.977

Table 1: Selected comparisons under the same 512-read budget: 800 trajectories of 3000 epochs, excluding the first 200; correctness standard errors below $.002$. The supplementary protocol retains all tested budgets and fault controls.

Agreement rises while the reliability of an agreeing pair falls. Among tested reference counts $4, 16, 32, 64, 128$ per sign, 64 gives the highest correctness; this is not a global optimum. The full common-bias scan and unequal-reference controls are retained with the protocol.

A 100-epoch fault starts at a random epoch and adds a clipped $.30$ bias or sets one instrument's gain to zero. With 16 references per sign, unweighted agreement transfers zero-gain correctness losses $.121, .123$ to the undamaged observer. Weighting by the partner's reported gain reduces them to $.014, .016$; 64 references reduce them to $.003, .004$. A matched no-coupling observer has zero transfer. A separately written implementation gives equal-weight harm $.110$ and weighted harm $.0036$. Isolation at a scorer-known fault time is a diagnostic, not a learned deployment policy. References remove bias; a zero-gain reading requires an additional gain-sensitive decision.

Communication costs 22 bits per epoch for agreement and 44 for weighted agreement. All observers start with 4096 reference reads. The world is uncontrolled, the rules are designed, and only two readers and the stated faults are tested. These data support calibration and fault transfer in this protocol, not a consensus-to-geometric law.

6 What makes a shared world testable

The intersection fixes common observables, restrictions of one actual state fix common state values, and instruments fix joint record laws. Correct public reports additionally require a reference that separates the public target's indistinguishability classes. This can leave private frames and old histories different. The algebraic intersection and the public-reference criterion answer complementary questions.

The common-pulse construction joins selected access to an actual source test. Its physical translation, preparation, source regularity, stress and collar readouts, and geometric consistency must coexist. Example 4.5 supplies that algebra/source combination in a conformal matter model. It retains the supplied coupling and gravitational data; the null generator in the chiral example is a different physical quantity. A sourced Einstein tensor does not determine free radiation, a metric gauge, or experience.

We propose that the operational extent of a shared world is selected partly by the references its participants can jointly sustain. Source changes alter reference availability and price; changed reference risk alters the legal common task family; the resulting actions change the next source. Some common capabilities can fail before private stored histories disappear. The finite reference witness establishes calibration and old-history preservation. Foundation III supplies a finite semiclassical mean-action return. One maintained reference/source/readout device remains a further requirement, rather than a consequence of a strict hull inclusion.

Failure conditions are concrete. A shared hull can be scalar, a reference can be conjugated with the faulty instruments, and a zero-gain instrument can erase the information needed to correct it. An energy reference cannot remove internal errors commuting with energy. Paid controls, output entropy and reusable memory determine the resource price; algebra size alone does not. Foundation III, T-DOME and Brain Anchor address the distinct return, calibration and formation mechanisms. Independent-source state equality and a common projected memory kernel retain their explicit appendix conditions.

A Independent sources: compatibility and its limits

The same-process construction uses restrictions of one actual state. For independent sources let M have a cyclic separating reference vector Ω, reference state ω_0, faithful normal states ϖ_i, and normal UCP semigroups $\mathcal{E}_t^{(i)}$. Suppose each source supplies an individual algebra $A_i \subseteq M$,

state $\omega_i = \varpi_i|_{A_i}$, and realized foreground space $V_i \subseteq M_{\text{sa}}$. The independently chosen ϖ_i need not agree on $A_1 \cap A_2$.

For a candidate $C \subseteq A_1 \cap A_2$, the independent-source conditions are:

JS-A. C is invariant under both semigroups and under the full reference modular group. The semigroups need not commute.

JS-B. A faithful normal state ψ on C is stationary under both restricted semigroups and satisfies $\omega_1|_C = \omega_2|_C = \psi$. Equality and stationarity are substantive inputs.

JS-C. When a joint projected record process is needed, its normal UCP idempotents P_1, P_2 commute; their product has von Neumann-algebra range containing C, fixes C, and the projected dynamics has one specified common predual memory kernel. This last clause is a dynamical condition, rather than a consequence of commuting projections [14, 15].

Proposition A.1 (Commuting expectations)**.** *If $E_i : M \to M_i$ are commuting normal conditional expectations, E_1E_2 is a normal conditional expectation onto $M_1 \cap M_2$. It supplies the algebraic clauses of JS-C when $C \subseteq M_1 \cap M_2$.*

Proof. The product is normal UCP and idempotent. Its range lies in both M_i, and every element of their intersection is fixed by it. The norm-one projection onto that subalgebra is a conditional expectation [18, 19]. No kernel identity follows. □

B An independent-source hull theorem

The state equality is an input in JS-B. The following conclusion constructs the minimal compatible hull inside a supplied candidate C; it does not derive agreement of independent states.

For a nonzero finite-dimensional $V_1 \cap V_2$, choose a basis $O_1, \ldots, O_k$ in C, and set $S = W^*(1, O_1, \ldots, O_k)$. The seed-space dimension is $k = \dim(V_1 \cap V_2) \leq \min(\dim V_1, \dim V_2)$. This bounds neither the algebraic dimension of S nor the size of its hull. A single observable $O = \operatorname{diag}(0, 1, 2)$ in M_3 spans a one-dimensional seed space but generates the three-dimensional diagonal algebra. One observable with infinite spectrum can likewise generate an infinite-dimensional algebra. The finite seed bound also says nothing about nontriviality. Let

$$\Phi = \bigcap \{N \subseteq M : N \text{ is a von Neumann algebra}, S \subseteq N,\ \sigma_s^{\omega_0}(N) = N,\ \mathcal{E}_t^{(i)}(N) \subseteq N\}.$$

Theorem B.1 (Independent-source compatibility)**.** *If C satisfies JS-A and JS-B and contains S, then*

$$S \subseteq \Phi \subseteq C \subseteq A_1 \cap A_2, \qquad \omega_1|_\Phi = \omega_2|_\Phi = \psi|_\Phi.$$

The state is faithful, and Φ inherits the reference modular flow on $\overline{\Phi\Omega}$. If a proper half-sided modular inclusion $N \subset \Phi$ is supplied there, with Ω cyclic and separating for N, Consciousness I's Cut correspondence applies. Standardness gives the net correspondence. A joint projected record process additionally requires JS-C.

Proof. JS-A makes C a competitor in the defining intersection, so the class is nonempty and $\Phi \subseteq C$. Intersections preserve the algebra, the seed inclusion and all specified invariances, proving minimality. Restriction preserves faithfulness, and JS-B gives equality of the states. Takesaki's theorem identifies the restricted reference modular group [17, 16]. The Cut and the projected kernel are the separate supplied inputs just stated. □

The Cut sits inside Φ in its reference representation; the hull is not asserted to be a Cut of the ambient algebra. The reference state $\omega_0|_\Phi$ need not equal $\psi|_\Phi$. The theorem does not turn agreement into an output when JS-B already assumes it.

Proposition B.2 (Finite compatibility witness)**.** *JS-A/B/C can coexist with disagreement elsewhere in the overlap.*

Proof. In $M_2(S) \otimes M_2(P_1) \otimes M_2(P_2)$ with tracial reference, let $D_X(a) = (a + Z_X a Z_X)/2$, $P_1 = D_S D_{P_1}$, $P_2 = D_S D_{P_2}$, and $\mathcal{E}_t^{(i)} = e^{-\lambda_i t}\mathrm{id} + (1 - e^{-\lambda_i t})P_i$, $\lambda_i > 0$. Choose $C = W^*(Z_S)$, $A_1 = A_2 = W^*(Z_S, Z_{P_1}, Z_{P_2})$, and

$$\varpi_1 = \rho(1/3)\otimes\rho(1/4)\otimes\rho(1/5), \quad \varpi_2 = \rho(1/3)\otimes\rho(-1/4)\otimes\rho(-1/5), \qquad \rho(a) = (1+aZ)/2.$$

All states are faithful. Both semigroups fix C pointwise, the reference flow is trivial, and the states agree on C, verifying JS-A/B. The projections commute and their product has diagonal range. On the fixed sector the common kernel is zero, verifying JS-C. The foregrounds $\mathrm{span}(Z_S, Z_{P_1})$ and $\mathrm{span}(Z_S, Z_{P_2})$ overlap in Z_S, while the state discrepancy on Z_{P_1} is $1/2$. This finite witness does not realize a Type III Cut. □

C Private content and common modular flow

A private complement $P_i^{\mathrm{priv}} \subseteq A_i$, when one is used, means $A_i = W^*(C \cup P_i^{\mathrm{priv}})$; no canonical tensor or free factorization is implied. A state-preserving expectation $E_i : A_i \to C$ is a separate regime condition.

Localization of discrepancy. For $\delta = \omega_1|_{A_1 \cap A_2} - \omega_2|_{A_1 \cap A_2}$, JS-B directly gives $C \subseteq \ker \delta$, so $\delta(x) \neq 0$ implies $x \notin C$.

The kernel need not be an algebra. In M_3, take states with densities $\rho = \mathrm{diag}(.5, .3, .2)$, $\sigma = \mathrm{diag}(.3, .4, .3)$, and $X = \mathrm{diag}(0, 1, -1)$. Then $(\rho - \sigma)(X) = 0$ but $(\rho - \sigma)(X^2) = -.2$. Nor do matching marginals specify a joint-record agreement rate.

Corollary C.1 (Common modular flow)**.** *If faithful normal $E_i : A_i \to C$ preserve ω_i, then*

$$\sigma_t^{\omega_1}|_C = \sigma_t^{\psi} = \sigma_t^{\omega_2}|_C.$$

It equals the reference flow used for a Cut if $\psi = \omega_0|_C$; equality of flows need not imply equality of states. For example, on $C = \mathbb{C}^2$, the faithful states with weights $(1/3, 2/3)$ and $(2/3, 1/3)$ differ but both modular flows are trivial. No factor condition is assumed here.

Proof. Apply Takesaki's theorem to each expectation and its preserved state. □

Example C.2 (A Type III shared sector with different source states)**.** Let $B = \mathrm{Vir}_1(0, \infty)$, with vacuum state ω_B; let $K, P_1, P_2 = M_2$ with normalized trace τ. Put $M = B\bar{\otimes}K\bar{\otimes}P_1\bar{\otimes}P_2$, $C = B \otimes 1$, and

$$A_1 = B\bar{\otimes}K\bar{\otimes}P_1 \otimes 1, \qquad A_2 = B\bar{\otimes}K \otimes 1\bar{\otimes}P_2.$$

With faithful η_i and $\rho_\pm = (1 \pm aZ)/2$, $0 < a < 1$, choose $\varpi_1 = \omega_B \otimes \rho_+ \otimes \eta_1 \otimes \tau$ and $\varpi_2 = \omega_B \otimes \rho_- \otimes \tau \otimes \eta_2$. They agree on C, differ by $2a$ on Z_K, and their partial state expectations onto C preserve the respective states. The tracial finite-factor reference makes the reference flow on C the vacuum dilation flow. Strong additivity of the $c = 1$ Virasoro net makes $\mathrm{Vir}_1(b, \infty) \subset B$, $b > 0$, a standard Cut on its own vacuum space [1, 5]. This witnesses the algebra, state and Cut conditions together; source semigroups and formation of foregrounds have not been specified.

D Finite families of independent sources

Theorem B.1 extends to any finite number of sources by requiring one candidate $C \subseteq \bigcap_i A_i$, one shared seed algebra in C, JS-A/B for every source, and one Cut inside the resulting hull. Its proof is the same intersection and restriction argument. If joint projected records are required, pairwise commuting normal UCP idempotents give an idempotent finite product; algebraic range and one common kernel still need to be supplied. Pairwise choices of overlap sectors and kernels do not establish one coherent joint process. The same-source theorem in the main text already gives the largest common admissible algebra without these independent-state conditions.

E Imported interfaces

Foundation I supplies the hull construction, stationary fixed algebras, restricted modular groups and conditional Cut inheritance [1]. Its projected stationary fixed-sector memory kernel vanishes; a general projected kernel remains extra dynamical data. Consciousness I supplies the half-sided modular inclusion/net correspondence, and Consciousness II supplies the bounded pulse and conditional higher-dimensional identification used in Section 4 [5, 6]. T-DOME's record value, history-selected frame and priced calibration are different functional interfaces [11, 12, 13]. They do not derive JS-B or preparation permission from a fixed algebra. The no-foreground-overlap example and the independent-state witnesses answer different questions and cannot replace one another.

F Simulation protocol for Section 5

The complete protocol is retained in the supplementary source record `FII_protocol.tex`, with the measured tables. Its implementation is `d2_v3.py`; the selected independent fault checks use `d2_checks.py`. The source map identifies the fixed scripts and outputs. All reads use legal clipped Bernoulli instruments; fault and no-fault runs share the world and per-reader random streams. Calibration uses current reference reads before the current report, while agreement updates the following epoch. Communication is quantized to eleven bits per scalar. The scorer's fault-time isolation is kept separate from deployable detection.

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Foundation III: The Perfuming Loop Held Content, Learned Frames and Physical Return

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Abstract

An observer's return to its source can change the accessible algebra, the evolving state, or both. We separate these effects and construct normal completely positive feedback lifts on a supplied Type III algebra. For a finite family of compact noise groups, repeated partial Haar dephasings converge strong* to the expectation onto their common fixed algebra whenever each group receives divergent cumulative exposure. The order, gaps and individual exposure strengths need not be fixed. Thus simultaneous sharing and long-time persistence select the same algebra in this model. A countable-family example shows why recurrence alone cannot replace the finite-family condition. Summable channel perturbations preserve convergence to the selected algebra, although they can change the state of the retained content.

For GNS-symmetric source and feedback semigroups, the fixed observables of a finite product are their common fixed observables, even without commutation. A uniformly mixing source has only scalar fixed observables; it cannot at the same time retain a nontrivial fixed algebra. Moving its stationary state requires a different return mechanism. We realize that mechanism in a controlled bank of diagonal qubits and bound its risk under persistent estimation error. The compact actions and physical controls are inputs, not a derivation of their origin. Physical returns compose on the complete retained history, with nonlinear continuity bounds where required. Q-RAIF C supplies the imported quantum history-control and receiver-renewal bounds: fixed controllers are nonexpansive across decision thresholds, and finite renewal preserves later bounded tasks and smooth tested material-field sources under the same Hamiltonian and retained history. These finite results do not imply infinite-time stability or renewal of the entire apparatus. In a finite semiclassical scalar/JT mean-action realization, an acquired record installs a protective encoding for held quantum content and changes the apparatus's own source. With the initial mass and geometry fixed, the resulting geometric response enters an X-basis acquisition of the old probe on a spent cell, after the first record has been acquired. Both readings retain the old content, references and physical cursor. Learning outperforms both fixed encodings under the declared unknown-sign prior. The stated example's sign-prior-averaged geometric signal is about -5×10^{-9}, an existence-level return rather than a practical gravitational performance gain. The hardware and diagnostic relation are supplied; the installed frame, rather than a new microscopic law, is formed.

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1 Introduction

The source does not merely produce records. The observer uses those records to act on the process that will produce its next records. Foundation III studies this return arrow. Foundation I [1] constructs selected algebras from supplied source dynamics and seeds; Foundation II [2] asks what different readers share. Here the question is what persists when the operations themselves change, and when a return modifies the world rather than only its description.

These are distinct questions. Adding a symmetric noise channel can remove fixed observables without moving the reference state. A reset command can instead move a stationary state while every diagonal observable remains in the same pointer algebra. Neither operation by itself establishes that a new observer or geometry has formed. Section 4 gives precise criteria for both effects. In particular, it connects sharing across readers to persistence across time: the common fixed algebra of a finite family of noise groups is also the strong* limit selected by any weighted-fair sequence of partial dephasings. This connects the shared-sector problem of Foundation II to an iterated source process without assuming cyclic visits, bounded gaps, a positive minimum exposure, or a uniform spectral gap.

Finite-dimensional projection convergence has a substantial history, including the paracontraction theorem of Elsner, Koltracht and Neumann [4]. The projection lemma below supplies the cumulative-weight version used here; compact-group representation theory then lifts it to normal noise maps on a von Neumann algebra. Our use of these tools concerns accessible content, not a new general principle of iterative numerical analysis. The countable-family construction identifies a real boundary of that use. Small errors are treated separately: summable implementation perturbations still converge into the common algebra, but need not preserve its original state.

The feedback signal is supplied by a specified calibration protocol. T-DOME III [8] provides reference and probe records, posterior-dependent maintenance decisions and bounded commands under drift. Its self-model parameter is written ϑ there and σ here. Residuals generated solely by the self-model need not identify their own calibration error; the reference permissions are part of the physical interface. A signal becomes a source operation only after a control channel has been specified. Sections 2 and 3 construct exact normal completely positive lifts and state their continuity and commutation conditions. A compact-group construction realizes the lift on the Type III algebra itself, while its available couplings and calibration-to-rate rule remain supplied controls.

Section 5 treats moving-source feedback in a bank of diagonal qubits. Every bounded command defines a legal channel; no autonomous-semigroup claim is needed for the adaptive sequence. Persistent estimation error gives an attracting risk ball. A smaller upper certificate need not mean a smaller actual loss, so the analysis separates the certificate from simulation and an exact open-loop comparison. This finite model implements part of the record–self-model–action loop; it is not an identification of the bank with a Type III observer or with experiential consciousness.

The algebraic results keep the group action, reference state, seeds and control permissions explicit. A nontrivial selected algebra can carry an internal Cut under the translation and standardness assumptions of Foundation I and Consciousness I [5]; persistence alone does not supply those assumptions. Source preparation and geometric identification belong to Consciousness II [6]. Physical actuation and its stress belong to the Q-RAIF/HAFF interface. Section 7 gives the composition rule for a single process that chooses access, produces a source and uses its response. Its finite example learns an installed encoding from actual records while keeping the old quantum content and preparation references. A second result uses the complete quantum

interface of Q-RAIF C [3]: a receiver renewed there can continue under the same interacting Hamiltonian with controlled next-window source and bounded-task error. The clock and old correlations remain in the comparison. Matching this quantum realization to the scalar/JT return is a further physical requirement.

The final sections state the formation hypothesis and its physical interfaces. Appendix A supplies the finite arrival estimate used in the joint return; Appendix B specifies the numerical protocol.

2 Feedback Lift and Regularity Conditions

We first specify the signal and the source operation it may control.

Definition 2.1 (Admissible calibration signal)**.** Fix a source datum S_{source} as in Foundation I, a bounded readable seed S, and a closure family Γ containing the source's modular group. The accessible object is

$$\mathcal{O}_{\text{acc}} = (\mathcal{A}_{\text{acc}}, \omega|_{\mathcal{A}_{\text{acc}}}, \Gamma, \mathsf{P}_{\text{phys}}), \qquad \mathcal{A}_{\text{acc}} = \text{Hull}_{\Gamma}(S).$$

Here P_{phys} specifies the actual probe instruments, preparations and resource budget. The ideal hull does not itself make every element measurable. Calibration records must be obtainable under these permissions. A stable hull additionally requires the stationary source and stable-seed assumptions of Foundation I; Cut inheritance requires its stated translation and standardness inputs. An *admissible calibration signal* is a vector

$$\delta\sigma \in T_{\sigma}\Sigma$$

in the local calibration parameter space of T-DOME III such that:

1. $\delta\sigma$ is obtained from the natural-gradient calibration step imported from T-DOME III, with a metric made positive definite by probe or reference records (read through the self-model alone, the metric vanishes at calibration);

2. $\sigma + \delta\sigma$ remains inside the Lyapunov neighborhood on which the T-DOME III tracking estimate is valid;

3. there exists a bound $\|\delta\sigma\| \le R_{\text{cal}}$ for some finite calibration radius $R_{\text{cal}} > 0$.

Remark 2.2 (Local signals and repeated returns). Definition 2.1 concerns one local signal, not convergence of calibration. Repeated partial dephasings have their own exact limit in Section 4; drift control has an attracting ball in Section 5; finite physical histories compose in Section 7. None follows by iterating the local signal definition alone.

For a source semigroup $\mathcal{E}_t$, the feedback package is a normal UCP semigroup $\mathscr{F}_r$ with $\mathscr{F}_r^{[0]} = \text{id}$, an every-observable weak derivative at zero, and strict commutation with $\mathcal{E}_t$. It must separately retain the named hull, stationarity, translation and standardness inputs. Appendix C states FB-A–FB-C and proves that this derivative gives a bounded normal generator. Strict commutation permits nontrivial decay reweighting, but does not mix the pre-existing joint modes of its finite example. The following lift acts directly on the source algebra.

Proposition 2.3 (A compact-group lift on the source algebra)**.** *Let a compact group G act continuously by normal automorphisms on $\mathcal{A}_U$, preserving its faithful normal reference state. For a closed subgroup $H \subseteq G$, write Q_G, Q_H for the Haar expectations and put*

$$\mathcal{E}_t = e^{-\gamma t}\mathrm{id} + (1 - e^{-\gamma t})Q_G, \qquad \mathscr{F}_r^{[\delta\sigma]} = e^{-\kappa q(\delta\sigma)r}\mathrm{id} + (1 - e^{-\kappa q(\delta\sigma)r})Q_H, \quad q(\delta\sigma) = \|\delta\sigma\|^2,$$

where $\gamma, \kappa > 0$. These are normal UCP semigroups on the source algebra, including when it is Type III. They satisfy FB-A, FB-B and the commutation clause of FB-C. Inputs supported on $\mathcal{A}_U^G$ are fixed by both semigroups. If $Q_H \neq \mathrm{id}$, $t, \eta > 0$ and $\delta\sigma \neq 0$, the composed source map differs from $\mathcal{E}_t$.

Proof. Each Haar expectation is a normal UCP idempotent, so $e^{a(Q-\mathrm{id})} = e^{-a}\mathrm{id} + (1 - e^{-a})Q$. This proves the semigroup and continuity statements and gives the bounded feedback generator $\kappa q(\delta\sigma)(Q_H - \mathrm{id})$. Haar invariance and $H \subseteq G$ give $Q_G Q_H = Q_H Q_G = Q_G$. Thus the two semigroups commute. For x with $Q_H x \neq x$, their difference is

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}\mathcal{E}_t(x) - \mathcal{E}_t(x) = e^{-\gamma t}(1 - e^{-\kappa q(\delta\sigma)\eta t})(Q_H x - x) \neq 0.$$

Elements of $\mathcal{A}_U^G$ are fixed at every step. □

This realizes a normal source-level lift without a finite-dimensional amplification. The group coupling, its availability to the controller and the response law q are still physical inputs. Preserving a full Foundation I chain requires its named inputs to lie in an invariant sector; a valid CP map alone does not ensure that.

The exact modified source is $\mathcal{E}_t^{[\eta]} = \mathscr{F}_{\eta t}^{[\delta\sigma]}\mathcal{E}_t$. At fixed t, its tangent is $\Gamma(\delta\sigma, \mathcal{E}_t) = t\mathfrak{G}_{\delta\sigma}\mathcal{E}_t$; a uniform perturbative use requires $\eta T \ll 1$ and a valid comparison window. Appendix C supplies the definitions and expansion proof. The next theorem concerns the exact CP operation.

3 Source-Datum Preservation

Theorem 3.1 (Source-datum preservation)**.** *Assume a source datum S_{source} and the feedback package FB-A–FB-C. For each admitted $\eta \geq 0$, $\mathcal{E}_t^{[\eta]} = \mathscr{F}_{\eta t}^{[\delta\sigma]}\mathcal{E}_t$ is a pointwise σ-weakly continuous semigroup of normal UCP maps. The unchanged Ω remains cyclic and separating for $\mathcal{A}_U$.*

Proof. Commutation in FB-C and the two semigroup laws give $\mathcal{E}_t^{[\eta]}\mathcal{E}_s^{[\eta]} = \mathscr{F}_{\eta(t+s)}\mathcal{E}_{t+s}$. Normality, complete positivity and unitality survive composition. Proposition C.2 gives norm continuity of $\mathscr{F}_r$. For fixed A and a normal functional φ, insert $\varphi(\mathscr{F}_{\eta t_0}\mathcal{E}_t A)$ between the expressions at t and t_0. The first difference is at most $\|\varphi\|\|\mathscr{F}_{\eta t} - \mathscr{F}_{\eta t_0}\|\|A\|$; the second tends to zero by σ-weak continuity of $\mathcal{E}_t$ and normality of $\varphi\mathscr{F}_{\eta t_0}$. The ambient standard-form triple has not changed. □

The source-datum format survives, not automatically its selected projection, foreground frame or manifest algebra. Rerunning a Foundation I chain requires the structural permissions in Appendix E. The first-order map $\Gamma(\delta\sigma, \mathcal{E}_t)$ in Definition C.8 is therefore the tangent of an exact CP perturbation, not a truncated map whose positivity is assumed.

4 Persistent Content under Changing Source Operations

4.1 Fixed observables and stationary states

Use $\mathrm{Fix}(\mathcal{E})$ for observables fixed at every time by a semigroup. Symmetry below refers to the GNS inner product $\langle x\Omega, y\Omega\rangle = \omega(x^*y)$, not to a chosen matrix representation.

Proposition 4.1 (Fixed content of a product)**.** *Let $\mathcal{E}^{(1)}, \dots, \mathcal{E}^{(m)}$ be normal UCP semigroups preserving one faithful normal state ω. Suppose each is GNS-symmetric:*

$$\omega(y^*\mathcal{E}_t^{(i)}x) = \omega((\mathcal{E}_t^{(i)}y)^*x).$$

For positive times t_i, the fixed observables of $\mathcal{E}_{t_m}^{(m)} \cdots \mathcal{E}_{t_1}^{(1)}$ are exactly $\bigcap_i \mathrm{Fix}(\mathcal{E}^{(i)})$. The maps need not commute.

If a source semigroup converges uniformly over all initial normal states to one state τ, its fixed observables are scalar. A CPTP map commuting with that source preserves τ.

Proof. Kadison–Schwarz and invariance implement each semigroup on GNS space by self-adjoint contractions $V_t^{(i)}$. Pointwise σ-weak continuity, together with contractivity, gives strong continuity on the dense set $\mathcal{A}_U\Omega$ and hence on GNS space. Since $V_t^{(i)} = (V_{t/2}^{(i)})^2$, they are positive. If their product fixes v, the chain of norm inequalities starts and ends at $\|v\|$, so each is an equality. A positive contraction preserves a vector's norm only on its eigenvalue-one space. Starting with the first factor therefore gives $V_{t_1}^{(1)}v = v$, then the same conclusion for each remaining factor. The semigroup spectral representation identifies this space with the vectors fixed at all times. Apply the result to $x\Omega$ and use the separating vector Ω. The reverse inclusion follows directly.

If $\mathcal{E}_t x = x$, uniform mixing gives $\rho(x) = \tau(x)$ for every normal state ρ, whence $x = \tau(x)\mathbf{1}$. For commuting feedback F, $\mathcal{E}_{t*}(F\tau) = F(\mathcal{E}_{t*}\tau) = F\tau$; uniqueness of the invariant state gives $F\tau = \tau$. □

Commutation remains useful for forming an autonomous composed semigroup, but is unnecessary for this fixed-content identity. Without GNS symmetry, opposite unitary rotations can compose to the identity and restore every observable although each individual commutant is smaller. Likewise, a unique mixing state and a nontrivial fixed algebra cannot be combined into one mechanism.

The reset bank illustrates another distinction. Its Heisenberg reset is $x \mapsto ax + (1-a)\,\mathrm{tr}(\tau_v x)\mathbf{1}$, which preserves $W^*(S)$ for every pointer seed S, even for a coherent target. With a fixed faithful diagonal reference state, this pointer algebra remains modular invariant as the target changes. If the reference state itself is replaced by a coherent faithful target, modular closure can enlarge the algebra. For a finite maximal diagonal algebra $\mathcal{D}$, $\rho^{is}\mathcal{D}\rho^{-is} = \mathcal{D}$ for all s holds exactly when the faithful density matrix ρ is diagonal: differentiation gives $[\log\rho, d] \in \mathcal{D}$, whose off-diagonal entries vanish for every diagonal d. Thus moving a state, changing a reference and changing operational access must be tracked separately.

4.2 Finite noise families with cumulative exposure

Lemma 4.2 (Weighted-fair projections)**.** *Let $P_1, \dots, P_m$ be orthogonal projections on a finite-dimensional Hilbert space and let $R_n = (1-\theta_n)I + \theta_n P_{i_n}$, $0 < \theta_n \le 1$. If*

$$\sum_{n:i_n=i} \theta_n = \infty \quad \textit{for every } i, \tag{1}$$

then $R_n \cdots R_1 v$ converges to the orthogonal projection of v onto $\bigcap_i \operatorname{ran} P_i$.

Proof. Write $v_n = R_n v_{n-1}$. The exact norm decrease is

$$\|v_{n-1}\|^2 - \|v_n\|^2 = \theta_n(2-\theta_n)\|(I - P_{i_n})v_{n-1}\|^2.$$

The sequence is bounded. For a cluster point z, let $I_0 = \{i : P_i z = z\}$ and suppose its complement J is nonempty. Put $\delta = \min_{i\in J} \|(I-P_i)z\| > 0$ and $R = \|v_0\|$. Take a late cluster visit N with $\|v_N - z\| < \delta/8$ and remaining norm-decrease budget E so small that $\delta/8 + 8RE/\delta^2 < \delta/2$. Steps in I_0 do not increase distance to z. Before any exit from the radius-$\delta/2$ ball, each step in J spends at least $\theta_n\delta^2/4$ of the budget and moves the vector by at most $2R\theta_n$. Their total movement is at most $8RE/\delta^2$, so no first exit is possible. Their cumulative weights are then at most $4E/\delta^2$, contradicting (1). Every cluster point is therefore in the common range. Its orthogonal projection is unchanged by every step, making the cluster point unique. □

Theorem 4.3 (Persistent algebra of finite compact noise families)**.** *Let a compact group G act continuously by normal automorphisms on a von Neumann algebra M, preserving a faithful normal state ω. For a finite family of closed subgroups $H_1, \ldots, H_m$, let Q_i be their Haar expectations and*

$$D_n = (1-\theta_n)\mathrm{id} + \theta_n Q_{i_n}, \qquad K = \overline{\langle H_1, \ldots, H_m\rangle}.$$

Under (1)*, for every $x \in M$,*

$$D_n \cdots D_1(x) \longrightarrow Q_K(x) \quad \textit{strong* in the faithful GNS representation}, \qquad M^K = \bigcap_i M^{H_i}.$$

No bound on visit gaps, minimum weight or common spectral gap is required.

Proof. State preservation implements the group by a continuous unitary representation on GNS space. Each Q_i acts on $x\Omega$ as its subgroup's Haar orthogonal projection. A compact-group representation is a Hilbert direct sum of finite-dimensional irreducibles with multiplicities. Every vector can be approximated by vectors in finite-dimensional invariant subspaces, taking finitely many multiplicity vectors. Apply Lemma 4.2 there and use contractions to control the approximation error. The limiting projection is the K-Haar projection, since invariance under every H_i is invariance under the closed group they generate.

Thus $x_n\Omega \to Q_K(x)\Omega$, with $\|x_n\| \le \|x\|$. The separating vector makes $M'\Omega$ dense, and $(x_n - Q_K x)a'\Omega = a'(x_n - Q_K x)\Omega$ for $a' \in M'$. Uniform boundedness extends convergence strongly to the full Hilbert space. Apply the same argument to x^* for strong* convergence. □

Partial dephasing for a duration $t_n > 0$ has $\theta_n = 1 - e^{-\kappa_{i_n} t_n}$. For a fixed positive rate per subgroup, condition (1) is equivalent to divergent total duration for each subgroup: small durations give comparable t_n and θ_n, while infinitely many durations bounded away from zero already give divergent weights. Full dephasing has $\theta_n = 1$. Choices based on past records can be used pathwise when their realized sequences satisfy the condition. The theorem describes the resulting noise-map product; additional measurement and control operations must not be silently omitted.

The algebra selected across time is therefore the algebra shared by all these noise sectors. In the SU(2) level-one model of Consciousness I [5], two nonparallel axis-rotation subgroups generate $SO(3)$; their fixed algebra is the Virasoro $c = 1$ half-line algebra. A sequence with cumulative exposure to both axes retains precisely that algebra. If the sequence stays on one axis, its current algebra can remain. The ambient action and the translations needed for an internal Cut are still supplied.

4.3 Implementation errors and the size of the noise family

Corollary 4.4 (Summable channel perturbations)**.** *Under Theorem 4.3, replace D_n by normal UCP maps $\widetilde{D}_n$ such that $\|\widetilde{D}_n - D_n\| \le \varepsilon_n$ and $\sum_n \varepsilon_n < \infty$, where the norm is the operator norm on maps $M \to M$. Then $\widetilde{D}_n \cdots \widetilde{D}_1(x)$ converges strong* to an element of M^K. If additionally $Q_K \widetilde{D}_n = Q_K$ for every n, the limit is $Q_K x$.*

Proof. Put $x_n = \widetilde{D}_n x_{n-1}$ and $e_n = (\widetilde{D}_n - D_n)x_{n-1}$. Contractivity gives $\|e_n\| \le \varepsilon_n \|x\|$. For a fixed N, the distance in norm between x_n and the unperturbed tail product $D_n \cdots D_{N+1} x_N$ is at most $\|x\| \sum_{j>N} \varepsilon_j$. The latter product converges strong* to $Q_K x_N$. Since $Q_K D_n = Q_K$,

$$Q_K x_N = Q_K x + \sum_{j=1}^{N} Q_K e_j$$

converges in norm to an element z of M^K. Letting $N \to \infty$ in the tail estimate proves strong* convergence to z, also for adjoints. The extra conservation condition makes every $Q_K x_N = Q_K x$. □

Persistent forcing need not be summable. It may hold a component away from the passive fixed algebra, which is one role of active protection. The corollary does not assign its implementation cost. Even a single finite control error can change the final retained content, so convergence into an algebra is weaker than conservation of every original record.

A finite family cannot be replaced by an arbitrary countable family using recurrence alone. On a qubit with trace state set

$$\sigma_\alpha = \cos\alpha\, X + \sin\alpha\, Y, \qquad Q_\alpha = (\mathrm{id} + \mathrm{Ad}_{\sigma_\alpha})/2.$$

At stage $k \ge 6$, visit in order $\alpha_j = j\pi/2^k$, $j = 1, \dots, 2^k$. Every dyadic axis receives infinitely many visits, and the common fixed algebra is scalar. Starting from Bloch vector $(1, 0, 0)$, the full projections give alternating signed X vectors at stage endpoints, with limiting length

$$\prod_{k \ge 6} \cos(\pi/2^k)^{2^k} \ge e^{-\pi^2/32} > 0.$$

The bound follows from $\log\cos t \ge -t^2$ on the interval used and $\sum_{k\ge 6} 2^{-k} = 1/32$. Replace each projection at stage k by the finite-duration map $4^{-k}\mathrm{id} + (1 - 4^{-k})Q_\alpha$. Telescoping the contractive Bloch maps bounds its distance from the full-projection sequence by $\sum_{k\ge 6} 2^k 4^{-k} = 1/32$. The alternating endpoints still do not converge, although every individual axis receives divergent cumulative exposure. This is a qubit counterexample to the unrestricted extension, not a reconstruction of a spacetime net.

5 A Legal Finite Feedback under Persistent Noise

The compact-noise limit retains observables but does not move the reference state. Here a bounded command changes the source reset target at every epoch. The resulting adaptive sequence is legal without strict commutation or an autonomous-semigroup description. The bank is finite and diagonal; its contraction bound concerns state risk, not Type III reconstruction.

5.1 The controlled bank

Take M qubits in Z-basis configurations $z_j \in \{\pm 1\}$ and write $x = M^{-1}\sum_j z_j$. For $|v| \le 1$ let $\tau_v = \frac{1}{2}(I + vZ)$ and let $R_v(\varrho) = \tau_v \operatorname{tr} \varrho$ be the reset to τ_v. With weights $a, c, r \ge 0$, $a+c+r = 1$, a command u and an environmental input d, the channel applied to each qubit in one epoch is

$$\Lambda_{u,d} = a\,\mathrm{id} + c\,R_u + r\,R_d. \tag{2}$$

The qubit keeps its state with weight a, is exchanged with a reference prepared at the command with weight c, and with a reference prepared at the environmental input with weight r. Legality, item (a) below, holds for all such weights. For the later statements we take $0 < a < 1$, $c > 0$, $r \ge 0$: $a = 1$ is the identity, and $a = 0$ is a complete reset, which is not a finite-time reset generator and, with $c = h = 1$, would give $|a - ch| = 1$.

Proposition 5.1 (Legality and mean dynamics)**.** *Let $|u| \le 1$ and $|d| \le 1$.*

(a) $\Lambda_{u,d}$ is completely positive and trace preserving, and its Heisenberg-picture dual is normal, completely positive and unital.

(b) If $a < 1$, $\Lambda_{u,d} = a\,\mathrm{id} + (1-a)R_v$ with $v = (cu + rd)/(1-a)$ and $|v| \le 1$. If also $a > 0$, then for $a = e^{-\gamma\Delta t}$ it is the time-Δt map of the reset generator $\gamma(R_v - \mathrm{id})$.

(c) If $u = -h\hat{x}$ with $\hat{x} \in [-1, 1]$ and $0 \le h \le 1$, then $|u| \le h \le 1$, so every epoch is legal whatever the estimate $\hat{x}$ is.

(d) Applying $\Lambda_{u,d}^{\otimes M}$ to a configuration of mean x gives a configuration of mean $x' = ax + cu + rd + \zeta$ with $\mathbb{E}[\zeta \mid x] = 0$ and $\mathbb{E}[\zeta^2 \mid x] \le M^{-1}$. The number of qubits at $+1$ after the step is a sum of two independent binomial variables.

Proof. (a) id and R_u, R_d are completely positive and trace preserving, and so is their convex combination. The dual is $\Lambda^*(A) = aA + c\operatorname{tr}(\tau_u A)I + r\operatorname{tr}(\tau_d A)I$, which maps I to I. (b) $cR_u + rR_d = (c+r)R_v$ because τ_v is affine in v, and $|v| \le (c+r)^{-1}(c|u| + r|d|) \le 1$. Since $R_v^2 = R_v$, $e^{\gamma t(R_v - \mathrm{id})} = e^{-\gamma t}\mathrm{id} + (1 - e^{-\gamma t})R_v$. (c) is immediate. (d) A qubit with value z ends at $+1$ with probability $\frac{1}{2}(1 + az + w)$, where $w = cu + rd$, independently of the others. Its mean is $az + w$ and its variance $1 - (az + w)^2$. Averaging over the bank gives mean $ax + w$ and variance $M^{-1}(1 - a^2 - 2axw - w^2)$, which is at most M^{-1} since $a^2 + 2axw + w^2$ is the average of the squares $(az_j + w)^2$. The qubits at $+1$ and at -1 each give one binomial variable. □

Remark 5.2 (Non-commutation). Let $E = a\,\mathrm{id} + (1-a)R_v$ with $a < 1$ be the reset of a source. Its only invariant state is τ_v, and the same holds for $E^{\otimes M}$ and $\tau_v^{\otimes M}$. If a completely positive trace-preserving map F commutes with E, then $E(F\tau_v) = F(E\tau_v) = F\tau_v$, so $F\tau_v = \tau_v$ and τ_v is invariant for $F \circ E$ as well. A feedback that commutes with the source reset therefore cannot move the stationary state of the source. The feedback of this section moves it from τ_v to $\tau_{v'}$ with $v' \ne v$ whenever the command moves the target. The strict commutation input FB-C of Definition C.1, which gives the semigroup law in Section 3, fails for it, and Theorem 3.1 does not apply. For two resets $\Lambda_i = a_i\,\mathrm{id} + (1 - a_i)R_{v_i}$ one finds directly $\Lambda_1\Lambda_2 - \Lambda_2\Lambda_1 = (1 - a_1)(1 - a_2)(R_{v_1} - R_{v_2})$, because $R_{v_1}R_{v_2} = R_{v_1}$. This concerns a fixed command. The history-dependent process is a sequence of legal channels with commands that change from epoch to epoch, and it is not a new autonomous semigroup. The next subsection replaces the exact time-ordering argument by a norm estimate.

5.2 An attracting ball instead of convergence

Here the estimate that sets the command may keep an error in every epoch, and summability of the feedback is not assumed. What can be proved is that the state stays in a ball.

Lemma 5.3 (Contraction with forcing and innovation)**.** *Let H be a Hilbert space and let $\xi_{k+1} = T_k\xi_k + f_k + \zeta_k$ with $\|T_k\| \le q < 1$, $\|f_k\| \le B_k$, and ζ_k orthogonal to $T_k\xi_k + f_k$ with $\|\zeta_k\| \le s$. Write $\varphi_B(X) = \sqrt{(qX+B)^2 + s^2}$. Then $\|\xi_{k+1}\| \le \varphi_{B_k}(\|\xi_k\|)$. If $B_k \le B$ for all k, then with*

$$\bar{X} = \frac{qB + \sqrt{B^2 + (1-q^2)s^2}}{1-q^2}$$

one has $\|\xi_k\| \le \bar{X} + q^k(\|\xi_0\| - \bar{X})_+$. If only $\limsup B_k \le B$, then $\limsup \|\xi_k\| \le \bar{X}$.

Proof. Orthogonality gives $\|\xi_{k+1}\|^2 = \|T_k\xi_k + f_k\|^2 + \|\zeta_k\|^2 \le (q\|\xi_k\| + B_k)^2 + s^2$. The map φ_B is increasing, $\varphi_B'(X) = q(qX+B)/\varphi_B(X) \le q$, and $\bar{X}$ is its fixed point, so $\varphi_B(\bar{X}+t) \le \bar{X} + qt$ for $t \ge 0$ and $\varphi_B(X) \le \bar{X}$ for $X \le \bar{X}$. Induction gives the bound. For the last statement, φ_B is increasing in B, so for any $\epsilon > 0$ the sequence is eventually below the orbit of $\varphi_{B+\epsilon}$, whose limit is $\bar{X}(B+\epsilon)$; let $\epsilon \to 0$. □

In a probabilistic reading $H = L^2(\mathbb{P})$, and conditional mean zero of ζ_k given everything determined before the step implies the orthogonality used. A Type III version would have to supply this hypothesis through a conditional expectation.

Proposition 5.4 (Bounded risk of the closed bank)**.** *Let $0 < a < 1$, $c > 0$, $r \ge 0$ and let the bank evolve by $\Lambda_{u_k,d_k}^{\otimes M}$ with $|d_k| \le D \le 1$ and $u_k = -h\hat{x}_k$, $0 < h \le 1$, where $\hat{x}_k \in [-1,1]$ is any estimate of x_k that is determined before the step, with $\|\hat{x}_k - x_k\|_{L^2} \le \varepsilon_k$. Put $k_c = ch$ and $q = |a - k_c|$. Then $q < 1$ for every legal h, and*

$$\|x_{k+1}\|_{L^2} \le \sqrt{\big(q\|x_k\|_{L^2} + k_c\varepsilon_k + rD\big)^2 + M^{-1}}.$$

If $\varepsilon_k \le \varepsilon_*$ for all k, then with $B = k_c\varepsilon_* + rD$ and $s^2 = M^{-1}$ in Lemma 5.3,

$$\mathbb{E}\,\tfrac{1}{4}x_k^2 \le \tfrac{1}{4}\big[\bar{X} + q^k(\|x_0\|_{L^2} - \bar{X})_+\big]^2.$$

Proof. By Proposition 5.1(d), $x_{k+1} = (a-k_c)x_k - k_c e_k + rd_k + \zeta_k$ with $e_k = \hat{x}_k - x_k$. The first three terms are determined before the step, and ζ_k has conditional mean zero and second moment at most M^{-1} given them, so Lemma 5.3 applies with T_k the multiplication by $a - k_c$ and $f_k = -k_c e_k + rd_k$. Finally $a - k_c > -1$ because $k_c \le c < 1 + a$. □

No independence between x_k and the estimation error is assumed, and no claim is made that $\varepsilon_k \to 0$. The radius grows with ε_*, and whether it is smaller than the radius without feedback depends on ε_*, as the next subsection shows.

5.3 What the ball certifies

Compare the closed-loop radius with the open-loop radius ($h = 0$, so $q = a$ and $B = rD$). Let $\bar{X}_0$ be the fixed point of $\varphi_0(X) = \sqrt{(aX + rD)^2 + M^{-1}}$.

Corollary 5.5 (Smaller certificate)**.** *Let $0 < k_c = ch \le a < 1$ and let $\bar{X}_h$ be the radius of Proposition 5.4 with the uniform error bound ε_*. Then, for every finite M,*

$$\bar{X}_h < \bar{X}_0 \iff \varepsilon_* < \bar{X}_0,$$

and $\bar{X}_h = \bar{X}_0$ exactly when $\varepsilon_* = \bar{X}_0$. As $M \to \infty$, $\bar{X}_0 \to rD/(1-a)$.

Proof. Both maps are increasing strict contractions. At $X = \bar{X}_0$ the quantity under the root for the closed loop is $(a - k_c)\bar{X}_0 + k_c\varepsilon_* + rD = (a\bar{X}_0 + rD) + k_c(\varepsilon_* - \bar{X}_0)$, so the closed-loop map takes $\bar{X}_0$ to a value below, at, or above $\bar{X}_0$ according to the sign of $\varepsilon_* - \bar{X}_0$. For a strictly contracting increasing map, that sign decides whether its fixed point lies below, at, or above $\bar{X}_0$. For $M \to \infty$, $\bar{X}_0 = rD/(1 - a)$. □

The corollary compares two certificates. It does not compare the losses of the two systems: an upper bound for one process being smaller than an upper bound for another says nothing about their actual losses. Take $M \to \infty$, $d_k \equiv 0$ and $x_0 = 0$, with the declared bound $D > 0$. Without feedback the state stays at 0. An estimator with a constant bias $\delta < rD/(1 - a)$ has $\varepsilon_* = \delta$ below the threshold, yet the closed loop settles at $x_* = -k_c\delta/(1 - a + k_c) \neq 0$.

In the bank used in T-DOME III's appendix on a drifting instrument in a closed loop [8] one has $(a, c, r) = (0.65, 0.25, 0.1)$, $h = 0.8$, $M = 4096$, $|d_k| \leq D = 0.85$, hence $q = 0.45$. The recursive estimate of that appendix, with a gain floor of 0.30 and the drift of its normal case, has $\limsup_k \varepsilon_k \leq \varepsilon_* = 0.4518$ by its propositions on memory under drift and on the error of the decoded state. The bound holds in expectation and not for all epochs at once, and it concerns the limit superior, which Lemma 5.3 accepts. Then $B = 0.1754$, $\bar{X}_h = 0.3195$, and $\limsup \mathbb{E}\frac{1}{4}x_k^2 \leq 0.02553$, below the trivial bound $\frac{1}{4}$. (The cruder closed form in that paper gives $\varepsilon_* = 0.5123$ and 0.02915.) The open-loop radius is $\bar{X}_0 = 0.2443$, which is the finite-M threshold ($rD/(1 - a) = 0.2429$ for $M = \infty$), and ε_* is above it. The certificate therefore gives a larger radius for the feedback than for the open loop, 0.3195 against 0.2443. The loss comparison rests on other evidence. For the same bath, initial state and epochs 200–1199, the open-loop world loss is 8.17×10^{-3}, obtained exactly from the mean and variance recursions of Proposition 5.1(d). The simulated persistent policy gives 3.60×10^{-3}, 56% lower. The epoch-averaged root-mean-square estimation error of the persistent policy is 0.135 (0.173 for the memoryless policy and 0.211 for the frozen one). An epoch average is not a uniform bound.

5.4 Scope

The model is a bank of qubits restricted to diagonal states. It uses no quantum effect, and the cost of non-commuting readout or of writing the memory is not included. The reference states used by the estimate are assumed to be prepared reliably. The bound on ε_* is an input from another paper and holds in expectation for one drift family. The mean and variance in Proposition 5.1(d) were checked against exact enumeration on a seven-qubit bank over a grid of commands and inputs (largest residual 4.4e-16). Nothing here constructs a normal completely positive lift on a Type III algebra.

6 From Persistent Content to a Priced Return

Theorems about persistence specify which observables a process retains; they do not specify the apparatus that retains them. The weights θ_n and rates κ_i describe channels, not joules. A physical account must separately price the generation of these channels, sensing, retained records, reference preparation, control and disposal of outputs. A proper fixed algebra by itself supplies no universal positive entropy-production rate.

In the bank, an epoch uses n sensor readings and on average cM commanded reference qubits. If those references are freshly prepared from degenerate thermal qubits at temperature

T, their minimum preparation charge, conditional on the known command, is

$$k_B T\, cM\, \mathbb{E}\left[\log 2 - H_2\left(\frac{1+u_k}{2}\right)\right].$$

This is an input-preparation charge, not the net cost of a complete cycle: outputs can carry work, entropy and correlations. The isolation, resetting and readout assumptions of the chosen T-DOME III protocol determine what further charge can be added. A record-dependent command is free neither because it is classical nor because its channel is CPTP.

These distinctions locate the remaining cross-series bridge. Foundation I selects an algebra from a seed and supplied operations. For the same compact action, the intersection of several fixed algebras is M^K; Theorem 4.3 identifies it with content that persists under cumulative exposure. An internal Cut and its geometric data require the additional translation and standardness inputs of Consciousness I. Consciousness II supplies conditions under which a prepared source identifies a geometric response. Q-RAIF and HAFF must realize the control as an interaction and account for its actual source. The T-DOME and Imprint loops specify how retained records choose the next operation; Brain Anchor asks how the channels and maintenance relations themselves are learned and restored.

A common fixed algebra can therefore be an output, rather than a separate sharing assumption. It does not turn independent source states into one state or shared carrier into shared content. Nor does passive convergence explain why a particular noise group, reference port or coupling became available. Closing the theory's return arrow requires a model in which that access is formed, its price affects decisions, and its response changes the next read or action. The persistence and legal-return results specify two of its interfaces. The next section supplies a finite composition theorem and a physical example with an acquired installed frame.

7 Returns on a Retained Physical History

7.1 Finite composition and its error budget

A round consists of acquisition, a retained record, a control decision, an interaction and the response used by the next acquisition. Its input includes the old content, preparation references, spent ports and the clock and geometric data needed to predict that response. Adding a fresh port enlarges this state; it does not replace a spent port by its initial state. In a semiclassical realization, the joint evolution of quantum and classical variables need not be one linear quantum channel.

Theorem 7.1 (Finite retained-history composition)**.** *For $n = 0, \dots, N-1$, let Φ_n and $\widetilde{\Phi}_n$ be an ideal and an implemented round from a metric state space (X_n, d_n) to (X_{n+1}, d_{n+1}). Suppose both trajectories remain in specified admissible sets C_n, and on those sets*

$$d_{n+1}(\Phi_n(x), \Phi_n(y)) \le \omega_n(d_n(x,y)), \tag{3}$$

$$d_{n+1}(\widetilde{\Phi}_n(x), \Phi_n(x)) \le \epsilon_n. \tag{4}$$

Here each ω_n is nondecreasing, continuous at zero, and $\omega_n(0) = 0$. Define $b_0 = d_0(\widetilde{x}_0, x_0)$ and

$$b_{n+1} = \omega_n(b_n) + \epsilon_n.$$

Then $d_n(\widetilde{x}_n, x_n) \le b_n$ at every round. For fixed finite N, $b_N \to 0$ as the initial discrepancy and all implementation defects tend to zero.

Proof. Insert $\Phi_n(\widetilde{x}_n)$ between $\widetilde{\Phi}_n(\widetilde{x}_n)$ and $\Phi_n(x_n)$ in the triangle inequality. Equations (3) and (4) give $d_{n+1} \le \epsilon_n + \omega_n(d_n)$. Monotonicity proves the recursive bound by induction; continuity at zero proves its finite-horizon limit, again by induction. □

The input sets and moduli carry the physical burden. In finite dimensions, a smooth mean-action evolution on a compact regular domain has a local Lipschitz bound. An unbounded material stress requires an energy-aware domain and a source certificate; ordinary trace distance does not provide one. For the number Hamiltonian, the states $\rho_m = (1-m^{-1})|0\rangle\langle 0| + m^{-1}|m^2\rangle\langle m^2|$ approach the vacuum in half trace distance m^{-1} while their mean energy grows as m. Clock positivity, a regular exterior region and the energy moments used by the source estimate must remain valid throughout the return.

Corollary 7.2 (Linear and Hölder budgets)**.** *Under Theorem 7.1, nonnegative vector bounds* $e_{n+1} \le A_n e_n + \varepsilon_n$ *give*

$$e_N \le A_{N-1}\cdots A_0 e_0 + \sum_{j=0}^{N-1} A_{N-1}\cdots A_{j+1}\varepsilon_j,$$

with an empty product equal to the identity. If instead $\omega_n(t) = Kt^\alpha$, $K > 0$, $0 < \alpha < 1$, *and implementation defects vanish, then*

$$d_N \le K^{(1-\alpha^N)/(1-\alpha)} d_0^{\alpha^N}.$$

For $\alpha = 1$ *this becomes* $d_N \le K^N d_0$.

Proof. Successive substitution proves the vector formula. Iterating $b_{n+1} = Kb_n^\alpha$ gives the geometric sum of exponents of K and the exponent α^N of b_0. □

These are sufficient precision budgets, not universal resource lower bounds. A Hölder estimate with $\alpha = 1/2$ can require an initial precision scaling as a target error to the power 2^N; finite composability is therefore not a claim of economical indefinite use. Uniform long-time control follows, for example, if an invariant domain has $\omega_n(t) \le qt$ with $q < 1$ and bounded defects, in which case $d_N \le q^N d_0 + \sum_{j<N} q^{N-1-j}\epsilon_j$. Neither finite continuity nor small one-round error supplies that contraction.

The comparison uses complete instruments or unnormalized record branches. A record of probability p can have two orthogonal conditional states even when the complete states differ in half trace distance only by p. Conditioning on that record divides away the small probability. Such conditional accuracy cannot be inserted as a uniform round certificate without a probability bound. A final bounded-loss measurement may discard unused systems; doing so at every intermediate round would change the retained process being proved.

7.2 Quantum continuation and receiver renewal

A fixed controller on the retained record defines one complete CPTP history map, even if its decision rule has posterior thresholds. Instrument errors therefore add over any fixed finite sequence, with the identity acting on old references. Q-RAIF C also supplies the positive-Hamiltonian smooth-source bound and a retained receiver–bank renewal family [3]. These are imported quantum interfaces, recorded with full assumptions and proofs in the source record identified by Appendix D. They preserve the spent receiver and untrimmed cursor; a later installed program is needed for another return. The finite quantum sequence does not make the semiclassical mean-action map below contractive or turn its return into a quantum JT equation.

7.3 Learning an installed frame for held content

Consider two body qubits B_1, B_2 with an unknown physical sign $s \in \{+1, -1\}$. Write $A_s = n_1 + sn_2$ for their occupation coupling. The odd encoding $E_o|0\rangle = |01\rangle$, $E_o|1\rangle = |10\rangle$ is protected when $s = +1$; the even encoding obtained by $X_{B_2}E_o$ is protected when $s = -1$. In the correct encoding A_s is scalar, so $\exp(-i\varphi A_s Z_E)$ cannot disclose or dephase its logical qubit. Each corner contains a full M_2 of logical operations. The decoder and its installed frame are needed to use that corner; storage alone is not a readout permission. This distinction is also used in Brain Anchor [9]. Protecting an observable algebra is an established error-correction framework [11]; the question here is how a record selects an installed encoding within the physical return.

One initially held logical qubit is entangled with an old reference and encoded in E_o. It is never re-prepared. Two independent, initially supplied Gibbs cells E_0, E_1 have gap $\Delta = 0.05$ and excitation probability 0.1, with their purifications retained. A probe P starts in $|+X\rangle$, and a record R is blank. We also write R for its occupied-state projection $|1\rangle\langle 1|$. The same unknown plant sign that controls A_s rotates the probe by $\exp[-is \arcsin(0.8) Z_P/2]$. A controlled flip of R from the positive Y_P projection then acquires a sign estimate that is correct with probability 0.9. The diagnostic/storage relation is an assumption about the plant, not a label supplied to the controller.

The actual program applies an early phase contact $\exp(-0.05 i A_s Z_{E_0})$, the diagnostic rotation, the acquisition, the record-controlled frame

$$W_R = R \otimes I + (I - R) \otimes X_{B_2},$$

the late contact $\exp[-i(\pi/4)A_s Z_{E_1}]$, and W_R again as a physical decoder. The early contact acts before the frame has been learned, so the held content can already be damaged. The fixed comparisons replace W_R by I or X_{B_2}, with the same probe, ports, early contact and time slots. Entanglement infidelity is measured against the original encoded Bell pair after decoding, without postselection on the record or cursor.

Proposition 7.3 (Value of the acquired frame)**.** *For this gate program, set*

$$e = \frac{1 - \cos(0.1)}{2}, \qquad q = \frac{1 + 0.8^2 \sin(0.1)}{2}.$$

The learned program's losses at $s = +1, -1$ *are respectively* 0.05 *and* $0.9e + 0.1q$*. The fixed odd losses are* $0, q$*; the fixed even losses are* $1/2, e$*. Under the uniform unknown-sign prior, learning therefore gives loss* 0.0527214*, below both fixed losses* 0.2659733 *and* 0.2512490*.*

Proof. For a Gibbs cell with $\langle Z_E \rangle = 0.8$, an occupation eigenvalue difference of two multiplies coherence by $\cos(2\varphi) \pm 0.8i \sin(2\varphi)$. For $s = +1$ the early odd contact is scalar. The correct odd decision leaves the content intact; the wrong even decision has late real coherence zero and infidelity $1/2$. For $s = -1$ the early odd contact gives infidelity e. Changing to the correct even code adds no late dephasing. Remaining odd multiplies both contact factors, whose real part is $-0.8^2 \sin(0.1)$, giving q. The diagnostic record is independent of the logical input in this gate program. Weight its correct and incorrect decisions by 0.9 and 0.1, then average the two plant signs. □

An uninformative diagnostic gives average loss 0.2586112. Reversing the physical relation between diagnosis and storage instead gives 0.4645009. Thus installing a rule is not enough: its relation to the actual body must be learned correctly. The proposition compares two permitted

fixed frames, not every possible quantum recovery or an observer allowed to inspect the spent cells. For a diagnostic with symmetric success probability p, the same calculation gives mean loss $(1-p)(1/4+q/2)+pe/2$. It beats the better fixed frame precisely when

$$p > \frac{1/4+q/2-\min(q/2,\,1/4+e/2)}{1/4+q/2-e/2} = 0.514303\ldots.$$

This threshold concerns the stated gate and contact budget; the cost of obtaining p remains part of the physical diagnostic.

7.4 The encoded body in the physical return

Put this program on a finite nearest-neighbour cursor, with gate labels on its hopping edges [12]. Its positive cursor Hamiltonian is a unitary dressing of $2JI-J$ times the path adjacency and lies between 0 and $4J$. After decoding, append a second actual acquisition from X_P into the already used cell E_0. This cell is not blank, independent or reset: its old reference and contact correlations are still held. The acquisition is $C_X(P,E_0) = (I-X_P)/2 + (I+X_P)X_{E_0}/2$. After the first Y acquisition the probe retains $\langle Y_P\rangle = 0.8s$; a subsequent Z_P rotation changes its X component to first order. No extra probe or pure-state preparation is needed. The first acquisition remains $C_Y(P,R)$, and the frame and decoder are unchanged. In the zero-order completed-gate comparison, $\langle X_P Z_{E_0}\rangle = 0$ and $n_{E_0} = 1/2$ for either plant sign: this acquisition is sign-blind at that order and probes the induced geometric response. Finite couplings can still produce sign-dependent readings. The near-half cell occupation is not a reset of its retained correlations, and its energy cost stays inside E_c. Its occupation n_{E_0} then acts on the mechanical spring alongside the first record. Unlike the earlier gates, this last acquisition need not commute with bare port energy. Its work is included in the full supported core; free cursor arrival cannot be assumed for that core. Add the sensor and body forces as

$$H_c(\Phi,y) = H_b + \lambda(\Phi-\Phi_0)Z_P + \frac{\chi}{2}\left[A_s^2(y-d_1)^2 + (R+n_{E_0})(y-d_2)^2\right]. \tag{5}$$

Here H_b contains the cursor, bare port energy and an offset $E_{\text{off}} > |\lambda(\Phi-\Phi_0)|$ on the operating domain. Both springs are positive. The first responds to the installed body's actual occupation coupling, not to a task score. The second uses the actual acquired record. Their force is the expectation in the complete retained quantum state.

Use the scalar/JT worldline action of HAFF D [10], with $i\langle\Psi,\dot\Psi\rangle - \langle H_c\rangle$ for this core, mechanical energy e_m, support potential $U(\Phi)$ and rest mass. Writing $F_y = \langle\partial_y H_c\rangle$ and $F_\Phi = \langle\partial_\Phi H_c\rangle$, its energy identities are

$$\begin{aligned} \dot e_m &= -P_+ - P_- - F_y\dot y, & \dot E_c &= F_y\dot y + F_\Phi\dot\Phi, \\ M &= m_{\text{rest}} + U + e_m + E_c, & \dot M &= (U'+F_\Phi)\dot\Phi - P_+ - P_-. \end{aligned} \tag{6}$$

The outgoing scalar power, both JT junctions, the supported mass and the sensor all use this same action. The finite equations are locally well posed away from zero mass and singular exterior boundaries. On a compact regular operating window their continuity supplies the modulus needed by Theorem 7.1. It does not supply a channel on arbitrary quantum geometries.

For 101 cursor sites, $J = 0.8$, six data qubits and retained-reference dimension 8, take $\lambda = 0.05$, $\chi = 0.01$ and a proper-time window $[0,30]$. The initial supported mass is 10. Take $\mathcal{C}_-(0) = 4.7975$, with the other junction fixed by the same initial mass. The initial core energy

1.655 includes the already active body spring; omitting its 0.005 would break the core ledger. Direct evolution of the full purified state gives

	learned frame	fixed odd	fixed even
mean entanglement infidelity	0.05883044	0.26585700	0.31215165

for the same uniform plant prior. This $T = 30$ run is one finite member of the model family, not a run in the $\varepsilon \to 0$ scaling used in the existence proof below. These losses are not the exact gate-program values of Proposition 7.3. The learned $s = -1, +1$ readings have final occupations $0.49963517, 0.49949711$; their bare cell-energy gains are $0.01998176, 0.01997486$, carried inside E_c rather than added again as exported heat. Completion probability is at least 0.9986007 across the six policy/sign runs, but the reported loss includes the entire cursor state.

Changing the initial geometry tests background sensitivity. For example, the matched-mass inputs $\mathcal{C}_-(0) = 4.6975, 4.8975$ give $s = -1$ occupations $0.49996625, 0.49927896$. The following result instead keeps the complete input fixed and follows the source produced by the acquired record itself.

Proposition 7.4 (A learned frame and its own physical return)**.** *Within the finite scalar/JT mean-action family above, there are finite positive clock, cell and coupling choices for which the same retained run protects the original entangled content better than either fixed frame and reads a nonzero response to its own record-actuated source. The unknown-sign prior and its physical diagnostic relation are those of Proposition 7.3. The source, support, clock and both old cell purifications are included in the initial account. The second acquisition uses the same spent E_0, without resetting the content, record or cursor.*

For a matched causal comparison, replace $R + n_{E_0}$ in (5) *by $\alpha R + n_{E_0}$, $0 \le \alpha \le 1$. All runs have the same quantum, mechanical and geometric input and the same mass junction. The case $\alpha = 0$ retains the record and its encoding decision but removes its mechanical use. The geometry-mediated component of the second reading is nonzero; the entire reading contrast need not vanish when sensing is removed.*

Proof. The exact gate-program margin over the better fixed frame is $m_0 = 0.19852756\ldots$. A finite wholly idle cursor packet can be chosen with incomplete-prefix amplitude below η, retaining every reference and the untrimmed cursor, by Lemma A.1. For the pure gate-dressed hopping comparison the full-vector error is at most 2η. The physical core is not freely factorized once the cell gaps and forces are restored. Duhamel instead bounds its vector error by

$$e_\Psi \le 2\eta + 2\Delta T + T\big(|\lambda| D_\Phi + \chi C_y\big), \qquad C_y = 3(Y + 1/2)^2,$$

where $|y| \le Y$ and $|\Phi - \Phi_0| \le D_\Phi$ on the common window. Choose a finite arrival time and then positive Δ, λ, χ small enough that $e_\Psi < m_0/4$; keep $\beta\Delta = \log 9$ and charge the supplied cell preparations. The two bounded-loss errors then preserve the strict task margin. At zero radiation and coordinate coupling, use the positive static support $\Phi = 3$, $M = 10, \mathcal{C}_- = 4.7975$, $\mathcal{C}_+ = 5.1975$. It has constant Φ and mass, while its mechanical energy bounds y. Finite-time continuity supplies a common positive operating tube for sufficiently small positive radiation and couplings. Its rest mass is fixed from the complete initial core energy, including the cursor, cells, offset and initially active body spring. Fixed finite J permits it to remain positive. Take $d_1 = -1/2$, $d_2 = y_0 = 1/2$, $\dot{y}_0 > 0$ and $\theta_0 = \dot{\theta}_0 = 0$.

The forwardmost packet site can remain -3. Its shortest path to the first acquisition has eight edges. With weight $w_f > 0$,

$$\langle R \rangle(t) = a_R t^{16} + O(t^{17}), \qquad a_R = \frac{w_f J^{16} p_s}{(8!)^2} > 0, \quad p_s \in \{0.1, 0.9\}.$$

At zero sensor coupling and $y_0 = d_2$, $\dot{y}_0 \neq 0$, variation at $\alpha = 0$ gives the radiation-path coefficient

$$\delta_\alpha P(t) = D_P t^{21} + \cdots, \qquad D_P = -\frac{2\gamma k^2 y_0 \chi \dot{y}_0 a_R}{I^2 J_m \, 18\,19\,20} \neq 0.$$

The balanced outgoing junction has $\ddot{\Phi} = -\kappa P/2 - (U' + F_\Phi)\beta_+\beta_-/M - \Phi/L^2$. At zero sensor coupling, its corresponding mixed radiation response is $Ft^{23} + O(t^{24})$, with $F = -\kappa D_P/(2 \cdot 22 \cdot 23) > 0$. The induced mass, invariant and support changes are included; their contribution to this term is higher order.

The second acquisition is sixteen edges from the same packet front. In the full retained state, inserting this geometric response along the Z_P sensor gives a t^{56} coefficient

$$K_X(s) = -\frac{(16/25)s\lambda F_s w_f J^{32}}{16!\,40!} \sum_{j=8}^{15} \frac{(j+23)!}{j!} \neq 0,$$

where F_s denotes the coefficient F for the specified plant sign. Here $j = 0, \ldots, 16$ indexes the shortest path from -3 to 13. Let W_j be its complete gate prefix and let ρ_0 include the old logical reference and both cell purifications. The insertion factor is

$$2\,\mathrm{Im}\,\mathrm{Tr}\big[\rho_0 W_{16}^\dagger n_{E_0} W_{16} W_j^\dagger Z_P W_j\big] = \begin{cases} 0, & 0 \le j \le 7 \text{ or } j = 16, \\ -(16/25)s, & 8 \le j \le 15. \end{cases}$$

Before the first acquisition its Y dephasing removes the final X response. Afterwards, C_X reads $(I + X_P Z_{E_0})/2$; the derivative of a Z_P rotation is $-\langle Y_P Z_{E_0}\rangle = -(0.8s)(0.8)$. The earlier phase contact commutes with Z_{E_0}, and the record-controlled body gates preserve $Y_P Z_{E_0}$. This uses the spent cell and the full prefixes, including encoder and decoder. The last position contributes zero because the subsequent sensor commutes with the measured cell occupation. Integrating the insertion at position j gives $(j+23)!/(j!40!)$; interference with the sixteen-hop output amplitude gives the displayed coefficient. Thus its leading sensor insertion is entirely after acquisition of the actuating record. Under the unobserved uniform sign prior, $\overline{K}_X = (K_X(+1) + K_X(-1))/2 \neq 0$: the record-production coefficients have $p_+ = 0.9$ and $p_- = 0.1$, so $F_+ = 9F_-$. The sign is a plant parameter, never a controller input. This coefficient is a geometry-edge response, not the total E_0 contrast or its leading term at arbitrary coupling. Let $\mathscr{E}_\alpha[y, \phi](t)$ denote the second-cell expectation under the same gates, springs and complete input with those histories. The geometry-edge diagnostic is

$$G_\alpha(t) = \mathscr{E}_\alpha[y_\alpha, \Phi_\alpha](t) - \mathscr{E}_\alpha[y_\alpha, \Phi^{(0)}](t),$$

where $\Phi^{(0)}$ is the same-input $\alpha = 0$ trajectory. Differentiating at $\alpha = 0$ cancels the common direct spring, cursor and y changes, leaving the geometric response above. At zero sensor coupling G_α vanishes identically even when the total cell contrast does not. This paired propagation is a causal diagnostic, not a free alternative source or controller input.

The full finite equations are analytic in their regular tube. Take $\Delta = \varepsilon d$, $\chi = \varepsilon c$, $\gamma = \varepsilon g$ and $\lambda = \varepsilon^2 \ell$ with fixed positive d, c, g, ℓ, and keep $\beta\Delta = \log 9$. At fixed finite time the radiation path $\chi \to y \to \theta \to P \to \Phi$ has order $\gamma\chi = O(\varepsilon^2)$; the sensor-surface path has order $\lambda\chi = O(\varepsilon^3)$. The final reading adds one factor λ. Hence

$$G_{\alpha,s}(t) = \alpha\varepsilon^4 g_{4,s}(t) + O(\varepsilon^5), \qquad 0 \le \alpha \le 1,$$

uniformly on a fixed regular window, with the nonzero t^{56} coefficient above in $\varepsilon^4 g_{4,s}$. Higher powers of α require another χ and enter the remainder. The mass and invariant changes do not

alter the t^{23} radiation coefficient; the sensor-surface path cannot cancel this order. In particular $G_{1,s}$ and its sign-prior average have nonzero leading analytic functions: the derivative at $\alpha = 0$ is not being substituted for the actual $\alpha = 1$ run. These positive choices retain the strict task and domain margins. The nonzero analytic response has isolated time zeros, whereas the strict task and arrival margins hold on an open completion interval. Choose one finite T there away from those zeros: both the task advantage and the second reading then hold in the same retained run and window. All actual runs obey (6); the second-cell population change remains inside E_c, not an additional heat export. HAFF D's coupled-return tolerance applies to this same core and reference space with $L_H = |\lambda| + \chi[4(Y + |d_1|) + 2(Y + |d_2|)]$. Finite errors smaller than the chosen task and return margins preserve both conclusions. The acquired cell continues to act on the mechanical spring, rather than being copied into an external log. □

For the stated $T = 30$ numerical parameters, paired propagation gives $G_{1,-1} = -3.18616 \times 10^{-9}$ and $G_{1,+1} = -6.76572 \times 10^{-9}$, and thus the unconditioned prior average is -4.97594×10^{-9}. The t^{56} coefficient establishes analytic nonzero response; it does not predict the sign or magnitude at $T = 30$. Keeping only post-acquisition sensor differences gives respectively -3.18656×10^{-9} and -6.76533×10^{-9}; the pre-acquisition differences in this numerical propagation are below 4.1×10^{-13}. No perturbative order or mechanism is assigned to these small finite-parameter differences. Removing sensing makes G_1 identically zero, although the total port contrast remains nonzero. The return is a small finite existence signal, not a practical gravitational performance gain.

The installed frame, its source and its later reading are thus one finite mean-action return. The classical geometry responds to the complete core expectation; the construction does not assign an independent quantum geometry to each record branch. The supplied plant relation and primitive gates remain inputs. The phase contacts preserve bare populations; the spent-cell acquisition changes one and includes its work inside E_c, not as additional heat export. It is not a renewed thermal outlet. Full quantum-material matching, sustained renewal and the experiential interpretation remain distinct questions.

8 What the Return Can Generate

The finite return generates a protective frame for content already present, not new content supplied after diagnosis. The reference diagnoses a property of the same body that later couples to noise. Its record installs the encoder, affects the physical source, and changes the next acquired record. The controls, diagnostic relation and resource bank are given; their historically acquired use is not. This is the mechanism Foundation III adds to static persistence.

A broader hypothesis is that accessible histories can select their own durable operating sectors: episodes reveal which physical relations protect an old task, retained records install those relations, and their changed source alters future episodes. The learned-frame example realizes one finite instance; the compact-noise theorem shows when a sequence selects a common sector. These are complementary mechanisms, not one theorem. A persistent control that moves a reset target need not obey compact-noise convergence, while a convergent protected algebra need not keep calibrating or acting.

The hypothesis is testable through three separations. Remove the diagnosis–body relation while keeping the record quality fixed; learning should lose its protective advantage. Remove actuation while keeping information; records may change but the geometry-dependent emitted source should not. Exhaust the actual reference or output bank while leaving an exact protected algebra; active revision can fail even though the old content persists. The first two separations

have explicit controls above. The third requires a finite resource protocol, not a declaration that all persisting information must decay.

9 Physical Interfaces and Their Remaining Burden

The results use different inputs. The Type III lift assumes a compact action, legal couplings, a calibration-to-rate rule and, for an exact modified semigroup, strict commutation. Its bounded generator follows from differentiability on every observable; an unbounded lift needs a specified core, not the same hypothesis under a new name. The persistence theorem needs one common compact-group action, a faithful invariant state, finitely many subgroups and divergent cumulative exposures. Peter–Weyl approximation is essential; arbitrary infinite-dimensional products of conditional expectations are not covered. Summable perturbations can change the final content inside the common algebra. Persistent forcing is handled by a separate risk bound.

Physical returns instead need actual instruments and admissible retained-state domains. Section 7.2 transports a proved quantum receiver renewal into its later bounded tasks and smooth interacting field source. It does not prepare the Gibbs state of an interacting body while leaving arbitrary held content untouched. Such universal preparation is obstructed by correlations, as shown in Q-RAIF C. The Gibbs cells of the scalar/JT example are used, not renewed; its second read deliberately uses a spent cell.

The strongest next connection would make a history-selected protective sector, a priced bank update and the later interacting quantum source parts of the same autonomous program. This requires actual clock arrival/readiness and local gate permission, finite interacting-resource preparation, and the controller/support stress matched to the geometry. A common classical action and receiver renewal are already available upstream; quantum-local matching, renormalized total stress entering JT and all-material thermal renewal are not. A formal thermal state, a receiver's fresh marginal, or a small output-field error cannot supply these missing structures. The scalar/JT return and the full quantum interface remain distinct implementations.

Foundation I supplies the selected hull and, under its translation conditions, an internal half-sided inclusion; standardness and the geometric dictionary remain separate inputs. Shared structure from Foundation II is not shared correctness without an identifiable reference. Neither functional calibration nor the physical return establishes experiential consciousness. These interfaces preserve the world, observer and first-person questions without deriving them from one qubit score.

A A Finite Arrival Bound at Fixed Hopping Energy

Lemma A.1 (Wholly idle input and a finite completion window)**.** *Consider a finite gate program ending at site $L \geq 0$, with identity gates before its first edge and throughout an output idle interval. For every fixed $J > 0$ and $0 < \eta < 1$, a finite path, a normalized input packet supported wholly before the program, and a finite time T can be chosen so that the cursor's incomplete-prefix amplitude is at most η. Its Hamiltonian is*

$$H_c = 2J\mathbf{1} - J\,\text{adjacency(path)}, \qquad 0 \leq H_c \leq 4J\mathbf{1}.$$

Gate dressing leaves this energy bound unchanged. If the completed prefix is W, the full data, cursor and any retained reference differ in vector norm by at most 2η from the same cursor wavefunction tensored with $W\psi$. The arrival error can be strictly below η on an open interval

about T. Path length, preparation and time are resources; the result makes no fixed-deadline claim.

Proof. On the infinite comparison path, choose momentum $k_0 = \pi/2$ and write $k = k_0 + q$. The dispersion is $2J(1 + \sin q)$. The envelope

$$\widehat{c}(q) = \sqrt{2\pi/\delta}\cos\Big(\frac{\pi q}{2\delta}\Big)\,\mathbf{1}_{[-\delta,\delta]}(q), \qquad 0 < \delta < \pi/2,$$

is normalized under $dq/(2\pi)$. Translate its position mean to an integer j_0; its position variance is $\pi^2/(4\delta^2)$. Linearizing the dispersion gives mean $j_0 + 2Jt$, the same variance, and vector error at most $Jt\delta^3/3$, since $|\sin q - q| \le |q|^3/6$. An idle interval with padding A around that translated mean therefore has outside amplitude at most $\pi/(2\delta A)$ for the linear evolution.

Truncate and normalize the input on $[j_0 - R, j_0 + R]$. The removed probability obeys $p_0 \le \pi^2/(4\delta^2 R^2)$, so the vector change is at most $\sqrt{2p_0}$. Put both path boundaries at least d hops from this support. In the hopping expansion, terms below order d cannot reach a severed edge. Duhamel and the exponential tail give the sufficient boundary estimate

$$e_{\mathrm{bd}}(T) \le 2JT\, e^{2JT}\frac{(2JT)^d}{d!}.$$

Consequently the actual finite cursor satisfies

$$\sqrt{p_{\mathrm{bad}}(T)} \le \frac{\pi}{2\delta A} + \frac{JT\delta^3}{3} + \sqrt{2p_0} + e_{\mathrm{bd}}(T).$$

Choose $j_0 = -R - 3$, leaving the forwardmost input site at -3. The output interval lies after site L if $2JT \ge L + R + A + 3$. With $S = JT$, take

$$\delta = (3\eta/(4S))^{1/3}, \quad R = \left\lceil \frac{2\sqrt{2}\pi}{\delta\eta} \right\rceil, \quad A = \left\lceil \frac{2\pi}{\delta\eta} \right\rceil, \quad S \ge \max\{L + 5, 100\eta^{-2}\}.$$

The first three errors are at most $\eta/4$. Moreover $R + A < 17S^{1/3}\eta^{-4/3} + 2$, so this choice satisfies arrival. A finite d makes the boundary term at most $\eta/4$ because the factorial dominates at fixed S. Applying the construction with a smaller target gives strict margins; enlarging the idle interval slightly gives an open completion window by continuity. The endpoint coefficient can be chosen nonzero: the inverse Fourier coefficient at $j_0 + R$ is analytic in δ and not identically zero. An arbitrarily small change of δ avoids its isolated zeros while retaining the strict margins and the finite support.

For gate dressing, let W_j be the prefix at site j. The diagonal gauge $\sum_j |j\rangle\langle j| \otimes W_j$ conjugates the gate-labeled hopping to H_c. The input is wholly idle, and $W_j = W$ on the output interval. The untrimmed joint state is $\sum_j c_j(T)|j\rangle W_j\psi$. Its difference from $\sum_j c_j(T)|j\rangle W\psi$ has squared norm at most $4p_{\mathrm{bad}}$, including an arbitrary old reference. No cursor postselection or erasure is used. □

This quantitative estimate specializes standard Feynman cursor transport to the retained-history return. Fixed J bounds the cursor energy while increased accuracy consumes time, space and preparation. The positive shift and all data/battery energies remain in the apparatus account; a finite path eventually reflects.

B Finite Return Protocol

The numerical realization of Section 7 uses body qubits B_1, B_2, probe P, record R and cells E_0, E_1, in that order. The old logical reference and the two cell purifications are retained throughout. The initial logical/reference state is $(|01\rangle|0\rangle + |10\rangle|1\rangle)/\sqrt{2}$. Each cell and its reference start in $\sqrt{0.9}|00\rangle + \sqrt{0.1}|11\rangle$; P is $|+X\rangle$ and R is $|0\rangle$.

The clock sites are $-30, \dots, 70$ and the forward-edge gates, at sites $0, 2, 4, 6, 8, 10, 12$, are respectively the early phase contact, plant rotation, $C_Y(P, R)$, the frame, late phase contact, decoder and $C_X(P, E_0)$. Every reverse edge carries the adjoint gate. Other edges carry the identity. At input, with $\delta = 0.6$ and $a = \pi/(2\delta)$, the clock amplitudes are proportional to

$$e^{i\pi j/2}\left[\operatorname{sinc}\big(\delta(j+10+a)\big) + \operatorname{sinc}\big(\delta(j+10-a)\big)\right]$$

for $|j + 10| \le 7$, and vanish elsewhere. Here $\operatorname{sinc}(z) = \sin z/z$; the finite vector is normalized. Its support is wholly before the program. The clock is not measured to select completed trajectories.

Mechanical parameters are $I = 10$, $J_m = 20$, $k = 1$, $K = 0.1$, with potential $k(\theta - y)^2/2 + Ky^2/2$ and $P_+ + P_- = \gamma\dot{\theta}^2$, $\gamma = 1/(2\pi)$. Take $L = 1$, $\kappa = 0.01$, $\Phi_0 = 3$, $M_0 = 10$, $e_m(0) = 0.95$, $y_0 = 0.5$, $d_1 = -0.5$, $d_2 = 0.5$ and $\theta_0 = \dot{\theta}_0 = \dot{\Phi}_0 = 0$. The velocity $\dot{y}_0$ supplies the remaining mechanical energy. Use $U(\Phi) = 50(\Phi - \Phi_*)^2$, where $\Phi_* = 3.0750469043\ldots$ is fixed by the reference junction equilibrium. The nominal task and self-source comparison uses $\mathcal{C}_-(0) = 4.7975$; the separate background tests use 4.6975 and 4.8975. In each case $\mathcal{C}_+$ is fixed by $\beta_- - \beta_+ = \kappa M_0$, not specified independently. Both backgrounds use the same initial core and mechanical state.

The initial body spring energy is 0.005, the bare cell energy is 0.01, and the offset is 0.04. With the clock this gives $E_c(0) = 1.655$ and $m_{\text{rest}} = 7.1133981076\ldots$. The six policy/sign trajectories, the same-input α pair, and the matched geometric comparison are integrated over proper time $[0, 30]$, retaining all purification columns. Work integrals for the mechanical force, sensor and support potential independently check Eq. (6) and both junction constraints. Removing the springs also removes their initial energy from $E_c(0)$ and adjusts the rest mass to keep M_0 fixed. The two code policies, initial state, finite path and endpoint score are unchanged in these controls. The complete numerical reconstruction, prefix calculation and paired propagations are in the source supplement `supplemental/finite_return/`. Actual trajectories evolve the mass equation and independent mechanical, core and support work integrals, so the mass account is checked against its conserved definition as well as both junctions. Two integrators and a tighter tolerance change the paired geometric reading by less than 1.1×10^{-15}. The tighter propagation has maximum absolute energy-account residual 7.2×10^{-11}; the nominal six-policy run has residual below 1.9×10^{-10}. These are numerical consistency checks, not certified interval bounds.

C Feedback background

Definition C.1 (Feedback regularity package)**.** Let S_{source} be a source datum and let $\delta\sigma$ be an admissible calibration signal. A *feedback regularity package* consists of a family

$$\mathscr{F}^{[\delta\sigma]} = \{\mathscr{F}_r^{[\delta\sigma]}\}_{r\ge 0}$$

of normal maps on $\mathcal{A}_U$ satisfying the following conditions.

1. **FB-A: CP source lift.** $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, with

$$\mathscr{F}_0^{[\delta\sigma]} = \mathrm{id}_{\mathcal{A}_U}, \qquad \mathscr{F}_r^{[0]} = \mathrm{id}_{\mathcal{A}_U} \quad \forall r \geq 0.$$

2. **FB-B: small-signal differentiability.** For every $A \in \mathcal{A}_U$, the map $r \mapsto \mathscr{F}_r^{[\delta\sigma]}(A)$ is σ-weakly differentiable at $r = 0$. We denote the derivative by

$$\mathfrak{G}_{\delta\sigma}(A) := \frac{d}{dr}\mathscr{F}_r^{[\delta\sigma]}(A)\bigg|_{r=0},$$

   and record the normalization $\mathfrak{G}_{\delta\sigma}(\mathbf{1}) = 0$, which follows from the unitality in FB-A because $\mathscr{F}_r^{[\delta\sigma]}(\mathbf{1}) = \mathbf{1}$ for all $r \geq 0$.

3. **FB-C: chain compatibility.** The feedback lift commutes with the ambient source semigroup:

$$\mathscr{F}_r^{[\delta\sigma]}\mathcal{E}_t = \mathcal{E}_t\mathscr{F}_r^{[\delta\sigma]} \qquad \forall r, t \geq 0.$$

   For a return through Foundation I, also specify which seed, physical permissions and hull family are retained. Every stationary, translation or standardness assumption used for that return must hold for the modified source. These are additional compatibility conditions, not consequences of commutation alone.

Proposition C.2 (The feedback derivative is bounded)**.** *The every-observable derivative in FB-B makes $\mathscr{F}^{[\delta\sigma]}$ uniformly continuous in map norm, with bounded normal generator $\mathfrak{G}_{\delta\sigma}$. It is stronger than differentiability on a generator core.*

Proof. Fix the signal and set $B_r = (\mathscr{F}_r - \mathrm{id})/r$ for $0 < r \leq 1$. For each x and each normal functional φ, FB-B bounds $\varphi(B_r x)$ near zero; contractivity bounds it away from zero. Uniform boundedness on the predual gives $\sup_{0<r\leq 1}\|B_r x\| < \infty$. A second uniform-boundedness argument on the Banach space $\mathcal{A}_U$ gives $\sup_{0<r\leq 1}\|B_r\| < \infty$. Hence $\|\mathscr{F}_r - \mathrm{id}\| = O(r)$. The uniformly continuous semigroup theorem yields a bounded generator, and its norm derivative equals the FB-B weak derivative. A map-norm limit of normal maps is normal. □

Remark C.3 (Meaning of the strict commutation input). For $\eta > 0$,

$$\mathcal{E}_t^{[\eta]} = \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t$$

can differ from $\mathcal{E}_t$. FB-C makes this composition an autonomous semigroup; noncommuting operations require time ordering. The first-order map below is a tangent of this exact CP operation.

Proposition C.4 (A nontrivial commuting-dephasing feedback lift)**.** *The package FB-A–FB-C has a finite-dimensional normal UCP realization that depends nontrivially on $\delta\sigma$ and changes the source dynamics beyond a relabeling.*

Proof. On $\mathcal{A}_U = M_2 \otimes M_2$, let Z_j act on factor j. For $\gamma, \kappa > 0$, put $q(\delta\sigma) = \|\delta\sigma\|^2$ and

$$\mathcal{L}_0 = \gamma(\mathrm{Ad}_{Z_1} - \mathrm{id}), \qquad \mathfrak{G}_{\delta\sigma} = \kappa q(\delta\sigma)(\mathrm{Ad}_{Z_2} - \mathrm{id}),$$

The feedback semigroup is

$$\mathscr{F}_r^{[\delta\sigma]}(A) = a_r A + (1 - a_r)Z_2 A Z_2, \qquad a_r = \frac{1 + e^{-2\kappa q(\delta\sigma)r}}{2}.$$

It is normal UCP and differentiable, with the stated generator, and is the identity at zero signal. The same formula with $(Z_2, \kappa q, r)$ replaced by (Z_1, γ, t) defines $\mathcal{E}_t$. The two Pauli conjugations commute, verifying FB-C. Both fix the joint diagonal algebra, where the retained structural inputs may be placed. For $A = \mathbf{1} \otimes X$,

$$\mathcal{E}_t(A) = A, \qquad \mathcal{E}_t^{[\eta]}(A) = e^{-2\eta\kappa q(\delta\sigma)t} A.$$

Thus nonzero feedback adds a decay channel in this finite-dimensional realization. The following proposition implements a lift directly on the source algebra. □

Remark C.5 (What strict commutation can and cannot add). Proposition C.4 shows that FB-C does not force the lift to be the identity. It also displays the limitation: the commuting generators are simultaneously diagonal on joint Pauli sectors, so feedback only reweights pre-existing joint modes and does not mix them. A time-ordered sequence need not obey FB-C. Its algebraic limit is treated in Section 4; its moving-state realization is treated in Section 5.

Definition C.6 (Small-coupling regime and modified semigroup)**.** Fix a feedback regularity package $\mathscr{F}^{[\delta\sigma]}$. A coupling strength $\eta \geq 0$ is in the *small-coupling regime* if $0 \leq \eta < \eta_*$ for some window $\eta_* > 0$ on which the later comparison inputs remain valid. For a first-order interpretation on a return horizon $[0, T]$, we additionally require $\eta T \ll 1$. The exact composed semigroup below exists without this perturbative estimate, but its tangent need not be uniformly small at long times. For such η, define the *feedback-modified semigroup* by

$$\mathcal{E}_t^{[\eta]} := \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t \qquad (t \geq 0).$$

The corresponding *modified source datum* is

$$S_{\text{source}}^{[\eta]} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t^{[\eta]}\}_{t \geq 0}).$$

Remark C.7 (Heisenberg-picture convention). All semigroups in Definition C.6 act on observables. The ambient source $\mathcal{E}_t$ is only assumed pointwise σ-weakly continuous. FB-B makes the auxiliary feedback $\mathscr{F}_r$ norm-continuous by Proposition C.2; it imposes no bounded-generator condition on the ambient source.

Definition C.8 (First-order feedback map)**.** Under FB-B, define the *first-order feedback map* by

$$\Gamma(\delta\sigma, \mathcal{E}_t)(A) := t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)), \qquad A \in \mathcal{A}_U.$$

Equivalently, for each fixed $t \geq 0$,

$$\mathcal{E}_t^{[\eta]}(A) = \mathcal{E}_t(A) + \eta\, \Gamma(\delta\sigma, \mathcal{E}_t)(A) + o(\eta)$$

in the σ-weak topology as $\eta \downarrow 0$.

Verification of the expansion. By Definition C.6,

$$\mathcal{E}_t^{[\eta]}(A) = \mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)).$$

Applying the $r = 0$ derivative from FB-B at $r = \eta t$ gives

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)) = \mathcal{E}_t(A) + \eta t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)) + o(\eta),$$

which is exactly the stated formula. □

Remark C.9 (The tangent is not uniform at long times). The factor t controls this tangent at fixed time, or on windows $\eta t \ll 1$ under a uniform generator bound. It gives no infinite-time estimate. An arbitrary normal map need not be an admissible CP tangent; unbounded perturbations require a specified core.

D Quantum continuation source record

The imported continuation and renewal statements, full assumptions and proofs are retained in `FIII_quantum_imports.tex` in the supplemental source notes, with Q-RAIF C as their primary paper [3]. They use a positive physical Hamiltonian, a common material rms bound, smooth tested source forms, retained Gibbs-bank purifiers and the complete cursor. They do not certify the mean-action return below.

E Conditional Source Interfaces

The source datum is a standard Type III_1 triple and a pointwise σ-weakly continuous normal UCP semigroup. Foundation I constructs a persistent fixed-algebra projection for a stationary source, the seed's modular stable hull, and, under an ambient half-sided translation action with unique invariant vacuum, an internal half-sided inclusion in each nonscalar translation-closed hull. The finite-coordinate rank-cost and strict-improvement assumptions belong to its foreground coding theorem. Standardness is still needed for the corresponding net. These are named inputs and conclusions of Foundation I, not a geometry derived by the feedback lift.

T-DOME III supplies local reference/probe calibration signals and particular drift-control estimates. Self-model residuals alone can make the local metric vanish at calibration. Each use here must remain in its declared calibration domain; the diagonal-bank result separately uses the stated uniform L^2 estimation bound.

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